



KSBI
KOREAN SOCIETY FOR
BIOINFORMATICS

한국생명정보학회

KSBi-BIML 2025

Best Practices in Spatial Omics Data Analysis

Part I

Jeongbin Park

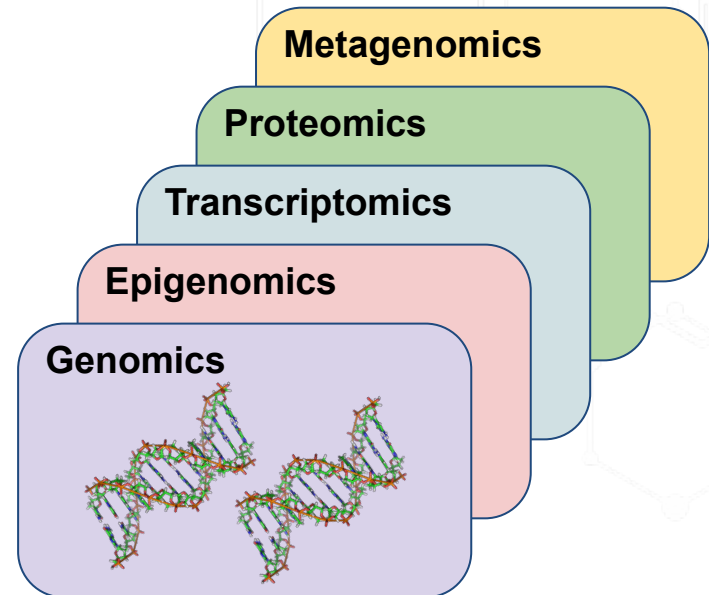
Pusan National University

Best Practices in Spatial Omics Data Analysis

- What is spatial omics?
- When should I consider spatial omics analysis?
- How should I design the experiments?

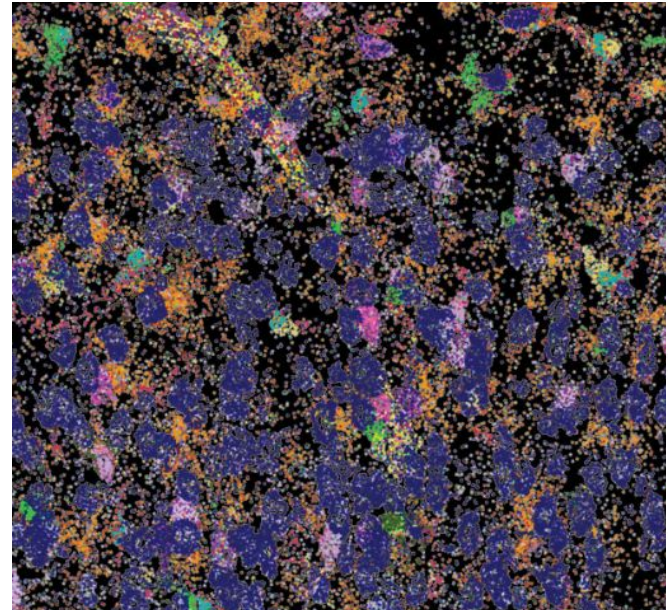
What is spatial omics?

- Methods that capture omics data in a spatial context
 - Transcriptomics, Genomics, Proteomics, Epigenomics, Metagenomics, etc.



Spatial Transcriptomics

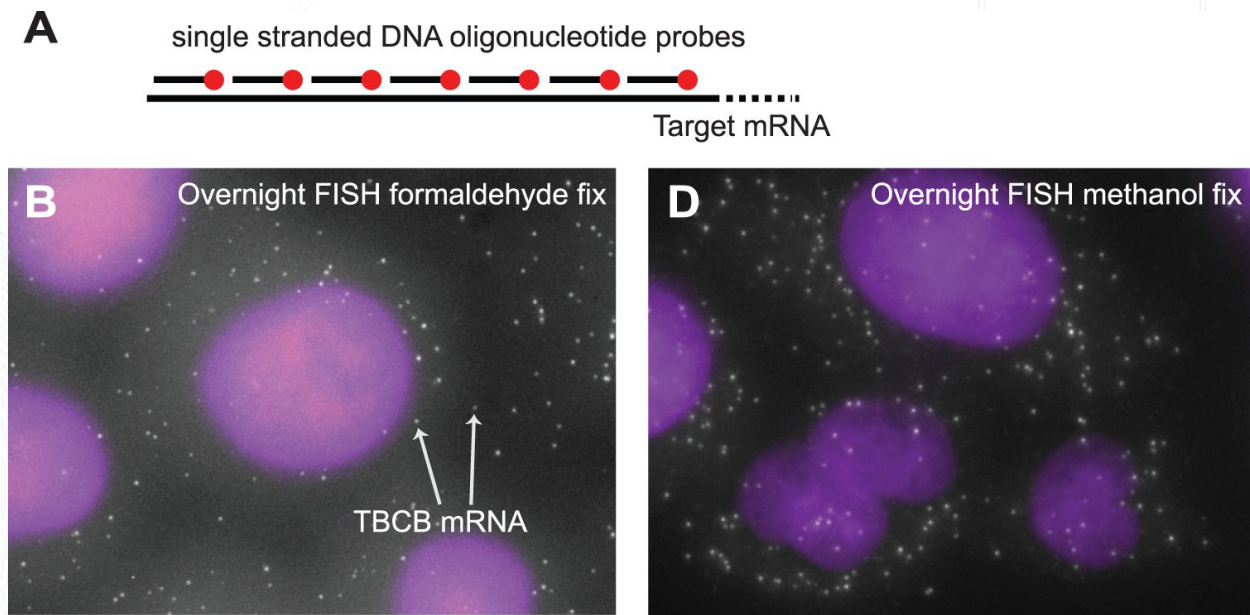
- Imaging-based methods
 - Fluorescence In Situ Hybridization (FISH)
 - In Situ Sequencing (ISS)



<https://linnarssonlab.org/osmFISH/>

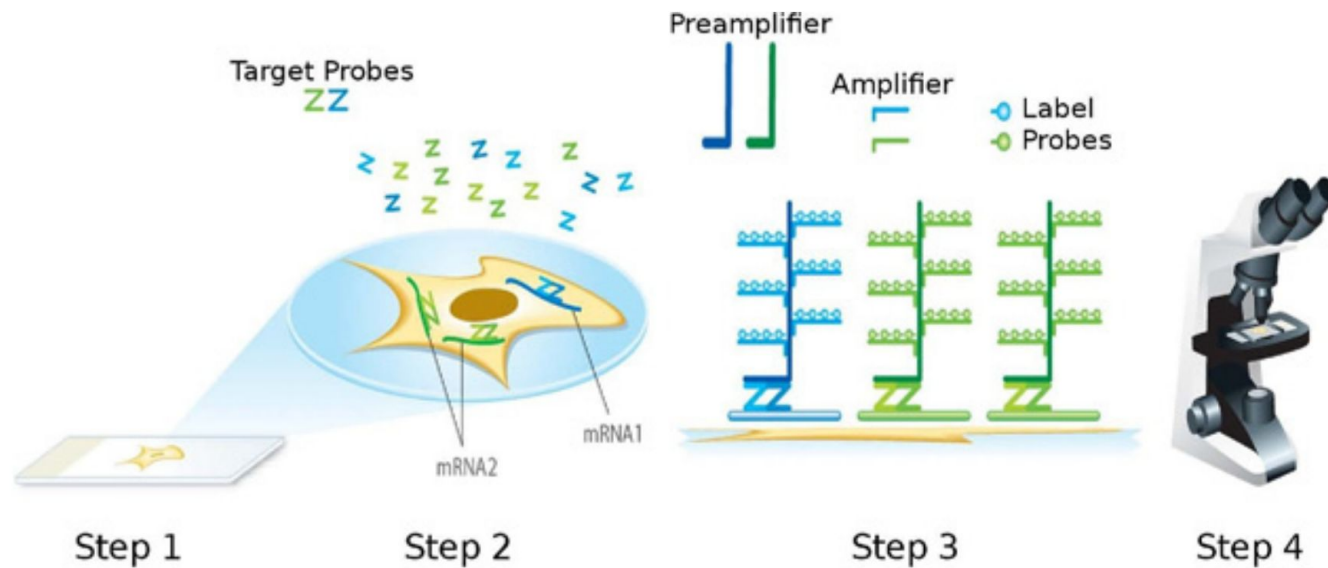
Spatial Transcriptomics

- cf. Fluorescence In Situ Hybridization (FISH)



Spatial Transcriptomics

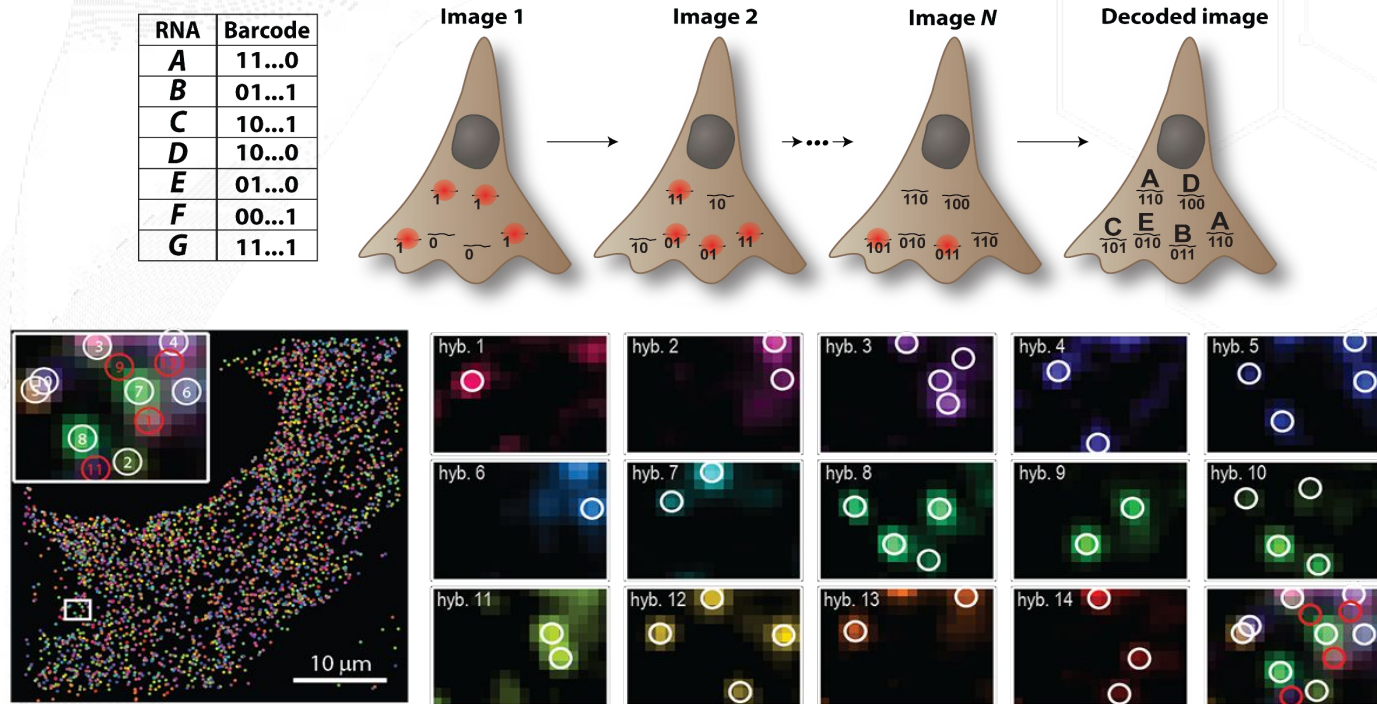
- RNAScope



[https://linkinghub.elsevier.com/retrieve/pii/S1525-1578\(11\)00257-1](https://linkinghub.elsevier.com/retrieve/pii/S1525-1578(11)00257-1)

Spatial Transcriptomics

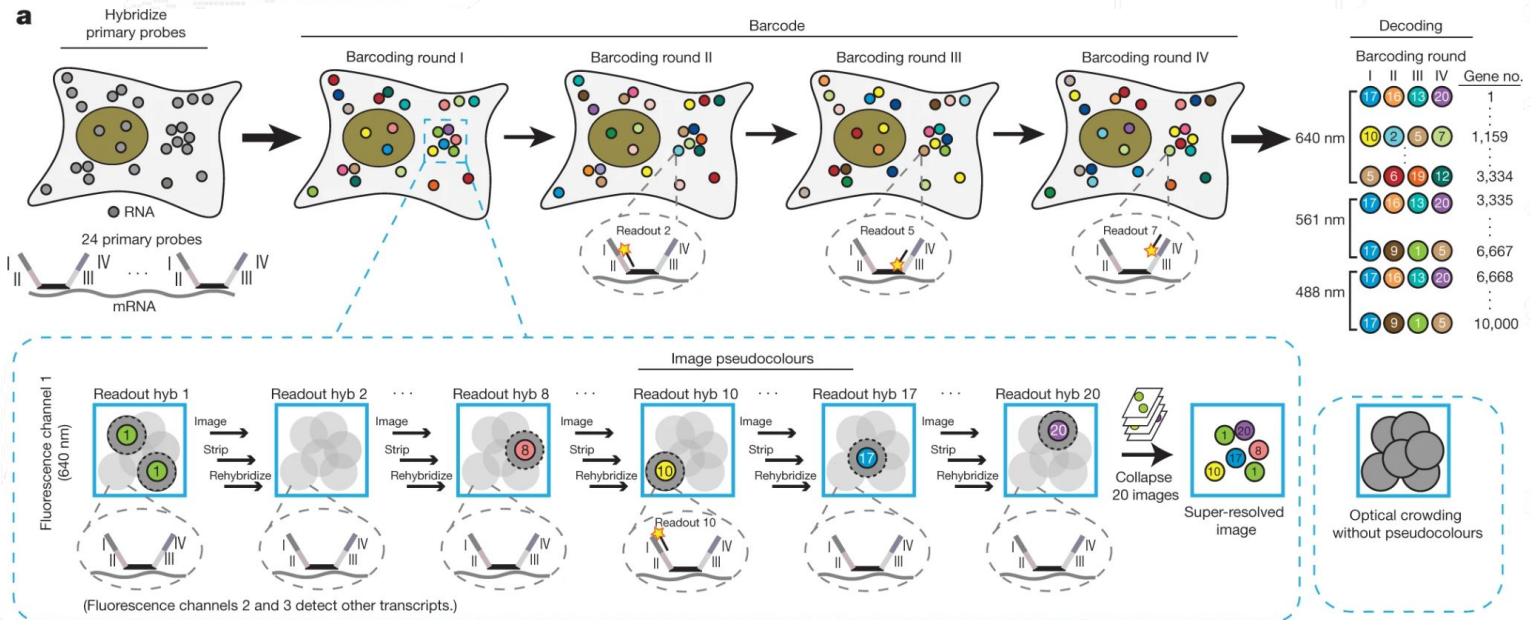
- MERFISH



<https://www.science.org/doi/10.1126/science.aaa6090>

Spatial Transcriptomics

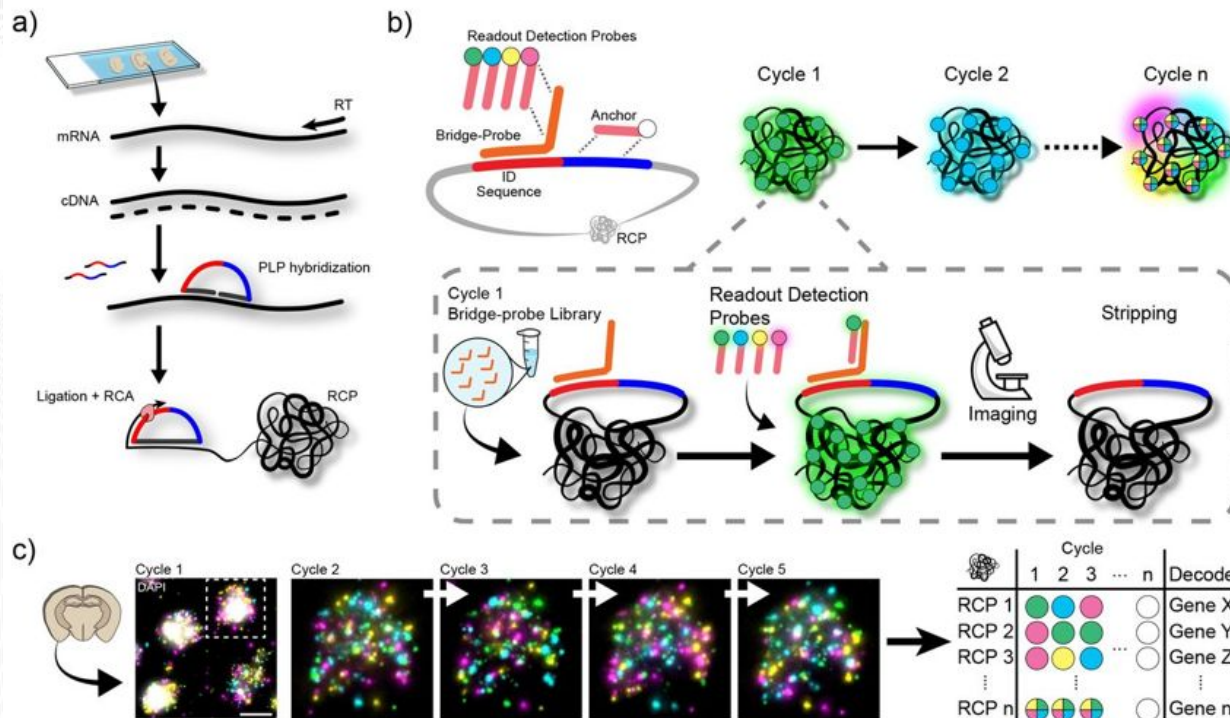
- SeqFISH+



<https://www.nature.com/articles/s41586-019-1049-y>

Spatial Transcriptomics

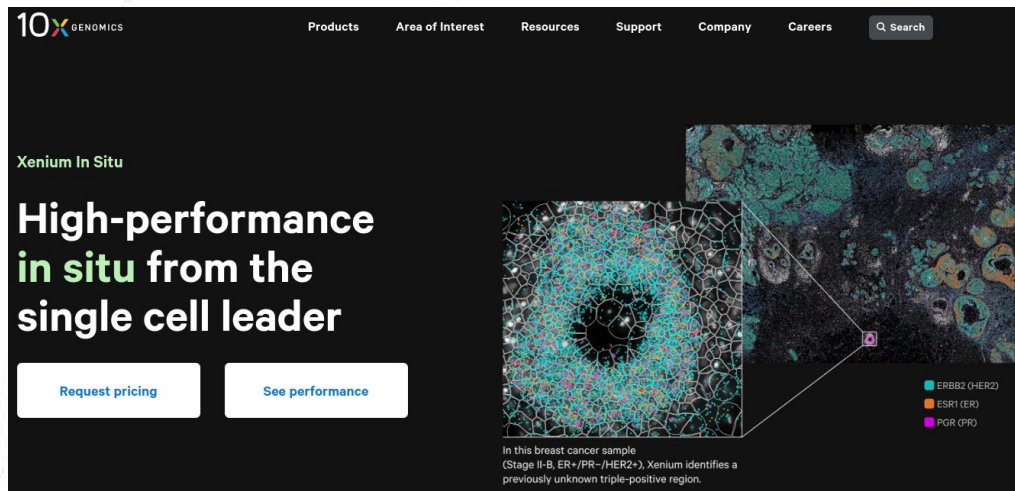
- cf. In Situ Sequencing (ISS)



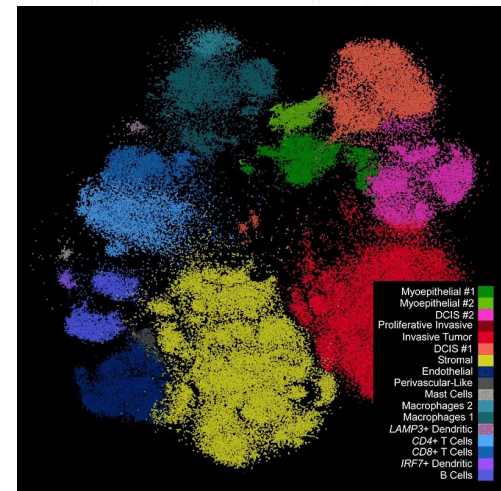
Gyllborg *et al*, Nucleic Acids Research (2020)

Spatial Transcriptomics

- 10X Xenium



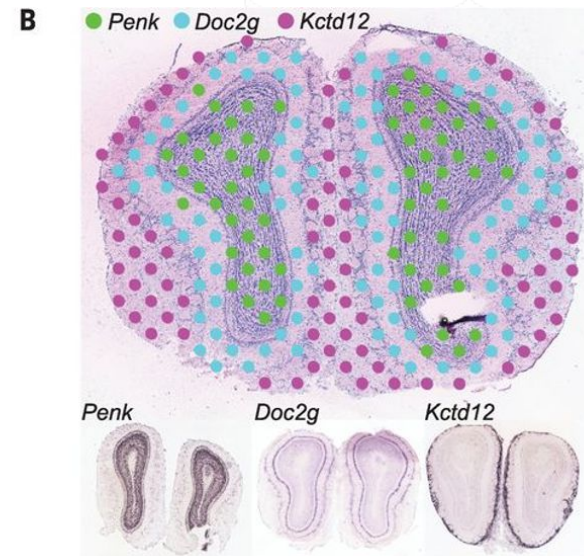
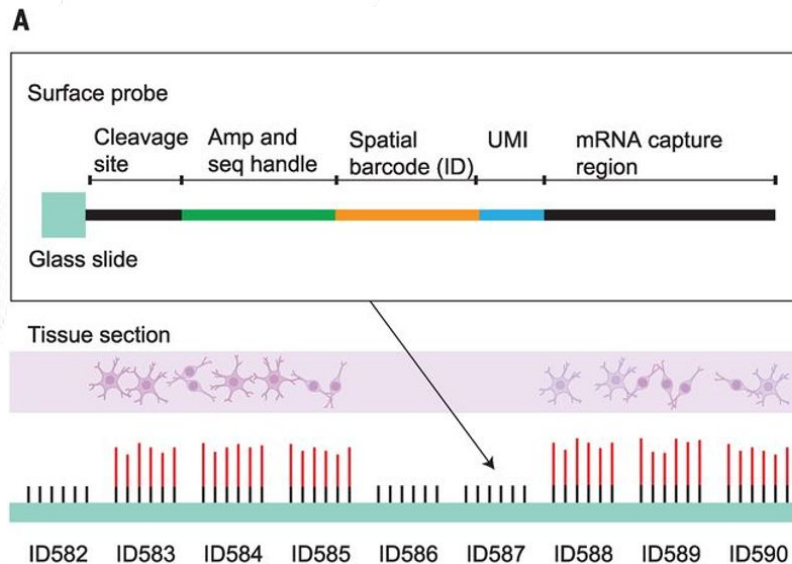
The screenshot shows the 10X Genomics website with a dark theme. The navigation bar includes links for Products, Area of Interest, Resources, Support, Company, and Careers, along with a search bar. The main heading is "Xenium In Situ" followed by the tagline "High-performance in situ from the single cell leader". Below this are two buttons: "Request pricing" and "See performance". The central image displays a spatial transcriptomics map of a breast cancer sample, with a legend on the right identifying cell types by color: ERBB2 (HER2) in teal, ESR1 (ER) in orange, and PGR (PR) in pink. A text box at the bottom of the image states: "In this breast cancer sample (Stage II-B, ER+/PR-/HER2+), Xenium identifies a previously unknown triple-positive region."



10X Genomics website

Spatial Transcriptomics

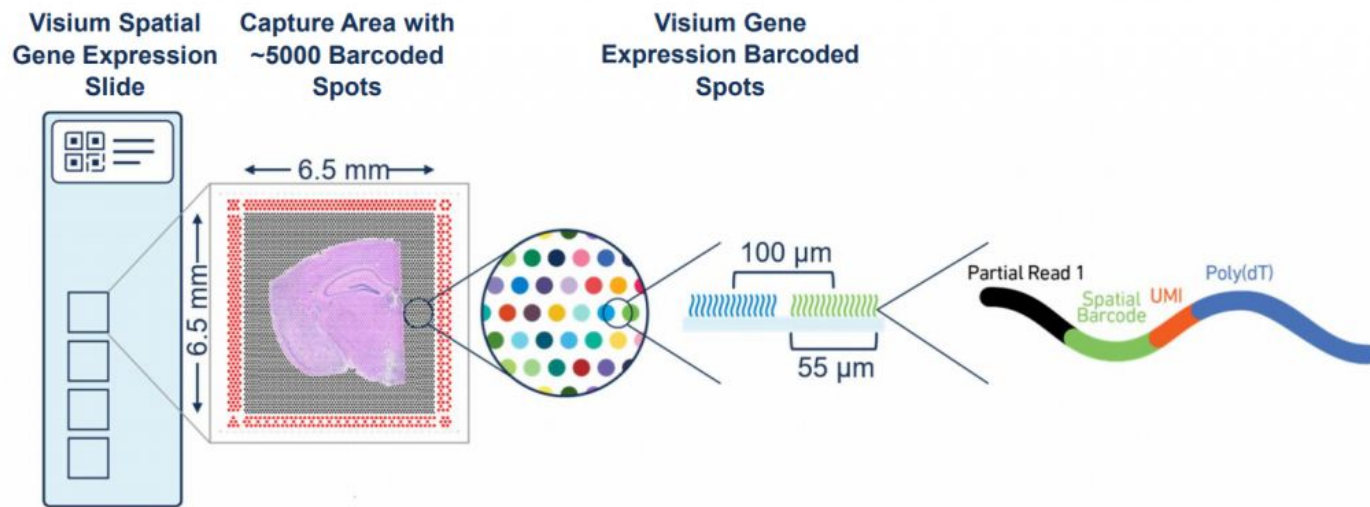
- NGS-based methods
 - “Spatial Transcriptomics” by Ståhl et al (2016)



<https://www.science.org/doi/10.1126/science.aaf2403>

Spatial Transcriptomics

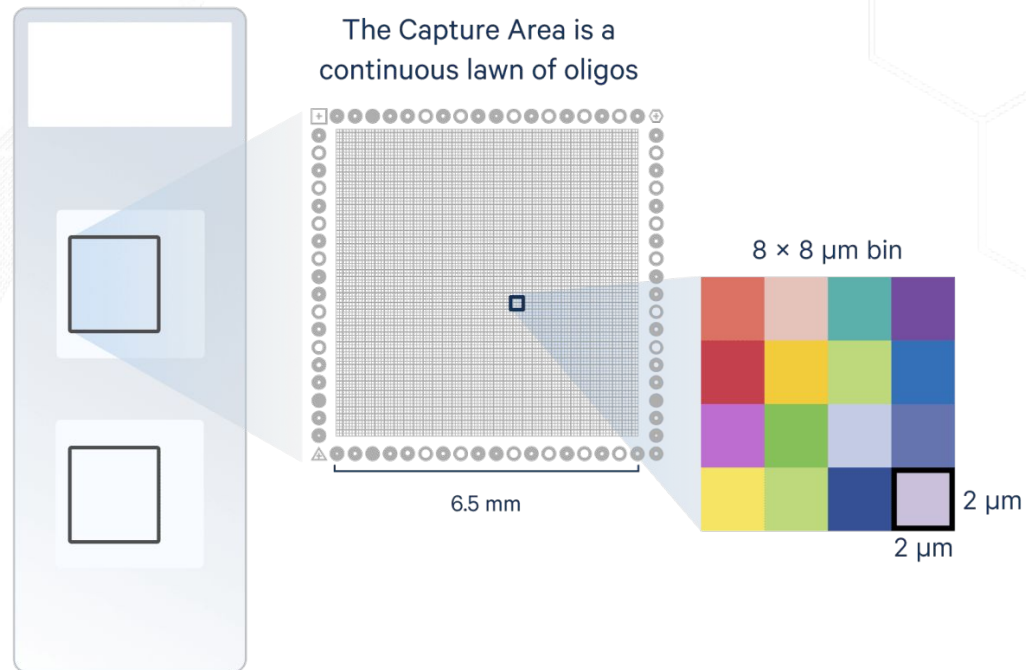
- 10X Visium



10X Genomics website

Spatial Transcriptomics

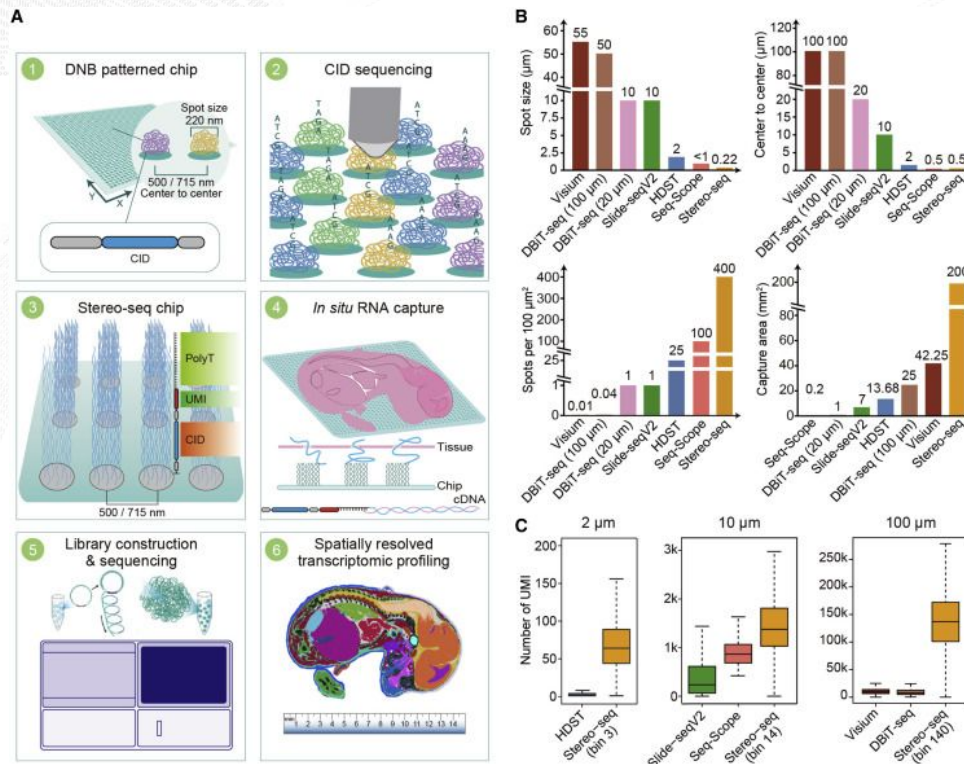
- 10X Visium HD



10X Genomics website

Spatial Transcriptomics

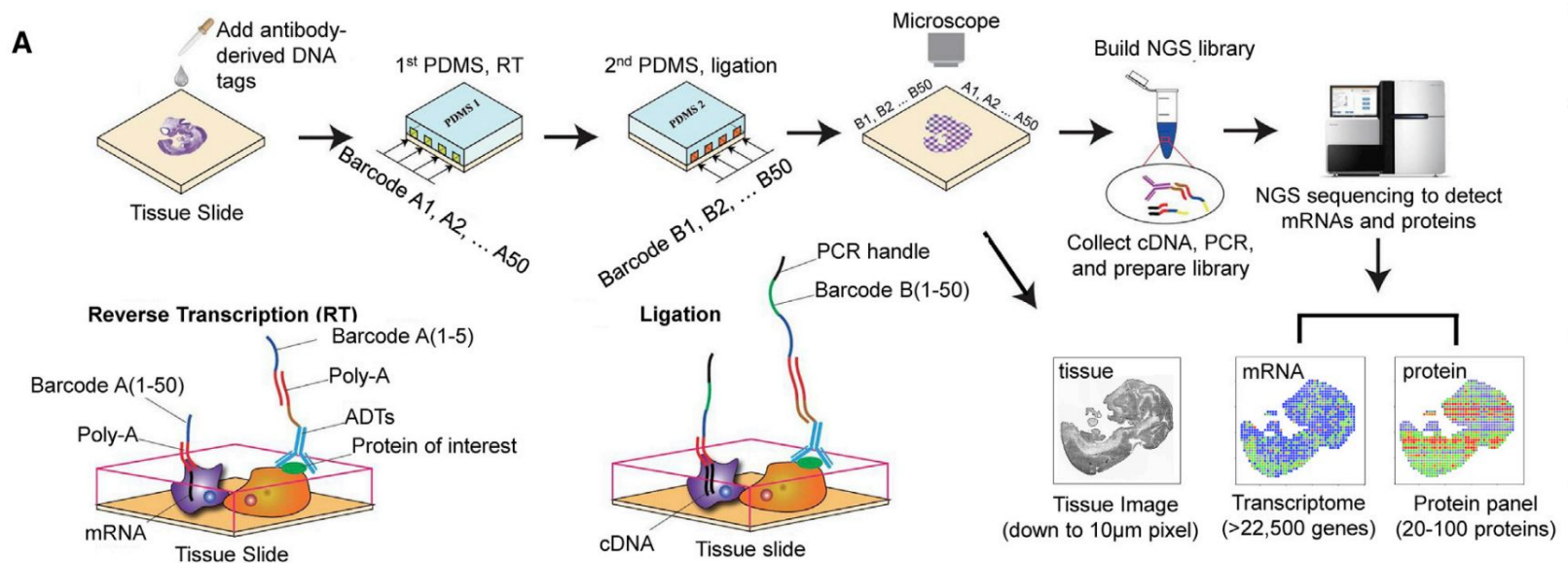
- Stereo-Seq (STOmics)



<https://doi.org/10.1016/j.cell.2022.04.003>

Spatial Transcriptomics

- DBiT-seq



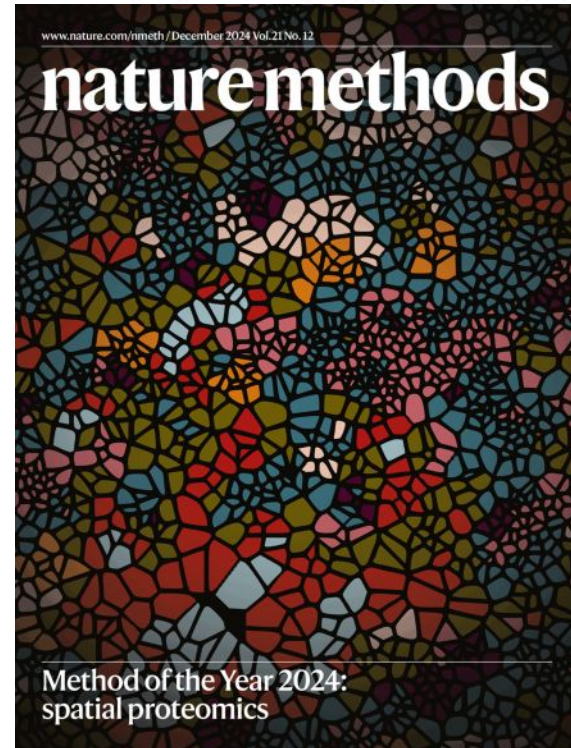
[https://www.cell.com/cell/fulltext/S0092-8674\(20\)31390-8](https://www.cell.com/cell/fulltext/S0092-8674(20)31390-8)

Spatial Transcriptomics

- Commercial products
 - ACD RNAScope
 - Vizgen MERSCOPE /MERSCOPE Ultra (i.e. MERFISH)
 - 10X Xenium / Xenium 5K (variant of ISS)
 - 10X Visium / Visium HD
 - MGI STOmics Stereo-seq / Stereo-seq OMNI
 - Nanostring GeoMx DSP / CosMx SMI
 - Spatial Genomics GenePS (i.e. SeqFISH)
 - Resolve Biosciences Molecular Cartography
 - AtlasXomics DBiT-Seq

Spatial Proteomics

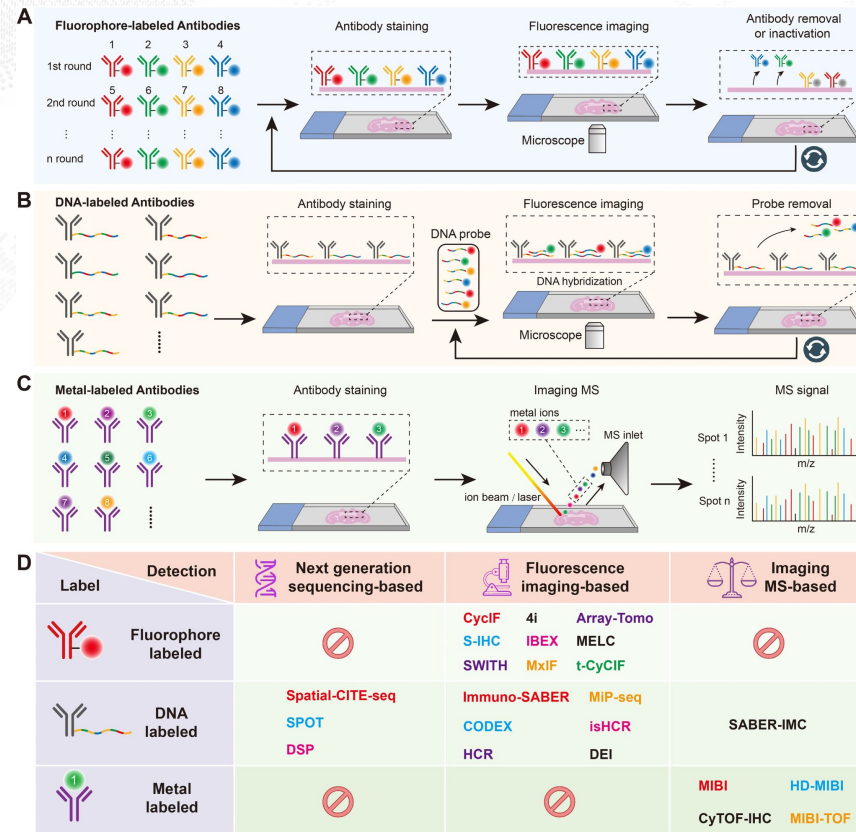
- Method of the year 2024!
 - Still lacks computational analysis methods
- Goals
 - Protein organization at single cell level
 - Protein-Protein Interactions (PPI) in the subcellular spatial context



<https://www.nature.com/articles/s41592-024-02565-3>

Spatial Proteomics

- Antibody-based methods

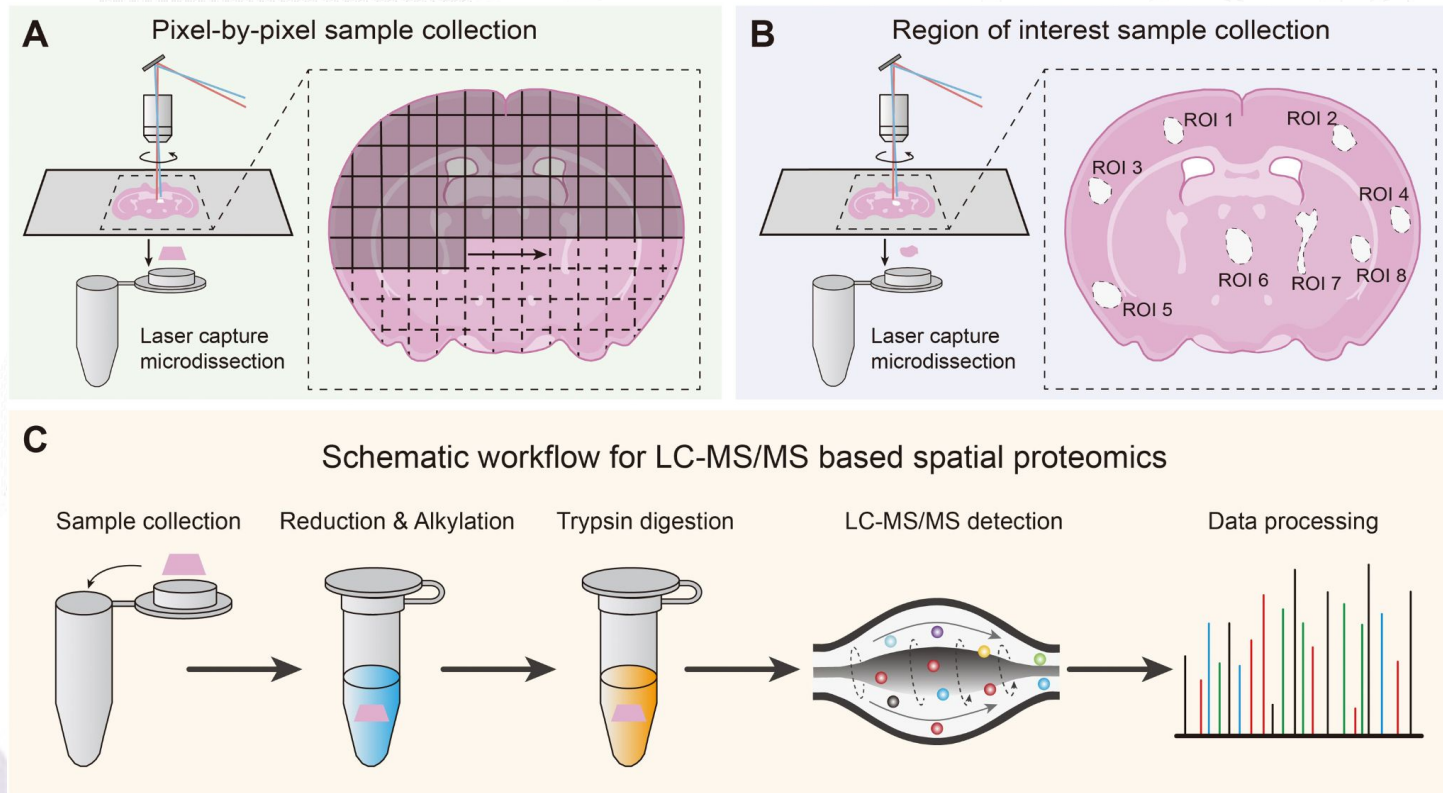


<https://www.sciencedirect.com/science/article/pii/S2667325824005132>

(very good review - recommend to read)

Spatial Proteomics

- LC-MS/MS based methods

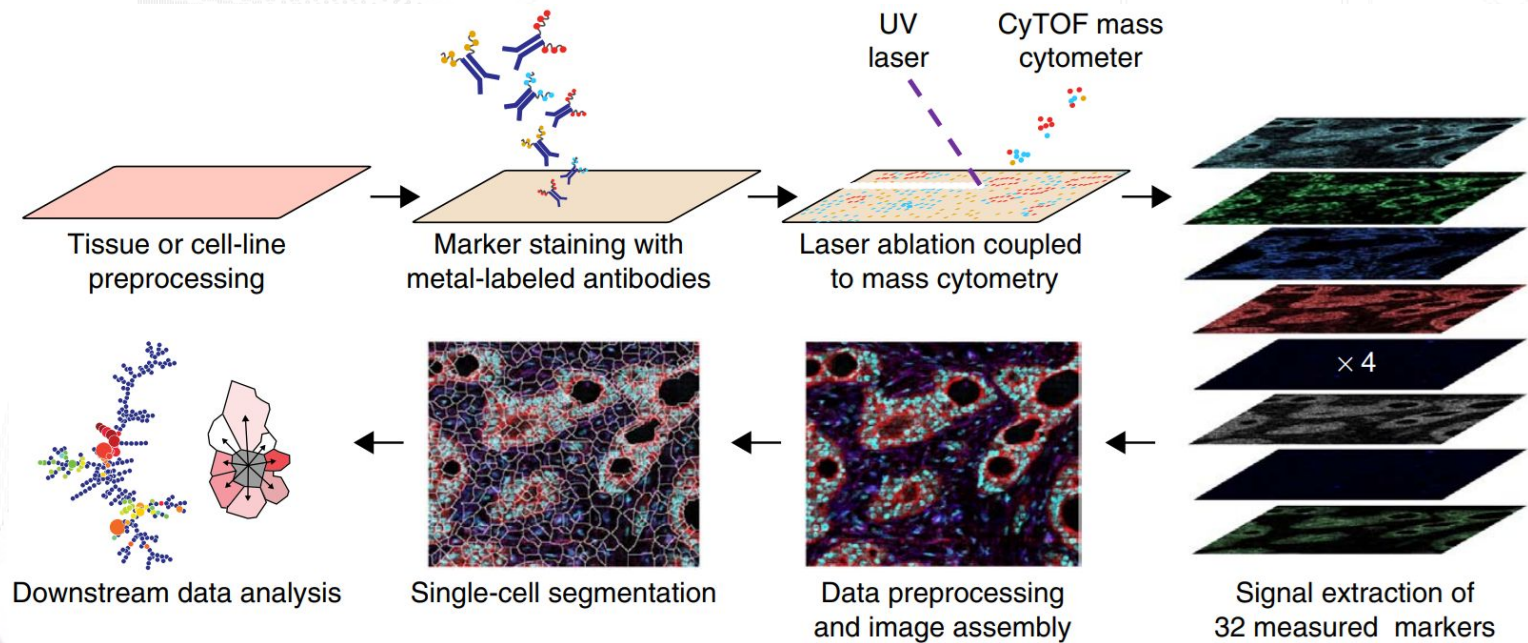


<https://www.sciencedirect.com/science/article/pii/S2667325824005132>

(very good review - recommend to read)

Spatial Proteomics

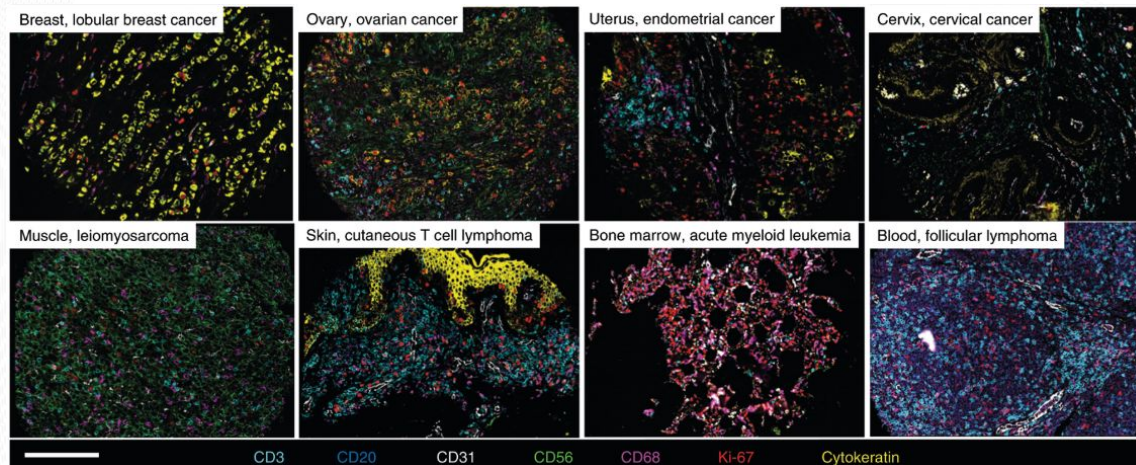
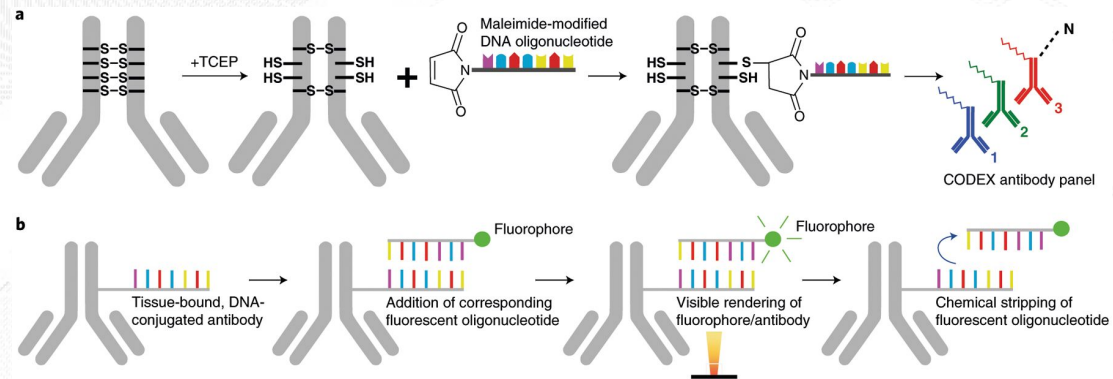
- cf. Imaging Mass Cytometry (IMC)



<https://www.nature.com/articles/nmeth.2869>

Spatial Proteomics

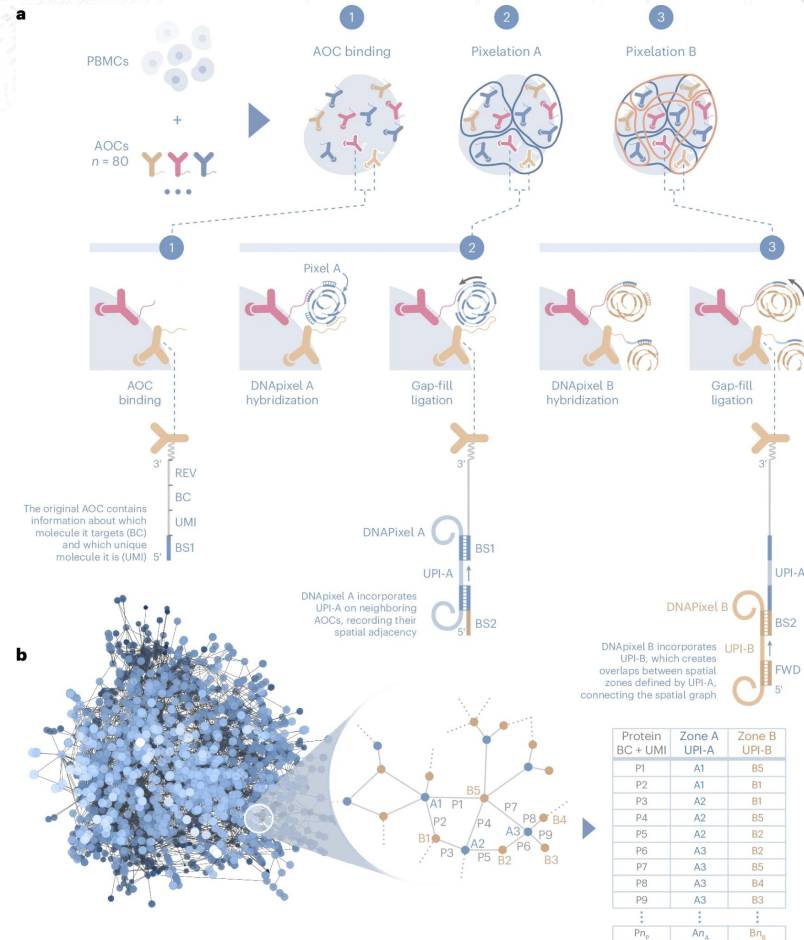
- cf. Co-Detection by Indexing (CODEX)



<https://pmc.ncbi.nlm.nih.gov/articles/PMC8647621/>

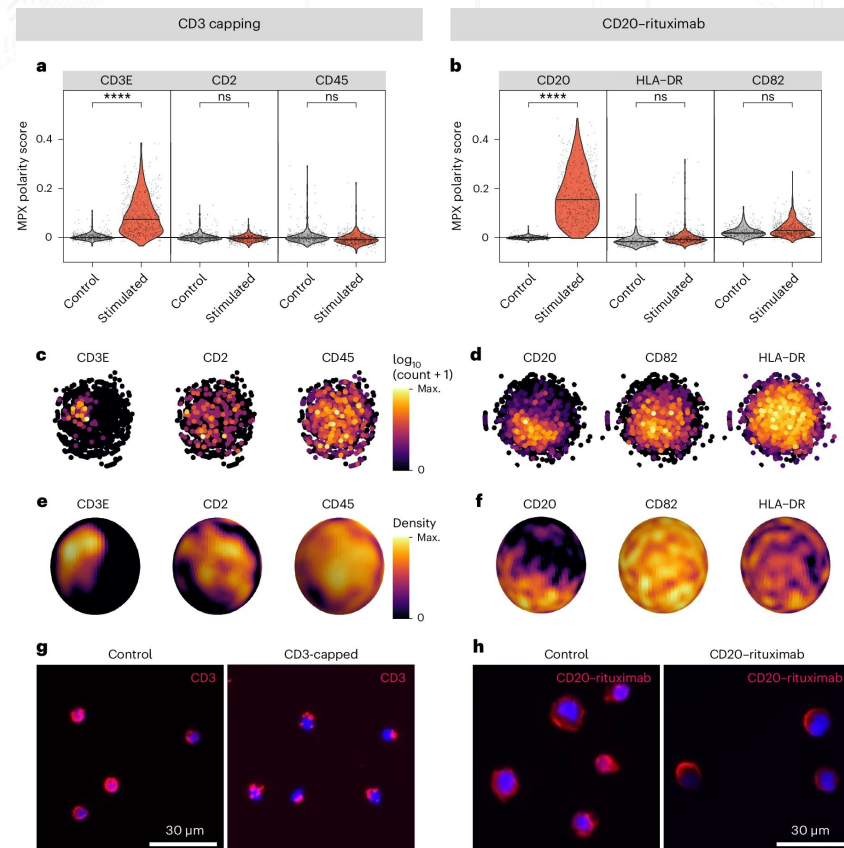
Spatial Proteomics

- cf. Molecular pixelation
 - Detects surface marker distribution on single cells



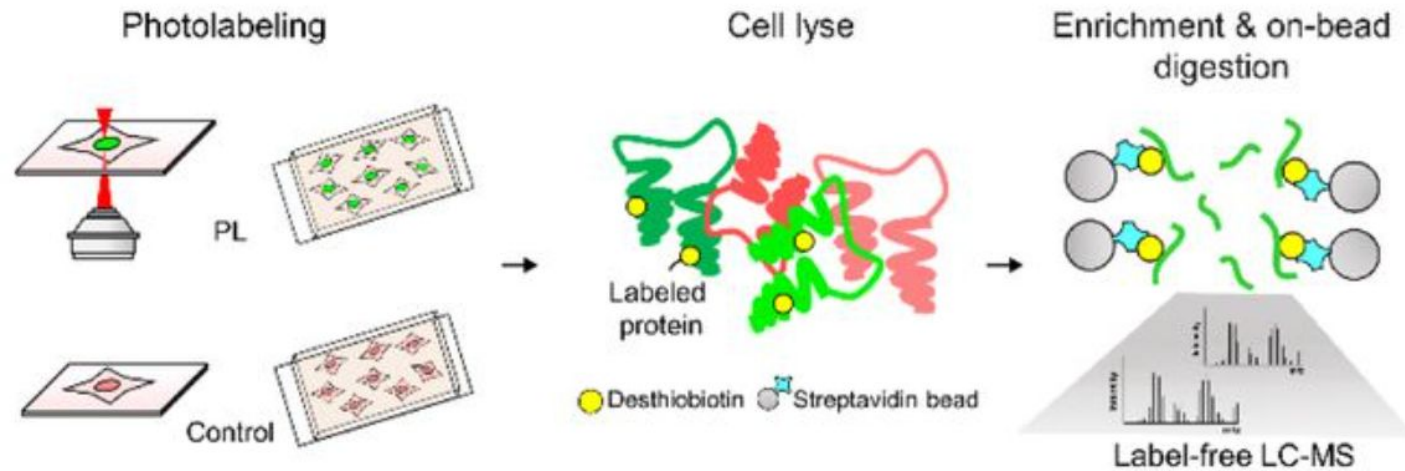
Spatial Proteomics

- cf. Molecular pixelation
 - Detects surface marker distribution on single cells



Spatial Proteomics

- cf. Targeted Photo-biotinylation



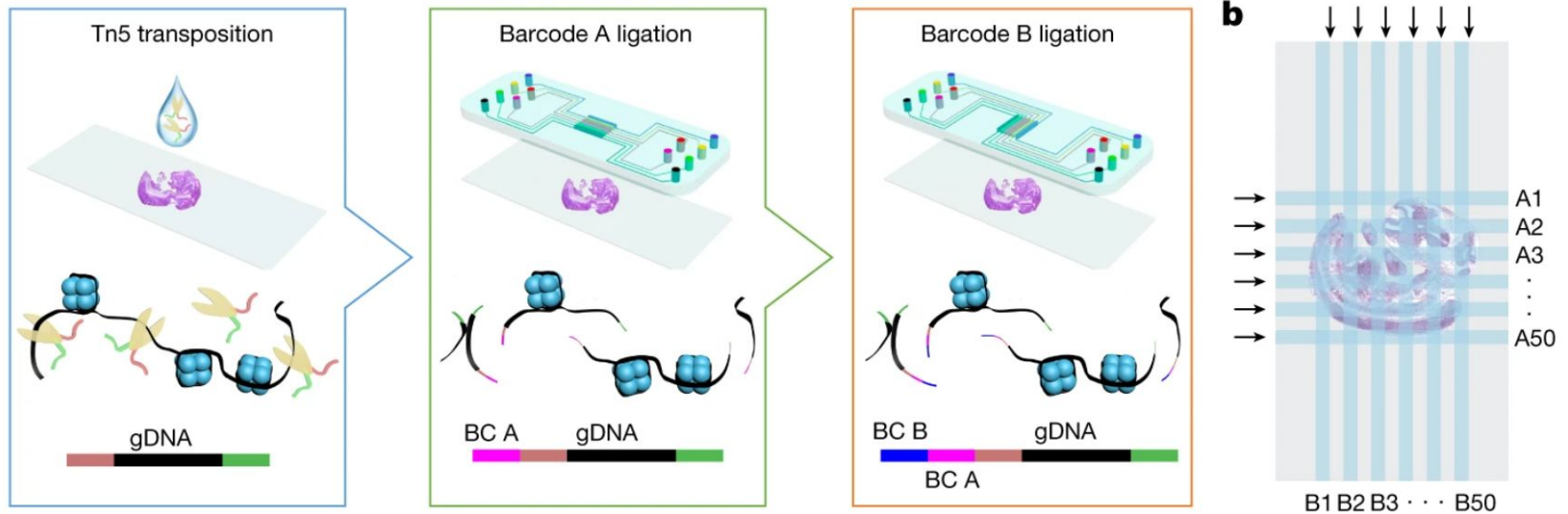
(resolution: ~200nm)

Spatial Proteomics

- Commercial Products
 - Akoya PhenoCycler (formerly CODEX)
 - Nanostring GeoMx IO Proteome Atlas / CosMx Protein Assays
 - IONPath MIBIscope
 - 10X Visium CytAssist Spatial Gene and Protein Expression assay
 - Canopy CellScape
 - Miltenyi Biotec MACSima Imaging Cyclic Staining (MICS)
 - Pixelgen Spatial Proteomics (i.e. Molecular Pixelation)
 - Syncell Microscoop (i.e. Targeted Photo-biotinylation)

Spatial Epigenomics

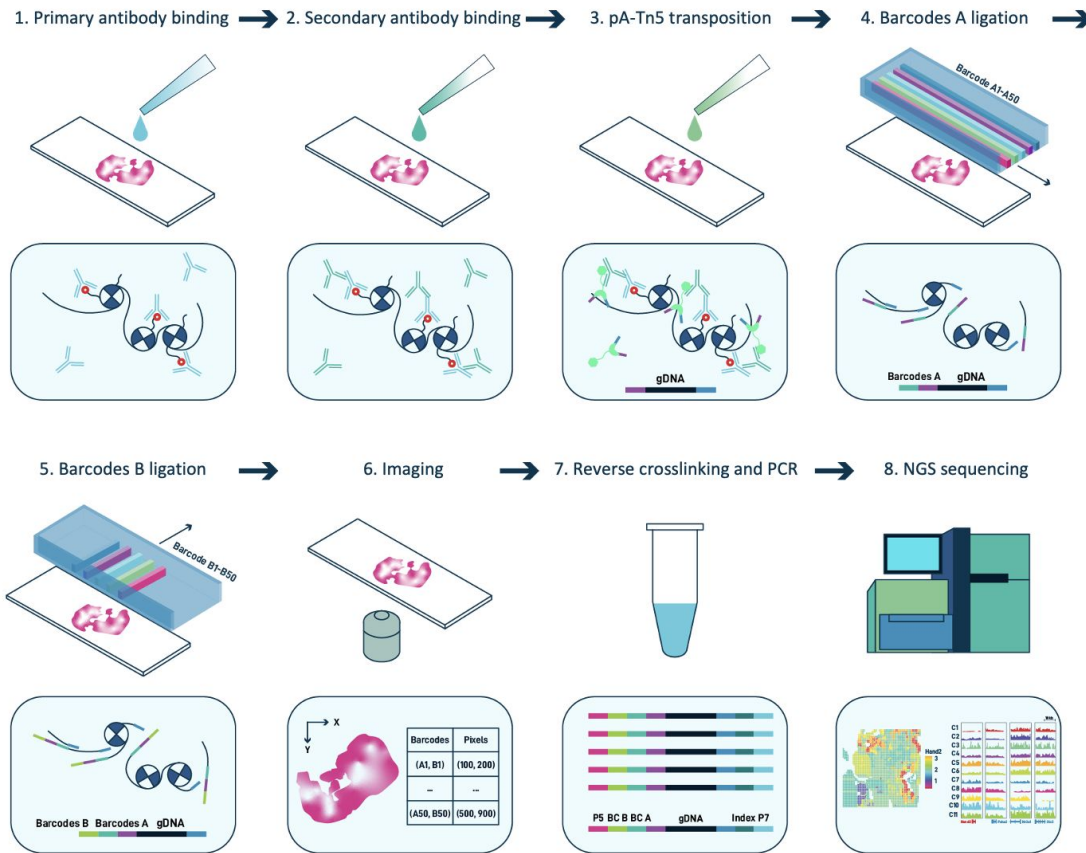
- AtlasXomics Spatial ATAC-seq



<https://www.nature.com/articles/s41586-022-05094-1>

Spatial Epigenomics

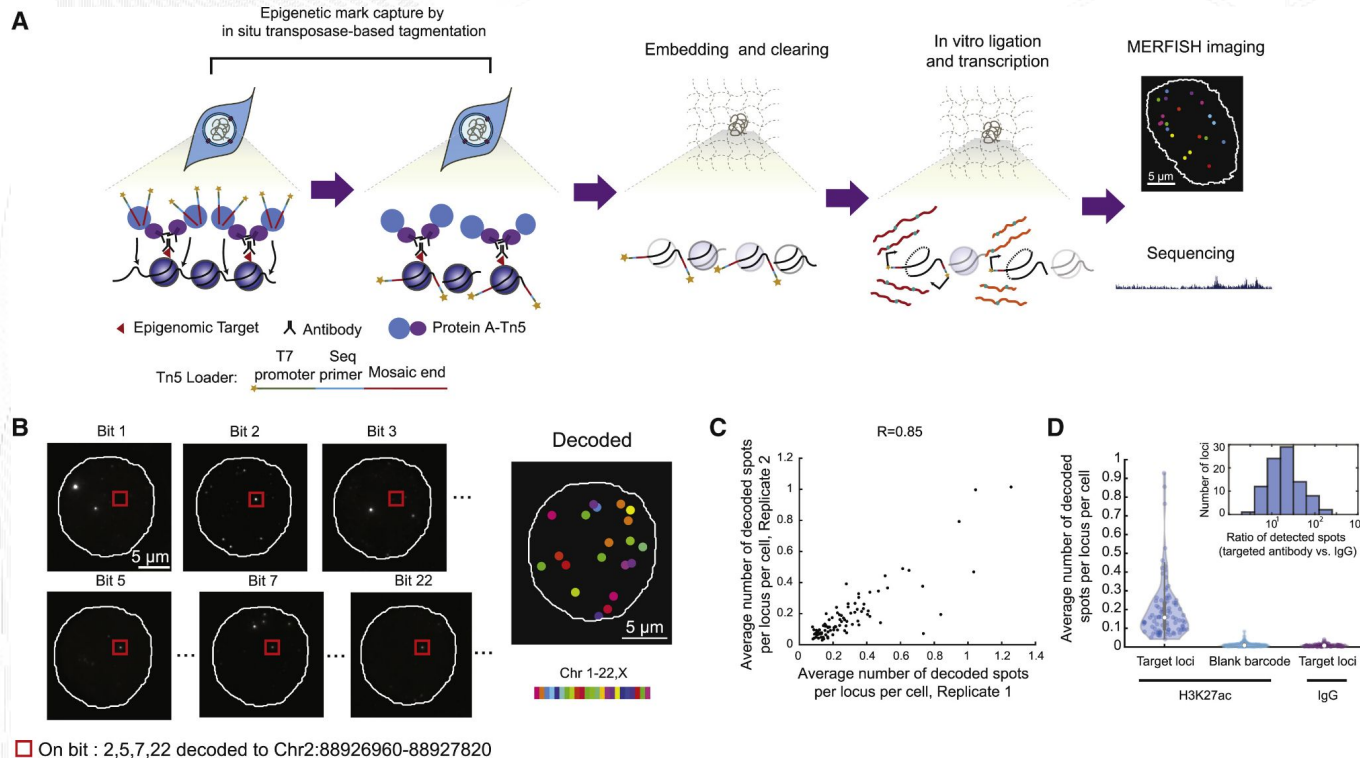
- AtlasXomics Spatial CUT&Tag (for chromatin modification profiling)



<https://www.atlasxomics.com/cuttag-spatial>

Spatial Epigenomics

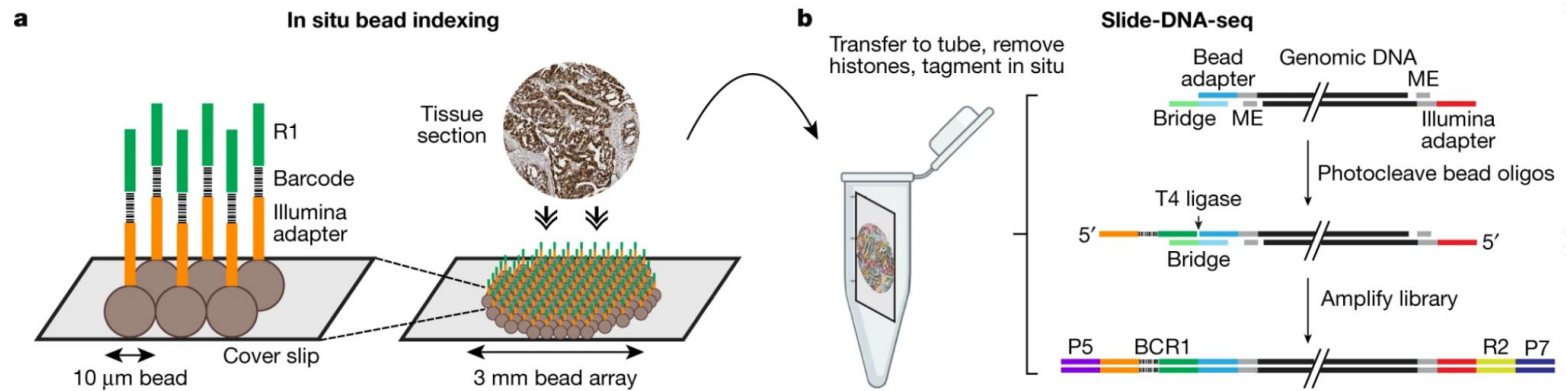
- Epigenomic MERFISH



<https://www.sciencedirect.com/science/article/pii/S0092867422012545>

Spatial Genomics

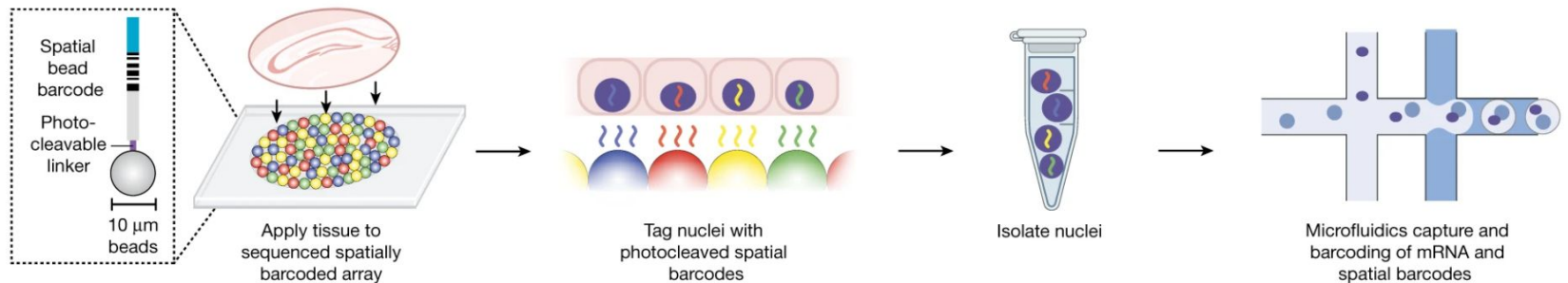
- Slide-DNA-seq



<https://www.nature.com/articles/s41586-021-04217-4>

Spatial Genomics (multimodal?)

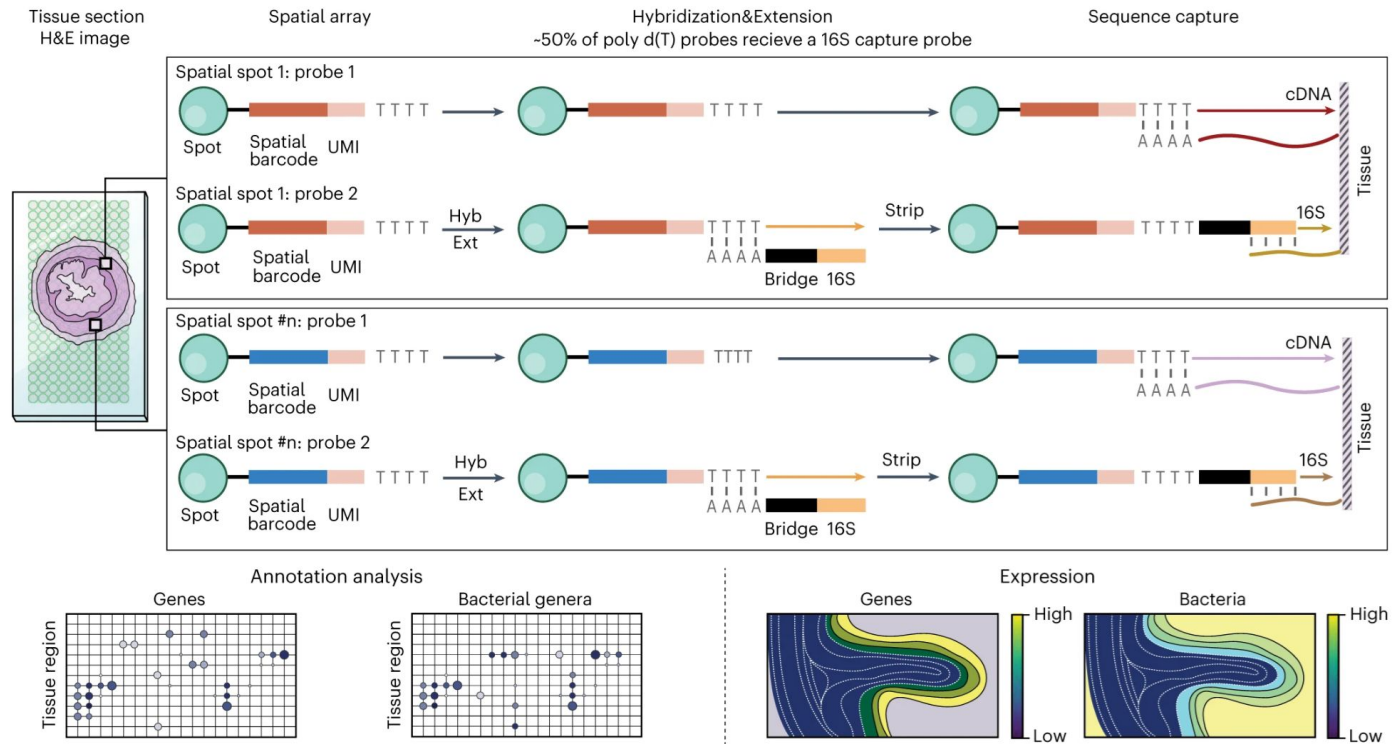
- Slide-tags



<https://www.nature.com/articles/s41586-023-06837-4>

Spatial Metagenomics

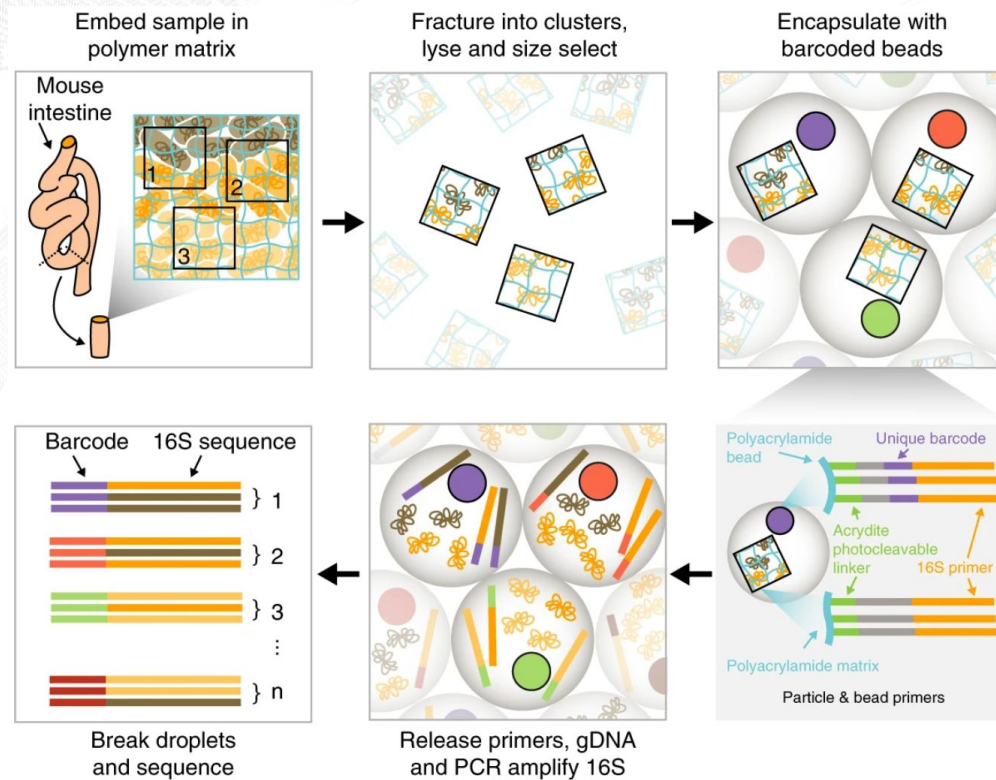
- SHM-seq



<https://www.nature.com/articles/s41587-023-01988-1>

Spatial Metagenomics

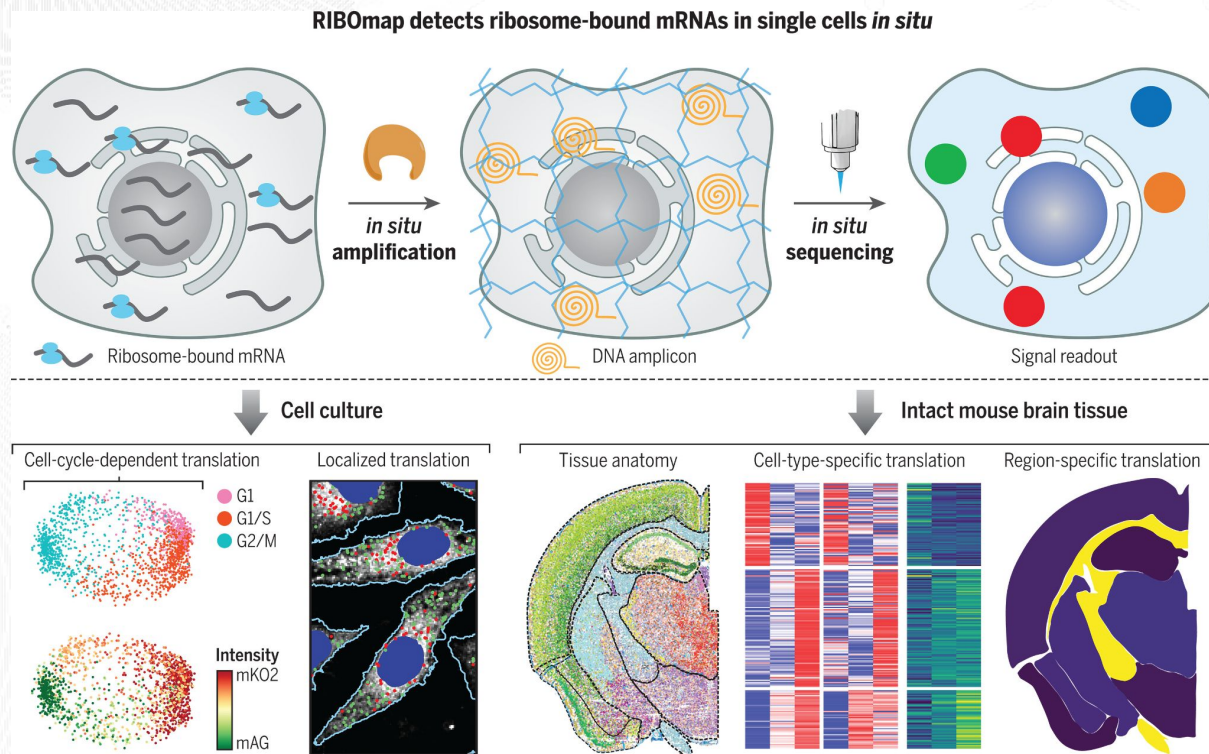
- MaPS-seq



<https://www.nature.com/articles/s41587-019-0183-2>

Spatial Translatomics (!)

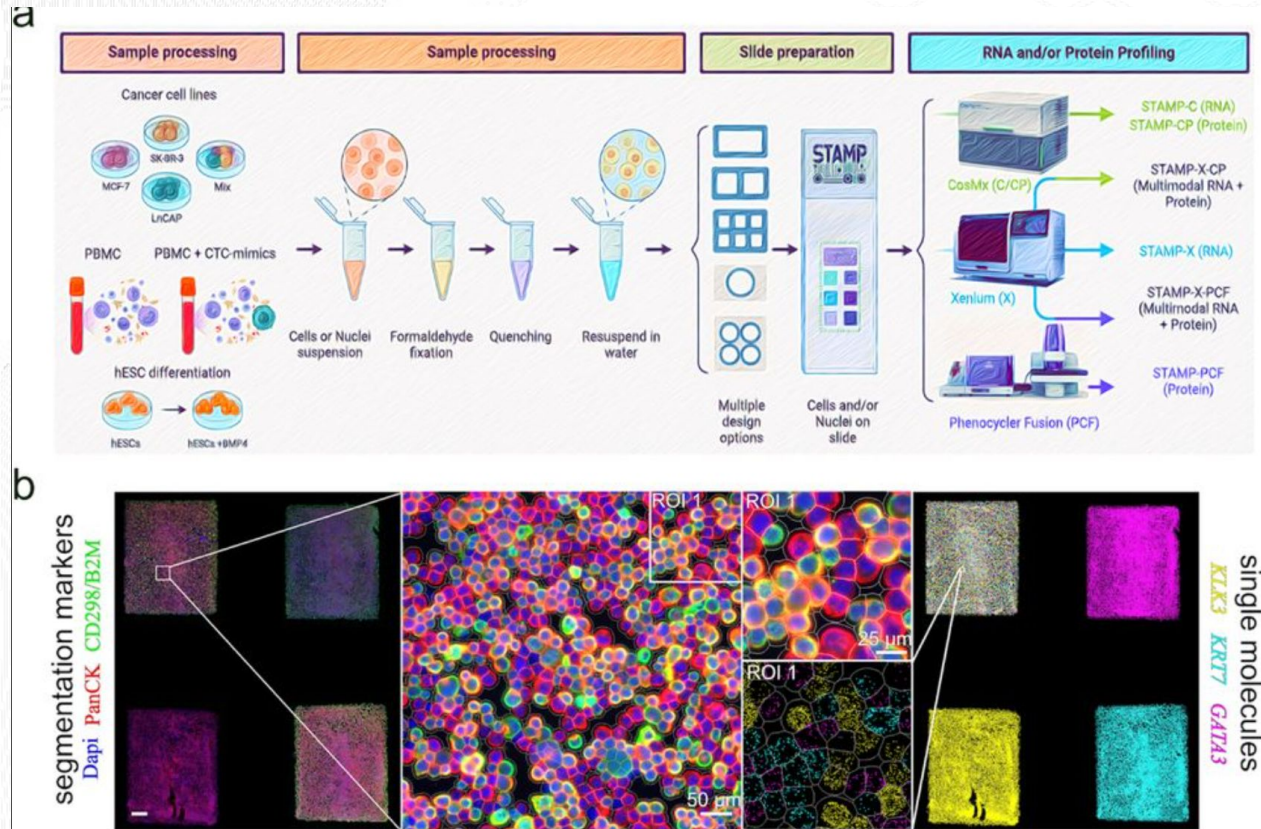
- RIBOmap



<https://www.science.org/doi/10.1126/science.add3067>

Replacing single-cell transcriptomics

- STAMP

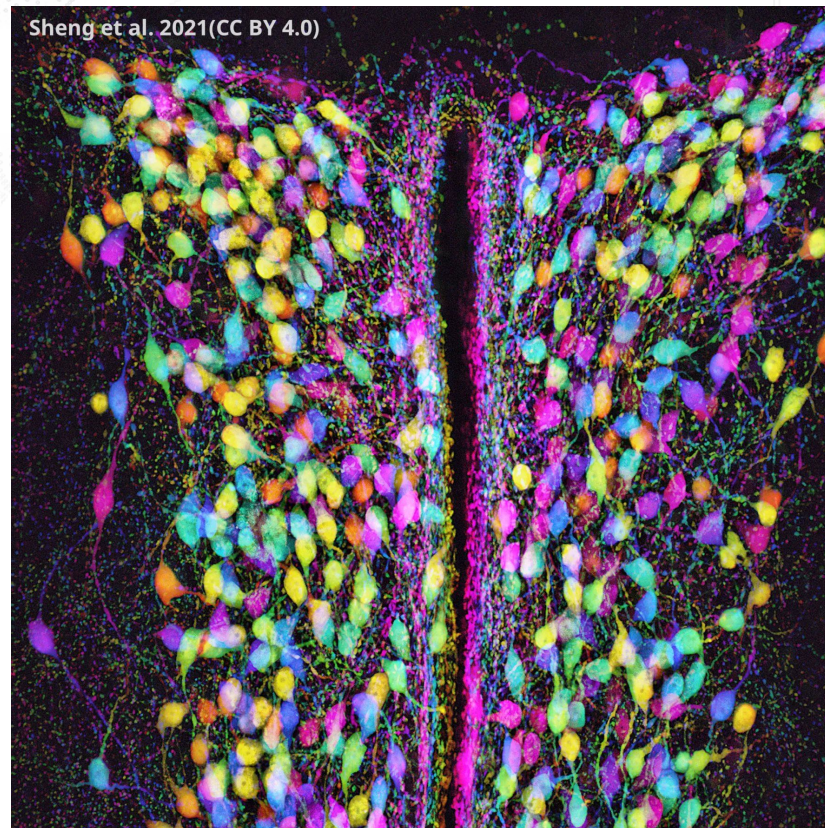


Spatial Transcriptomics

- In many cases, you would consider commercial solutions for ST
 - Visium / Visium HD
 - STOmics
 - Xenium
 - MERSCOPE

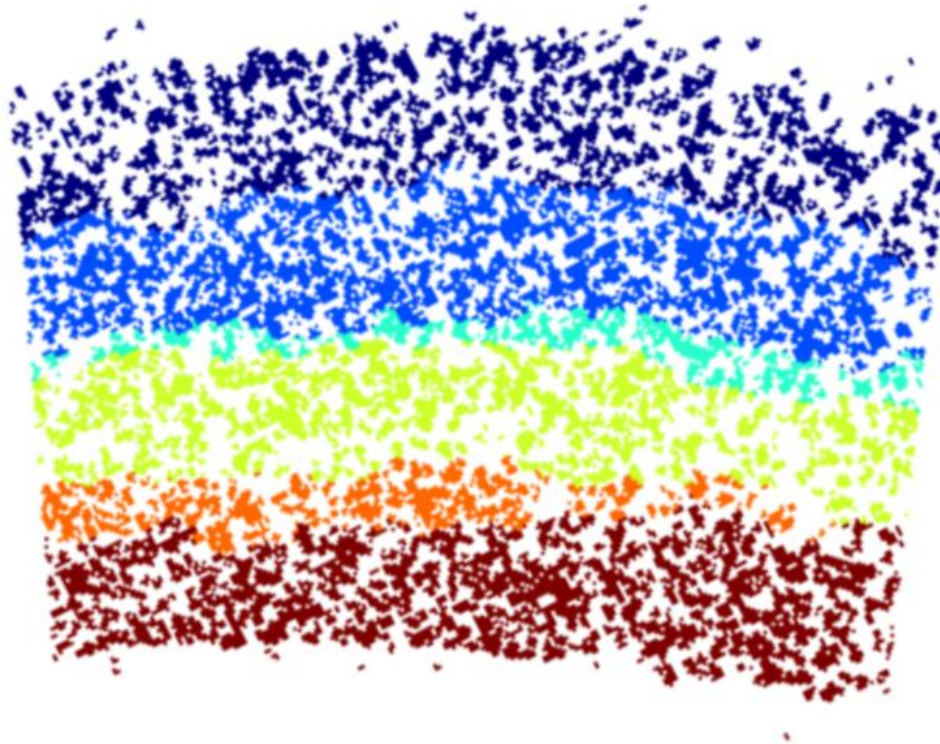
When should I consider spatial omics analysis?

- When scRNA-seq is not easily applicable



When should I consider spatial omics analysis?

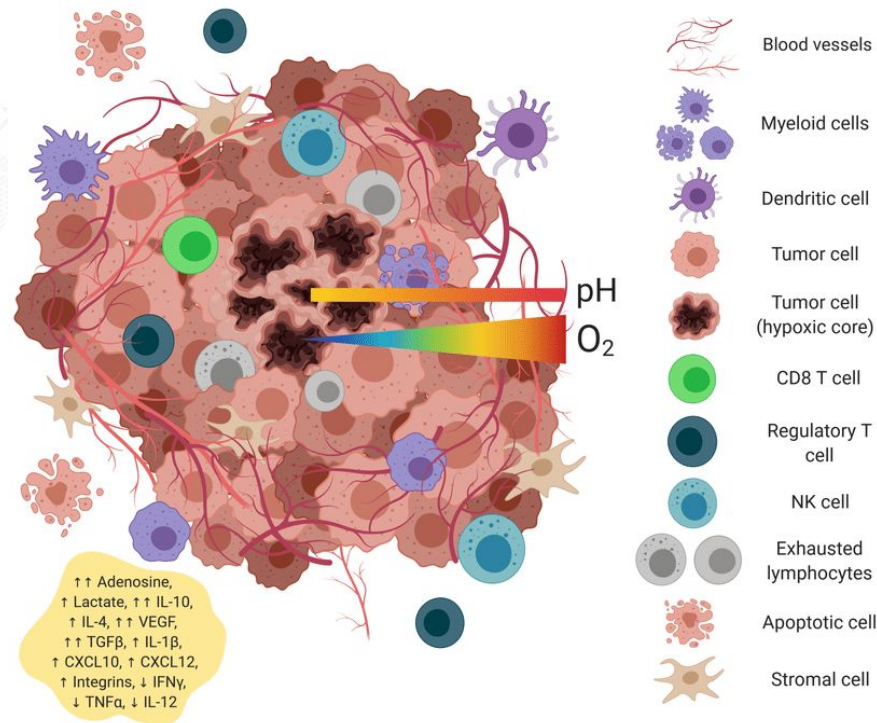
- When comparing differences between tissue regions



<https://ssam.readthedocs.io/en/develop/userguide/11-domain.html>

When should I consider spatial omics analysis?

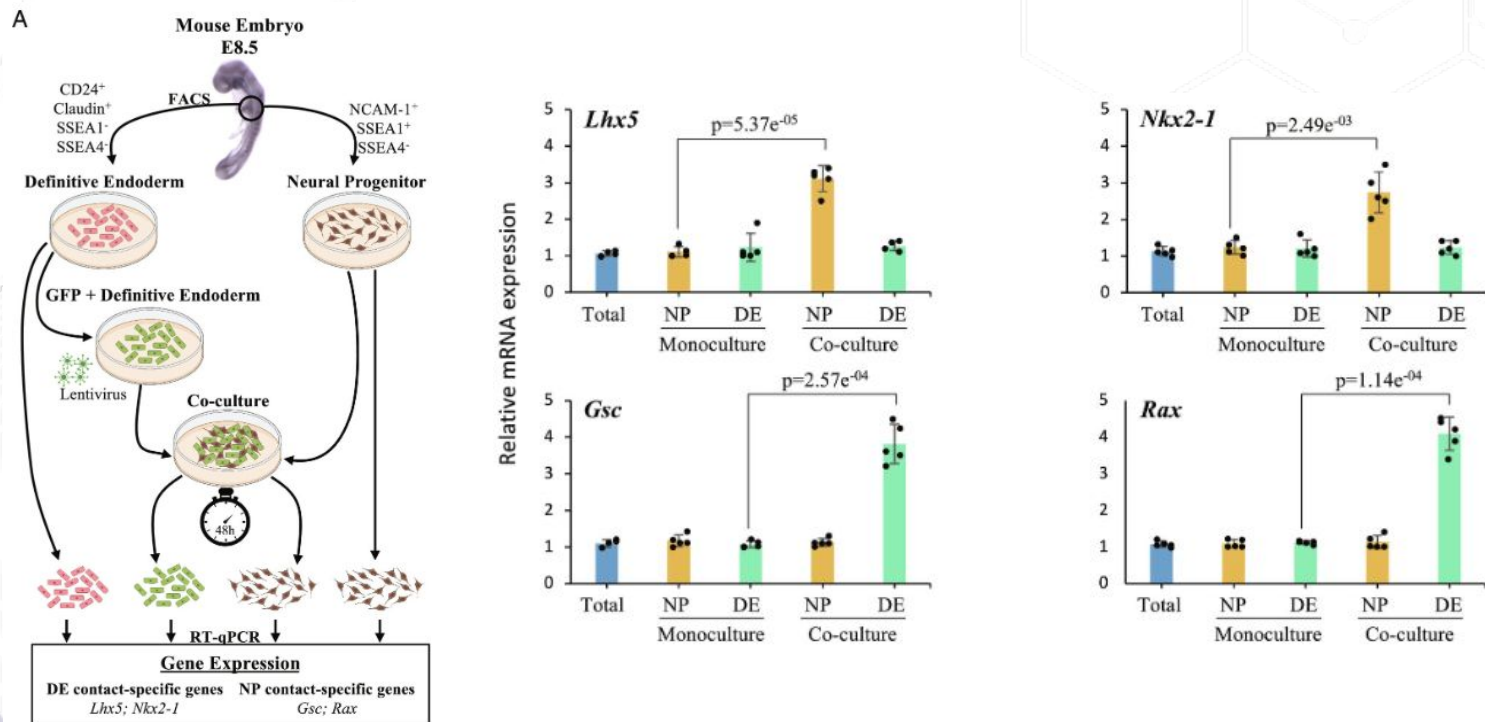
- When you expect high spatial heterogeneity (e.g., TME)



from Wikipedia

When should I consider spatial omics analysis?

- When you suspect dissociation of cells would disrupt the gene expression profiles



When should I consider spatial omics analysis?

- When you are interested in:
 - Identifying genes having spatial patterns
 - Spatial organizations of genes of interest
 - Spatial mapping of cell types
 - Interactions between the cells/mRNAs/proteins in proximity

When should I consider spatial omics analysis?

- Possible, but not usually recommended:
 - Differential expression analysis between samples
(expensive!)

FFPE or Fresh Frozen?



https://www.reddit.com/r/toolgifs/comments/144y72f/sectioning_umbilical_cord_on_microtome/

FFPE or Fresh Frozen?



<https://www.youtube.com/watch?v=UnCLhowHucU>

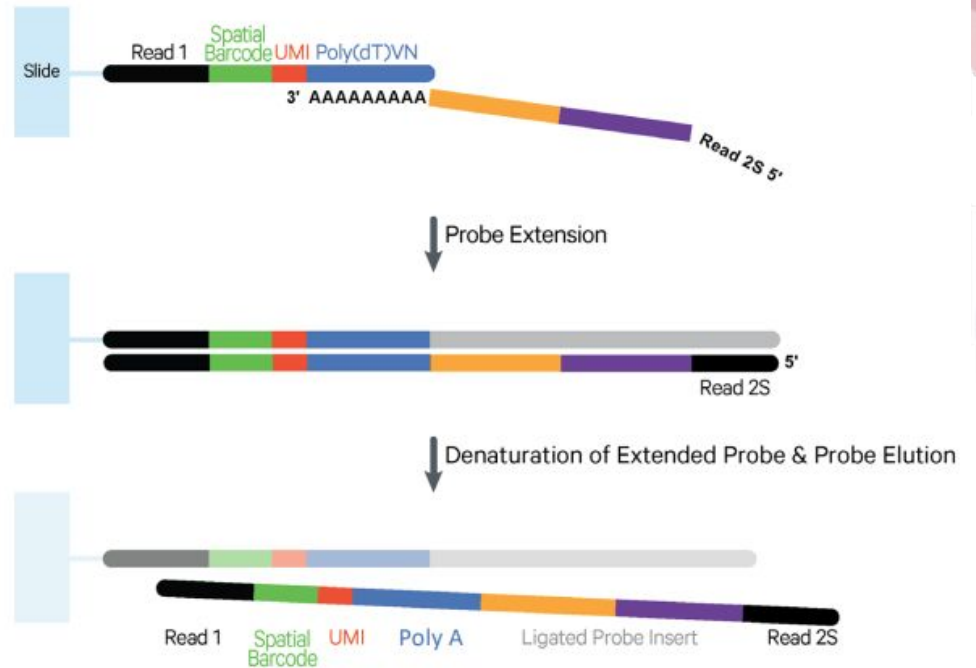
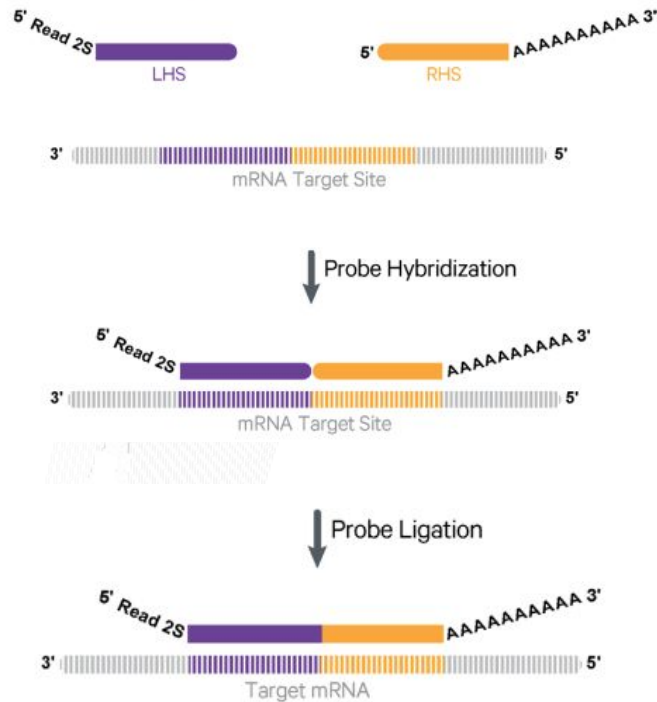
FFPE or Fresh Frozen?

- FFPE
 - High availability, easy to handle, but lower RNA quality
- FF
 - Low availability, difficult to handle, but higher RNA quality

Spatial Transcriptomics of FFPE

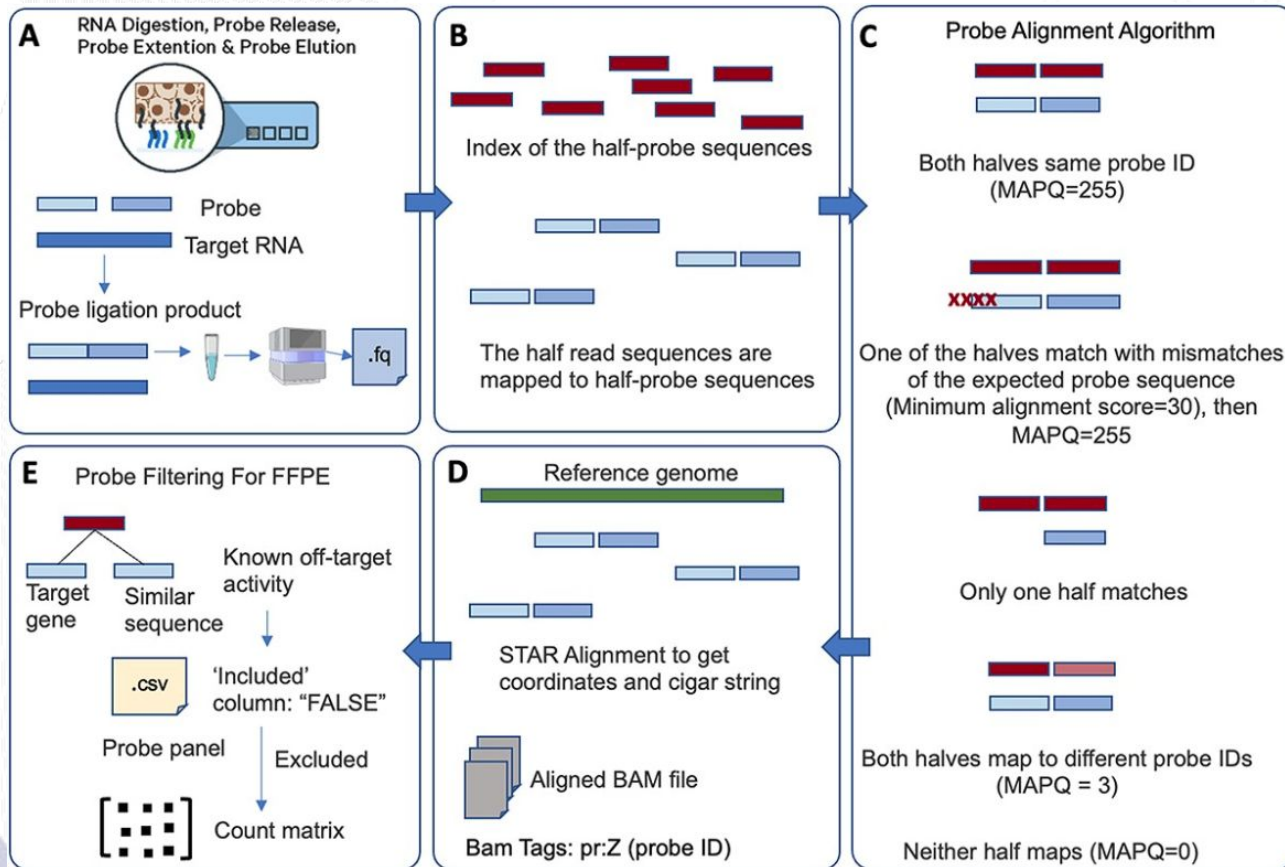
- Fluorescence-based methods
 - As they are targeted approaches, it should work
 - Xenium, MERSCOPE, CosMx SMI, etc.
- NGS-based methods
 - Often based on targeted probes, due to poly-A degradation
 - Potential off-target problem
 - Visium FFPE, Visium HD, STOmics OMNI, etc.

Spatial Transcriptomics of FFPE



Spatial Transcriptomics of FFPE

- Potential off-targets (10X Visium case)



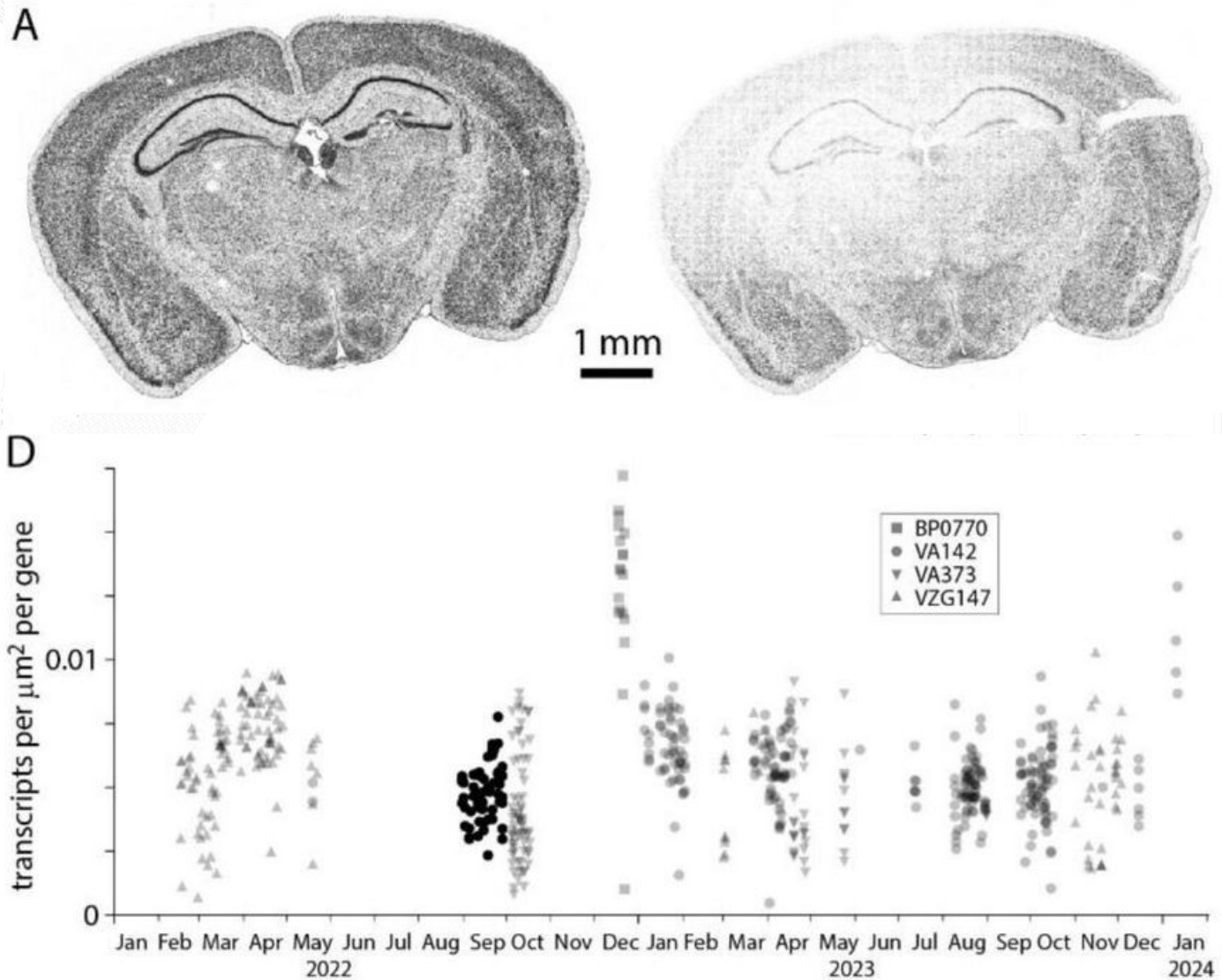
Spatial Transcriptomics of FFPE

Solution Overview

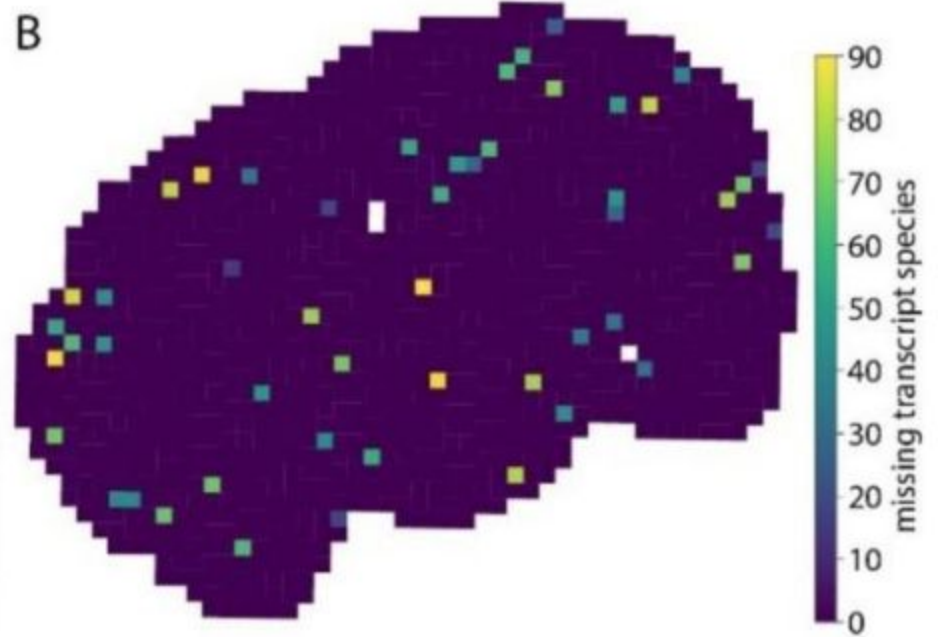
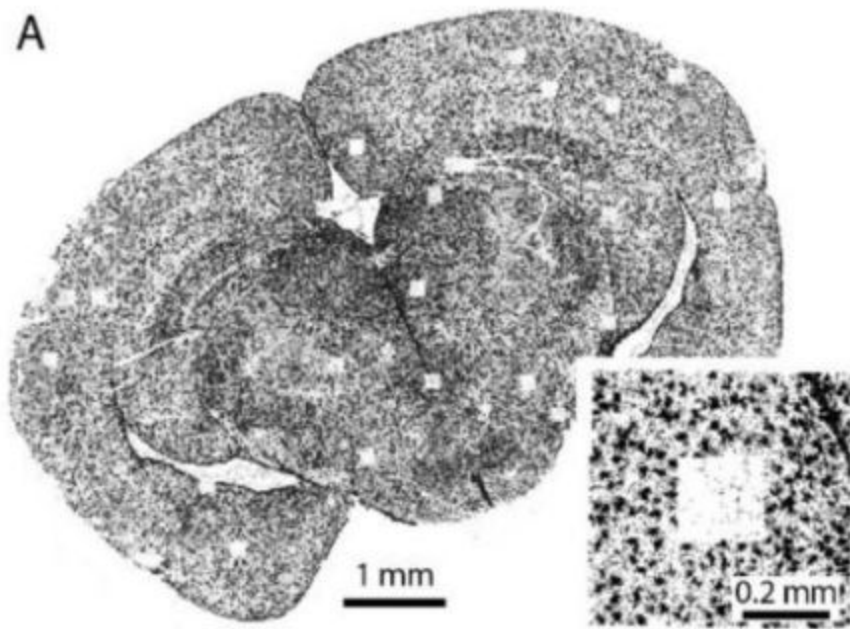
Stereo-seq OMNI, a revolutionary sequencing-based spatial transcriptomics solution to study your FFPE samples.

Stereo-seq OMNI provides true spatial single-cell total RNA information with accuracy and precision with the innovative 'Random Probe' design. Covering a continuous tissue area of 10mm by 10mm, Stereo-seq OMNI delivers best-in-class data and bioinformatics workflow for analyzing spatial total RNA information.

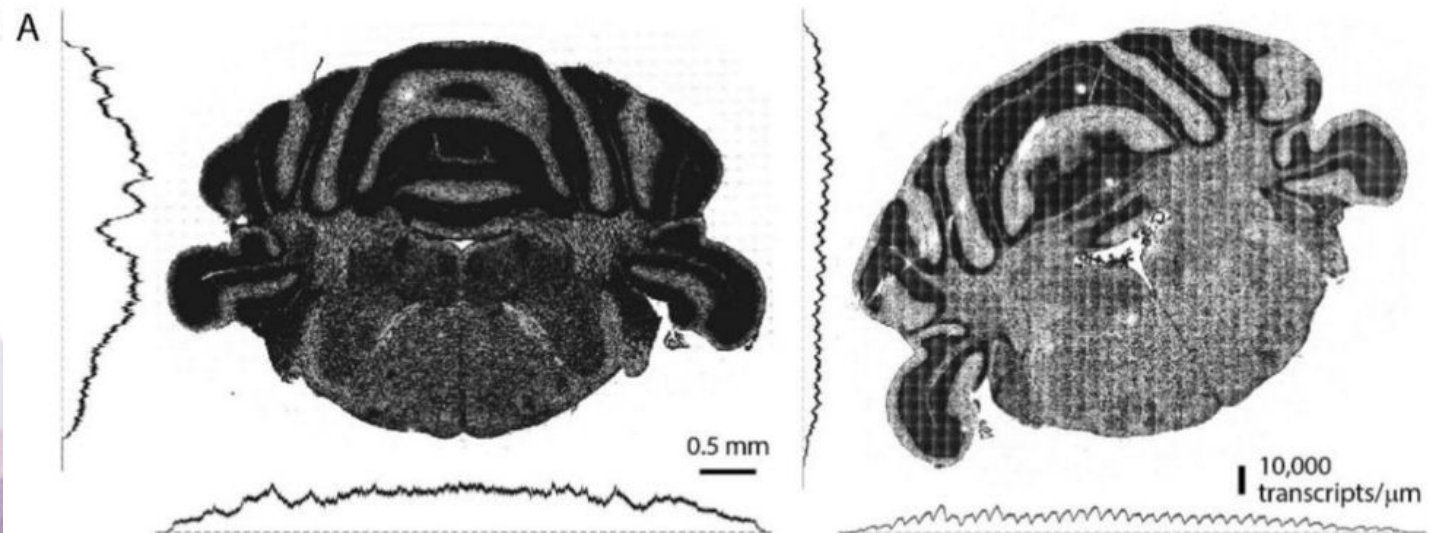
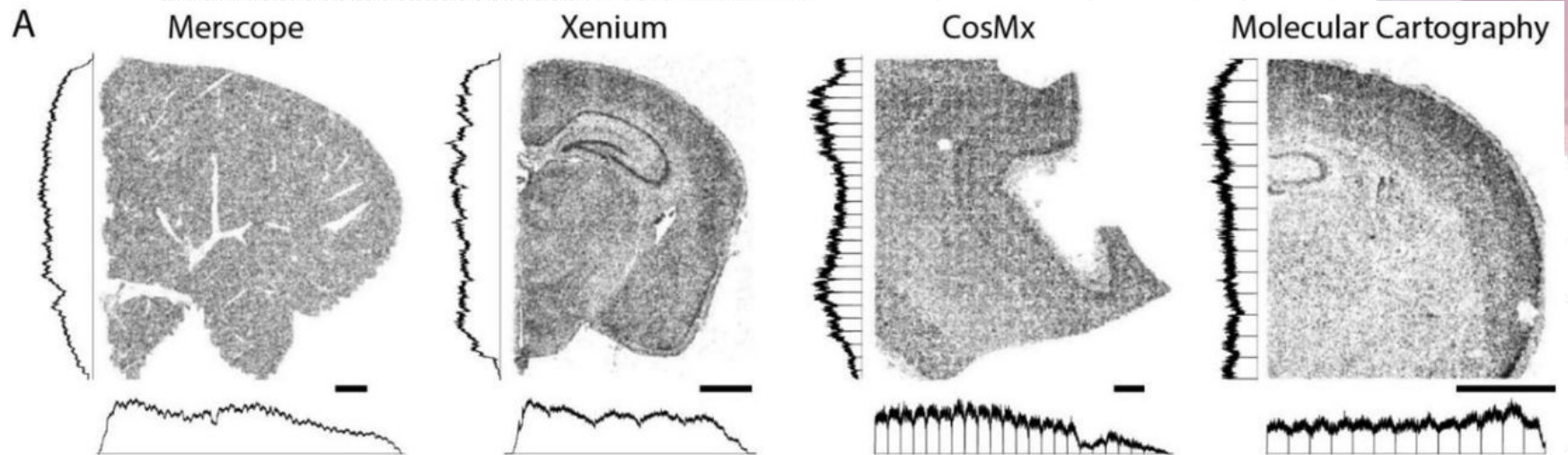
Does it always work?



Does it always work?



Does it always work?

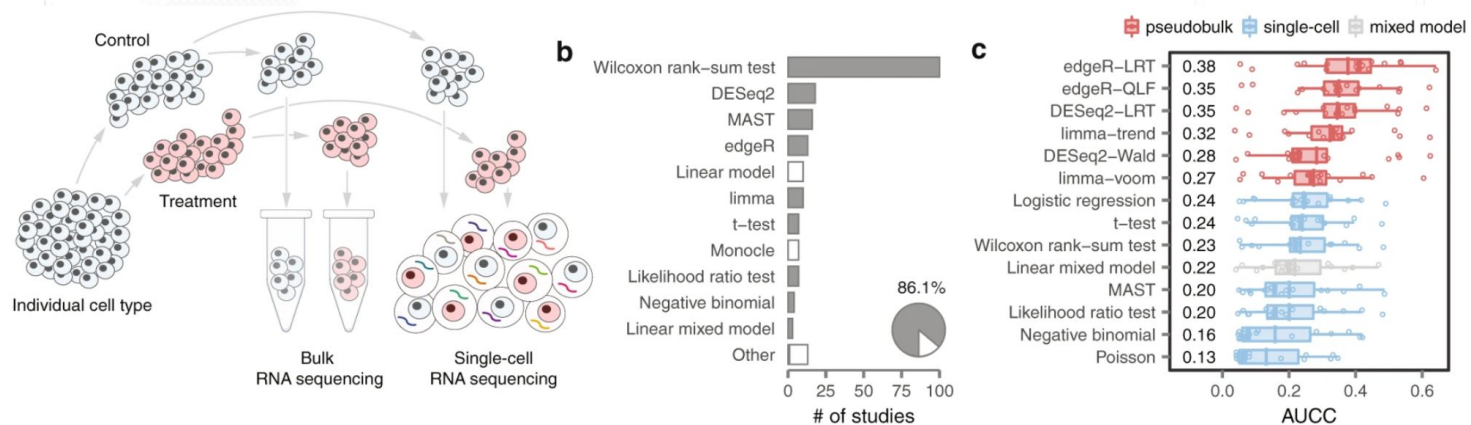


How should I design the experiments?

- Typically, I often recommend the following protocol:
 - Set up hypothesis (often comparison between two cases)
 - Do traditional RNA-seq or scRNA-seq to find differentially expressed genes
 - Find the spatial organization of mRNAs and cells in the spatial context

How should I design the experiments?

- Why not DEG with spatial transcriptomics?
 - Treating cells as samples is prone to false discoveries, thus you need to use “pseudobulks”



<https://www.nature.com/articles/s41467-021-25960-2>

(This is a **MUST-READ** article for anyone interested in scRNA-seq)

How should I design the experiments?

- Why not DEG with spatial transcriptomics?
- And for bulk analysis, you will need at least 6 - 12 biological replicates for DEG!

How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use?

NICHOLAS J. SCHURCH,^{1,6} PIETÀ SCHOFIELD,^{1,2,6} MAREK GIERLIŃSKI,^{1,2,6} CHRISTIAN COLE,^{1,6} ALEXANDER SHERSTNEV,^{1,6} VIJENDER SINGH,² NICOLA WROBEL,³ KARIM GHARBI,³ GORDON G. SIMPSON,⁴ TOM OWEN-HUGHES,² MARK BLAXTER,³ and GEOFFREY J. BARTON^{1,2,5}

¹Division of Computational Biology, College of Life Sciences, University of Dundee, Dundee DD1 5EH, United Kingdom

²Division of Gene Regulation and Expression, College of Life Sciences, University of Dundee, Dundee DD1 5EH, United Kingdom

³Edinburgh Genomics, University of Edinburgh, Edinburgh EH9 3JT, United Kingdom

⁴Division of Plant Sciences, College of Life Sciences, University of Dundee, Dundee DD1 5EH, United Kingdom

⁵Division of Biological Chemistry and Drug Discovery, College of Life Sciences, University of Dundee, Dundee DD1 5EH, United Kingdom

ABSTRACT

RNA-seq is now the technology of choice for genome-wide differential gene expression experiments, but it is not clear how many biological replicates are needed to ensure valid biological interpretation of the results or which statistical tools are best for analyzing the data. An RNA-seq experiment with 48 biological replicates in each of two conditions was performed to answer these questions and provide guidelines for experimental design. With three biological replicates, nine of the 11 tools evaluated found only 20%–40% of the significantly differentially expressed (SDE) genes identified with the full set of 42 clean replicates. This rises to >85% for the subset of SDE genes changing in expression by more than fourfold. To achieve >85% for all SDE genes regardless of fold change requires more than 20 biological replicates. The same nine tools successfully control their false discovery rate at ≤5% for all numbers of replicates, while the remaining two tools fail to control their FDR adequately, particularly for low numbers of replicates. **For future RNA-seq experiments, these results suggest that at least six biological replicates should be used, rising to at least 12 when it is important to identify SDE genes for all fold changes.** If fewer than 12 replicates are used, a superior combination of true positive and false positive performances makes *edgeR* and *DESeq2* the leading tools. For higher replicate numbers, minimizing false positives is more important and *DESeq* marginally outperforms the other tools.

Keywords: RNA-seq; benchmarking; differential expression; replication; yeast; experimental design; statistical power

How should I design the experiments?

- Why not DEG with spatial transcriptomics?

How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use?

This rises to >85% for the subset of SDE genes changing in expression by more than fourfold. To achieve >85% for all SDE genes regardless of fold change requires more than 20 biological replicates. The same nine tools successfully control their false discovery rate at $\lesssim 5\%$ for all numbers of replicates, while the remaining two tools fail to control their FDR adequately, particularly for low numbers of replicates. For future RNA-seq experiments, these results suggest that at least six biological replicates should be used, rising to at least 12 when it is important to identify SDE genes for all fold changes. If fewer than 12 replicates are used, a superior combination of true positive and false positive performances makes *edgeR* and *DESeq2* the leading tools. For higher replicate numbers, minimizing false positives is more important and *DESeq* marginally outperforms the other tools.

Keywords: RNA-seq; benchmarking; differential expression; replication; yeast; experimental design; statistical power

replicates should be used, rising to at least 12 when it is important to identify SDE genes for all fold changes. If fewer than 12 replicates are used, a superior combination of true positive and false positive performances makes *edgeR* and *DESeq2* the leading tools. For higher replicate numbers, minimizing false positives is more important and *DESeq* marginally outperforms the other tools.

Keywords: RNA-seq; benchmarking; differential expression; replication; yeast; experimental design; statistical power

How should I design the experiments?

- Why not DEG with spatial transcriptomics?
 - Well... These days we think at least triplicates as minimum.

Number of replicates

The number of replicates in an RNA-seq experiment depends on:

1. Technical variability in the RNA-procedures. Technical replicates are replicates where the biological material is the same but the technical steps used to measure gene expression are repeated (Figure 9). Technical reproducibility in RNA-seq is usually high at the level of sequencing but RNA-seq library preparation includes a number of steps (RNA fragmentation, cDNA synthesis, adapter ligation and PCR amplification) that may introduce biases in the data.
2. Biological variability of the system under study. Biological replicates consist of different biological samples that are processed separately (Figure 9). **Biological replicates are required if inference on the population is to be made, with three biological replicates being the minimum for any inferential analysis.**
3. Desired statistical power, that is the capacity for detecting statistically significant differences in gene expression between experimental groups.

<https://www.ebi.ac.uk/training/online/courses/functional-genomics-ii-common-technologies-and-data-analysis-methods/rna-sequencing/performing-a-rna-seq-experiment/design-considerations/number-of-replicates/>

How should I design the experiments?

- Why not DEG with spatial transcriptomics?
 - And, ST with several samples will cost a lot
 - If you design custom gene panel, then even more
 - And often the number of genes you can image is limited to ~1000
 - Therefore, I recommend “cheaper” analyses with at least triplicates to identify your genes of interest prior to ST
 - Do bulk RNA- or scRNA-seq first!
 - If scRNA-seq is not applicable, unfortunately, you should consider cheap ST methods instead
 - And then finally, visualize the genes you found using ST!

How should I design the experiments?

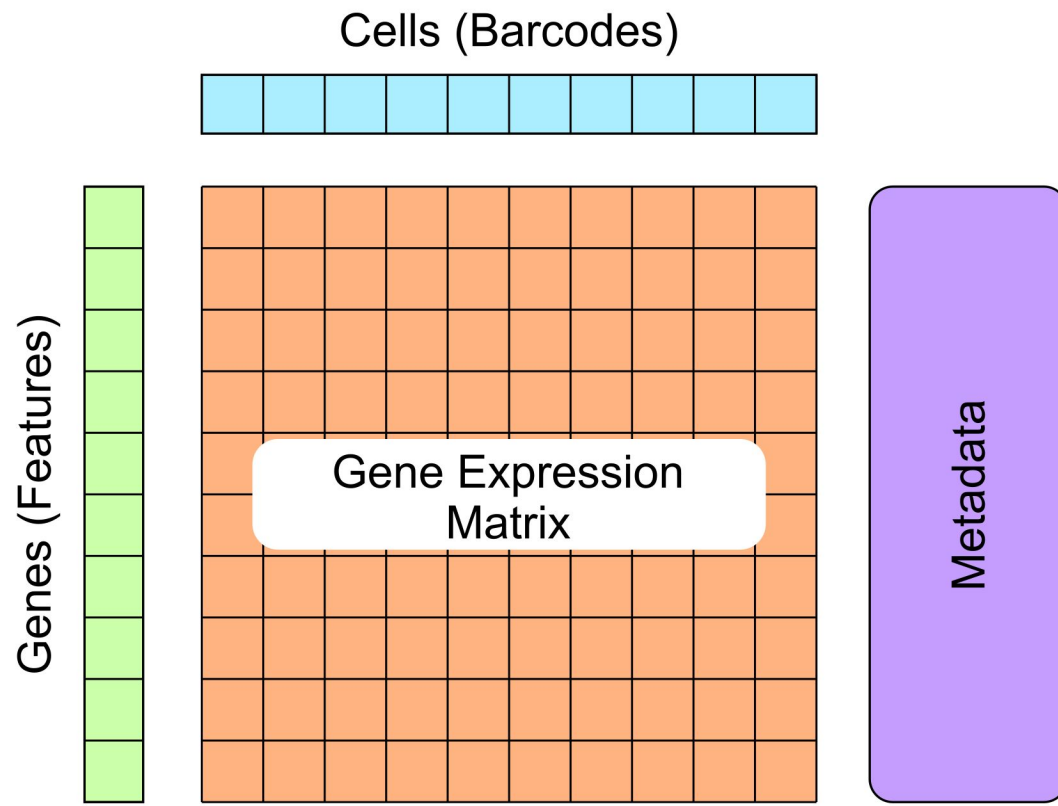
- Why not DEG with spatial transcriptomics?
 - Still, you can identify genes between regions and clusters within one tissue
 - However, this is not recommended

How should I design the experiments?

- Which genes should I consider for probe design?
 - Traditional cell-type markers
 - Differentially expressed genes identified by scRNA-seq or bulk RNA-seq
 - Genes with expression patterns that vary across space/time
 - If you need, housekeeping genes as controls
 - Any other genes related to your biological question
- You can try “Spapros” for an automatic gene panel selection!
 - <https://www.nature.com/articles/s41592-024-02496-z>

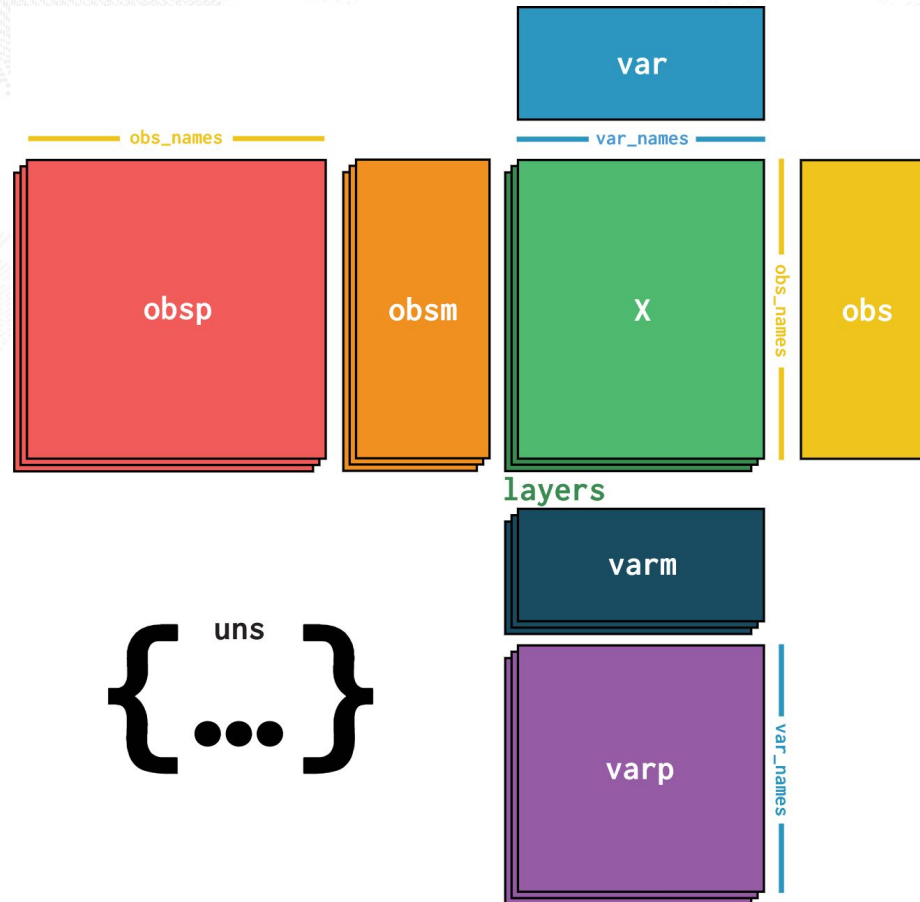
How are the data stored?

- The “cell x gene matrix”



How are the data stored?

- AnnData



How are the data stored?

- AnnData
 - obs: Observation annotations (cell metadata)
 - var: Variable/feature annotations (gene metadata)
 - obsm: Observation matrices (cell x embedding), stores dimensional reductions like PCA, UMAP
 - varm: Variable matrices (gene x annotation)

How are the data stored?

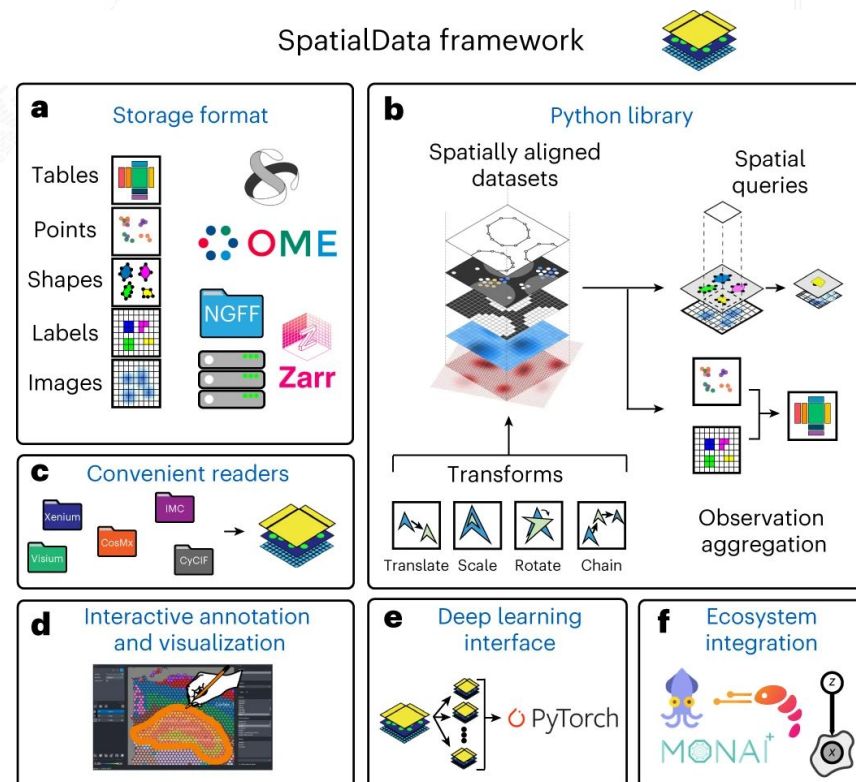
- AnnData
 - obsp: Observation pairwise (cell x cell), stores cell-cell graphs/distances
 - varp: Variable pairwise (gene x gene), stores gene-gene relationships
 - uns: Unstructured data, stores misc data like clustering parameters, plotting configs

How are the data stored?

- AnnData
 - obsp: Observation pairwise (cell x cell), stores cell-cell graphs/distances
 - varp: Variable pairwise (gene x gene), stores gene-gene relationships
 - uns: Unstructured data, stores misc data like clustering parameters, plotting configs

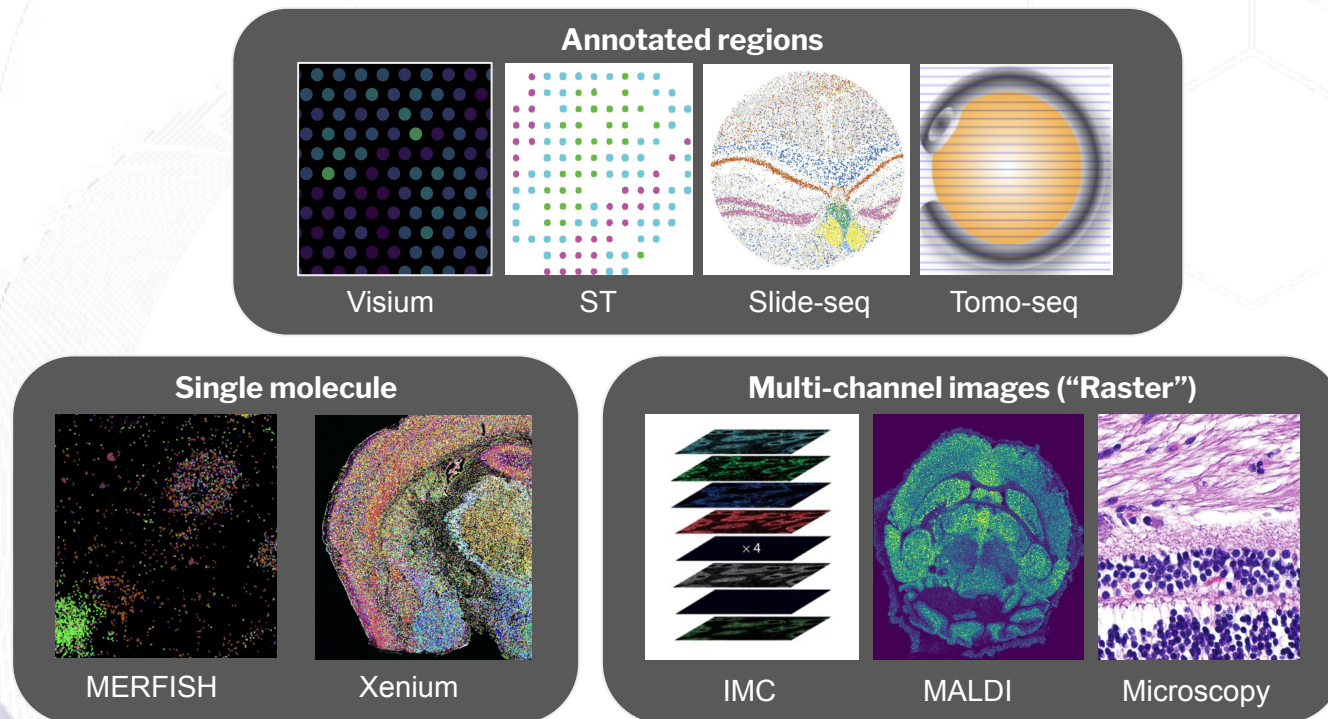
How are the data stored?

- SpatialData
 - A data storage format specially designed for ST analysis



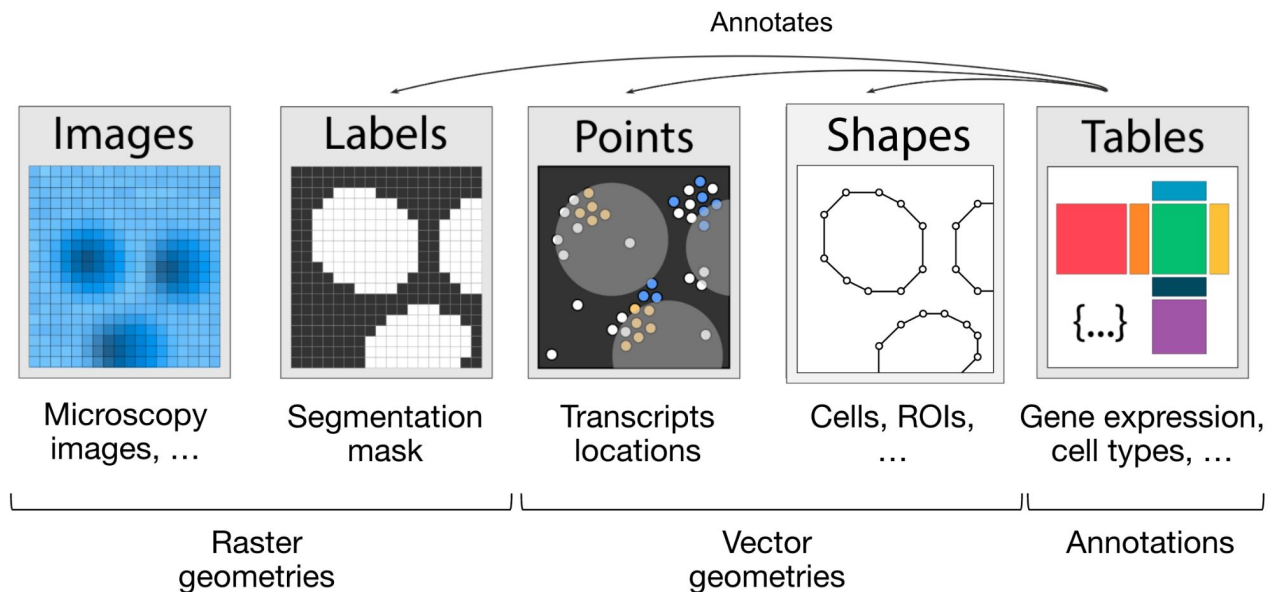
How are the data stored?

- SpatialData
 - A data storage format specially designed for ST analysis



How are the data stored?

- SpatialData
 - A data storage format specially designed for ST analysis



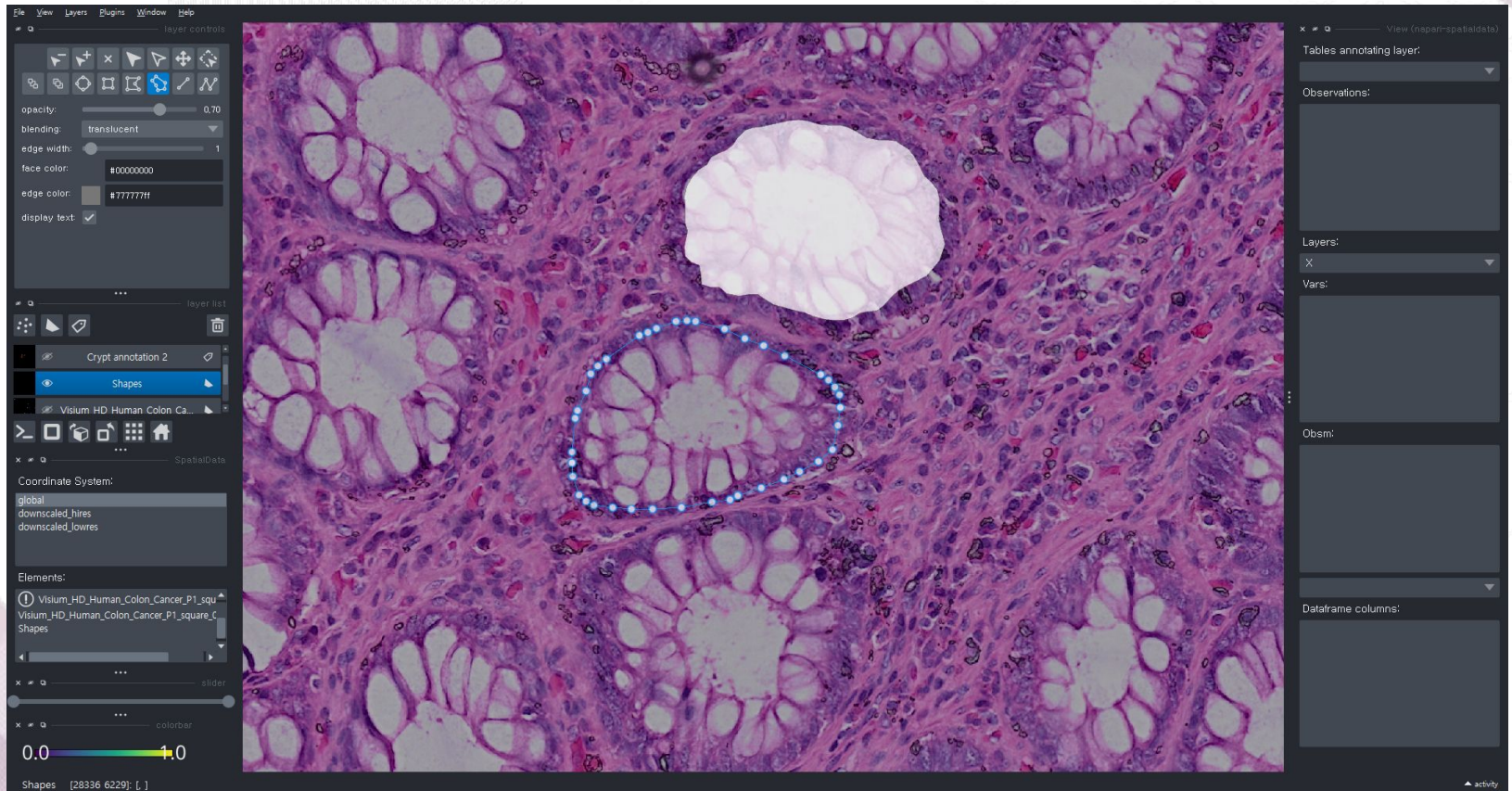
How are the data stored?

- Napari
- An interactive viewer for visualization of SpatialData



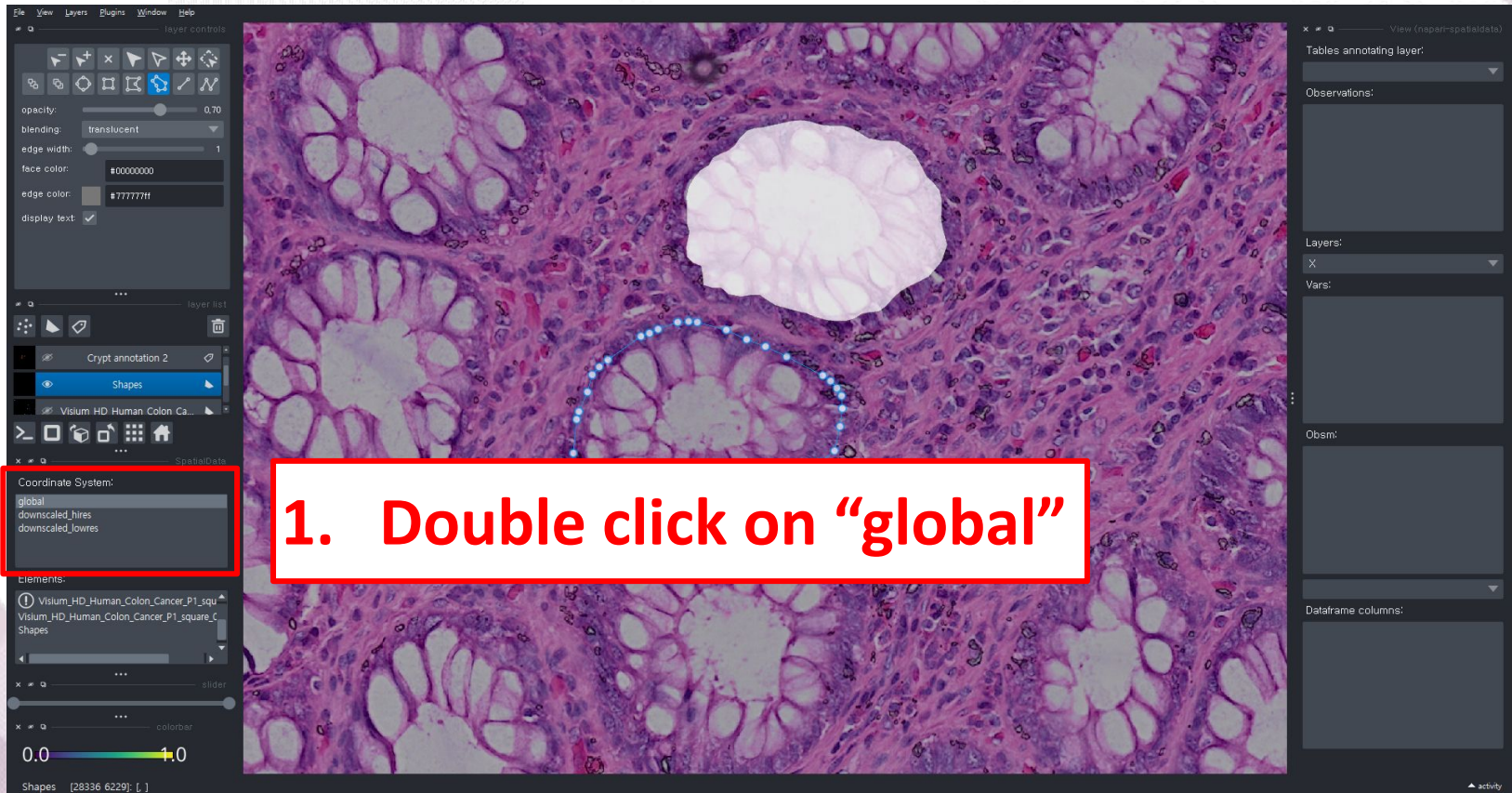
How are the data stored?

- Napari



How are the data stored?

- Napari



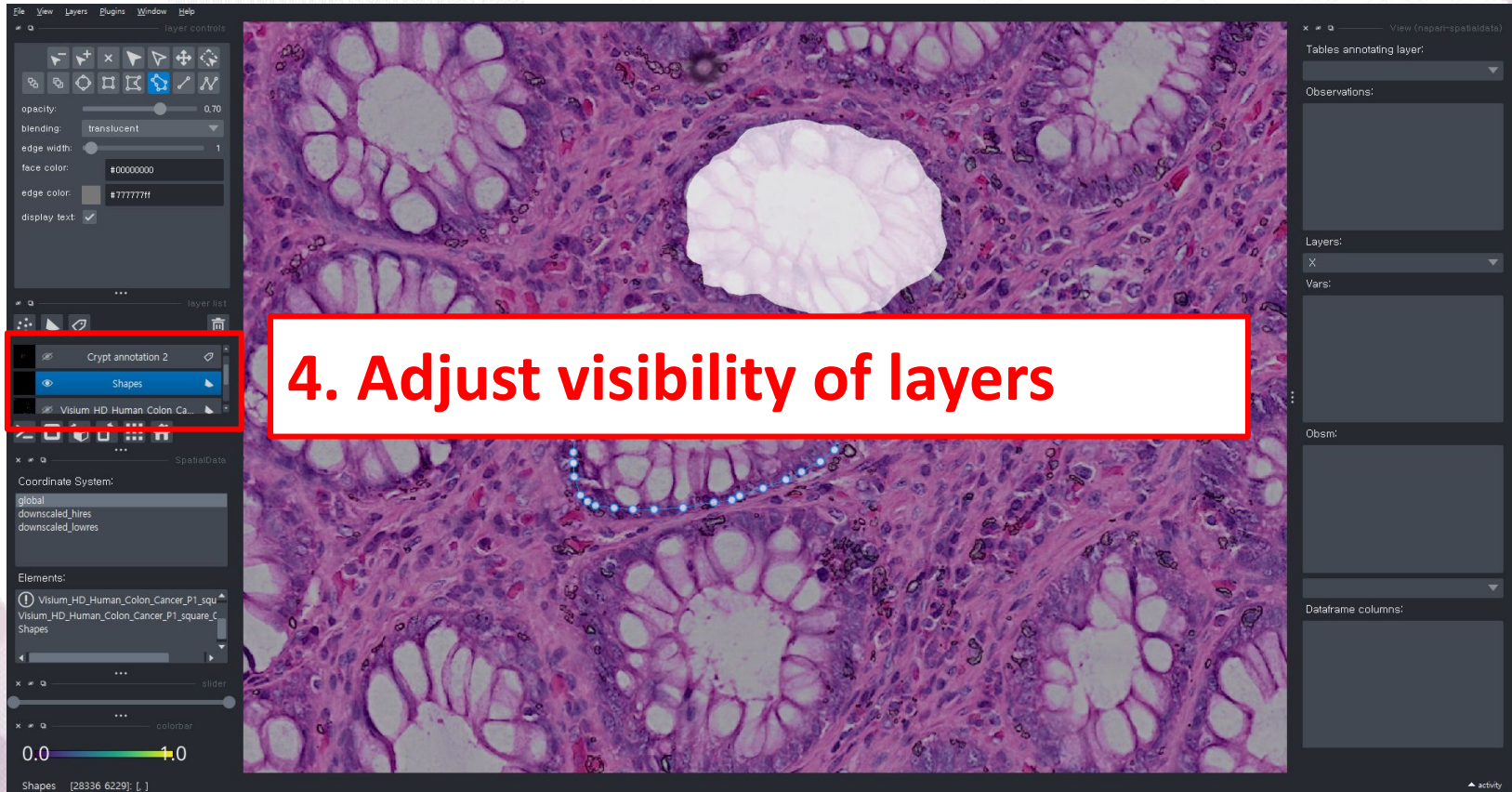
How are the data stored?

- Napari



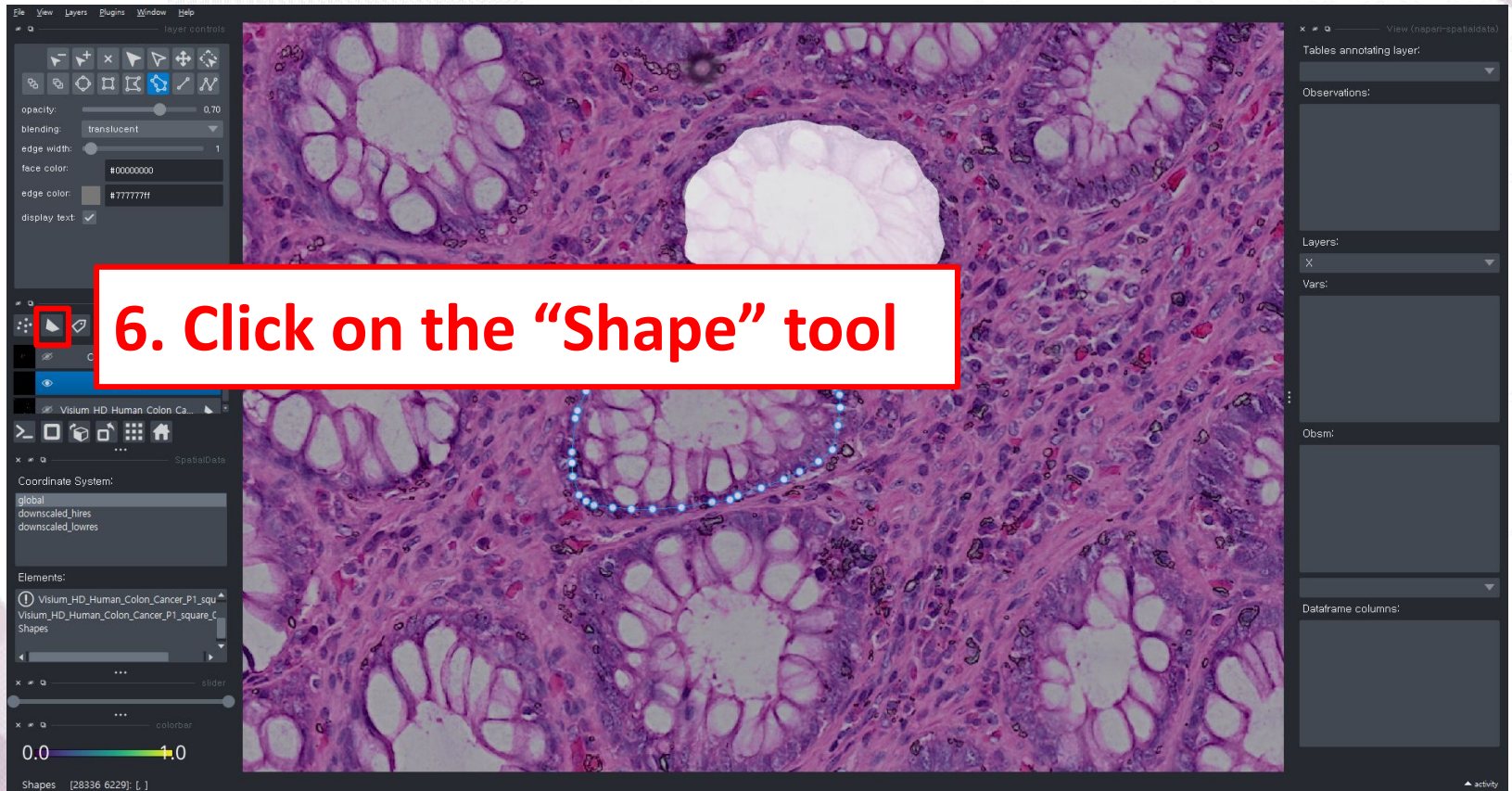
How are the data stored?

- Napari



How are the data stored?

- Napari



How are the data stored?

- Napari

7. Use these three tools to annotate



How are the data stored?

- Napari





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KSBi-BIML 2025

Best Practices in Spatial Omics Data Analysis

Part II

Jeongbin Park

Pusan National University

Hands-on Tutorial on SpatialData & Visualization

- Conda
 - Isolated environments to manage multiple python versions
 - For this tutorials, you need to create and manage multiple conda environments

Hands-on Tutorial on SpatialData & Visualization

- Conda
 - If conda is not installed, try “micromamba”

```
"${SHELL}" <(curl -L micro.mamba.pm/install.sh)
```

- Micromamba is a faster alternative drop-in replacement of conda
 - Just run “micromamba” instead of “conda”

Hands-on Tutorial on SpatialData & Visualization

- Conda
 - Modify conda config via:

```
nano ~/.condarc
```

- Paste below, and “ctrl-x” and “enter” to save:

```
channels:  
- conda-forge  
- bioconda  
- defaults
```

Hands-on Tutorial on SpatialData & Visualization

- Jupyter Lab
 - We will go through all tutorials with Jupyter Lab
 - To install Jupyter Lab, do the following:

```
micromamba create -n jl jupyterlab
```

- Finally, run Jupyter Lab:

```
micromamba activate jl  
jupyter lab --ip 0.0.0.0
```


Hands-on Tutorial on SpatialData & Visualization

- Tutorials are available at:
 - <https://github.com/pnucolab/BIML-2025>



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Best Practices in Spatial Omics Data Analysis

Part III

Jeongbin Park

Pusan National University

Spatial Omics Data Analysis

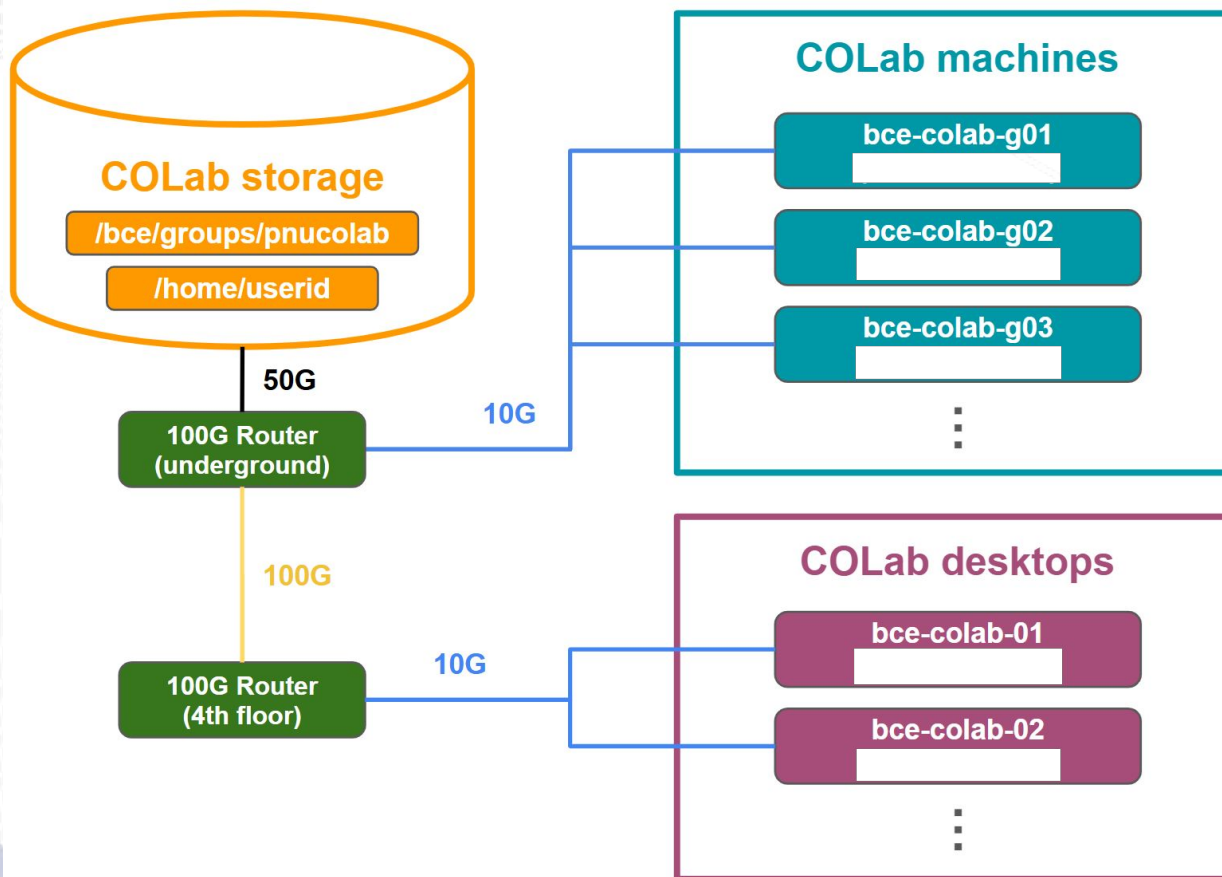
- We will address the following questions:
 - What would be the recommended server requirements?
 - How does the data look like?
 - Which algorithms/computational methods should I use?

What would be the recommended server requirements?

- Requirements
 - Analysis servers
 - Consumer GPUs (RTX 4090, 5090, etc.)
 - High memory (>500GB)
 - Storage servers
 - After 3 years, we collected 60T of datasets already
 - Fast Network
 - I recommend faster connections between desktops and servers

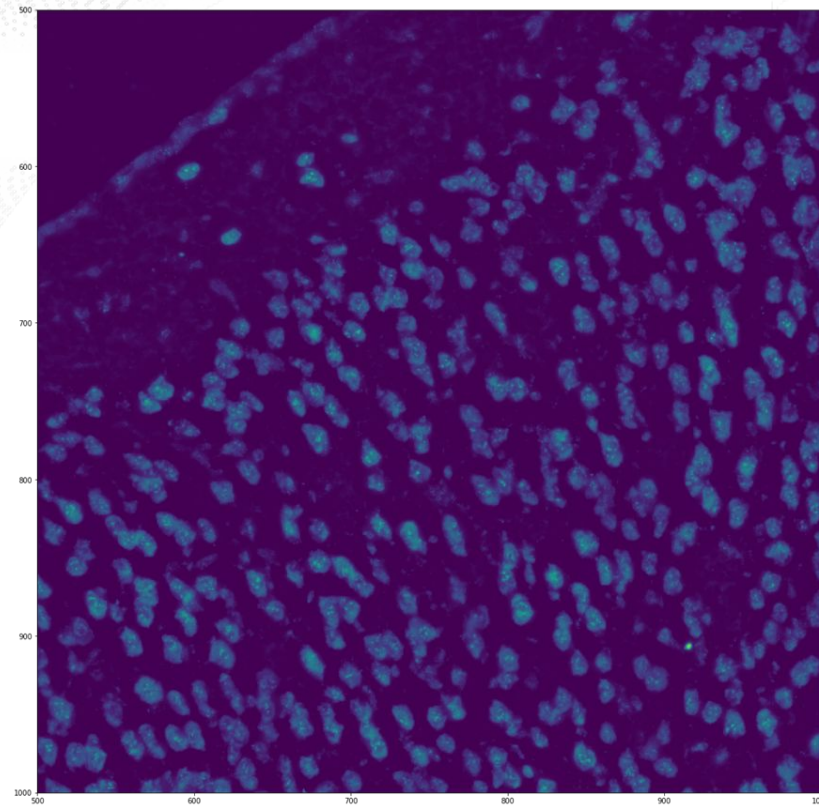
What would be the recommended server requirements?

- Network diagram (example)



How does the data look like?

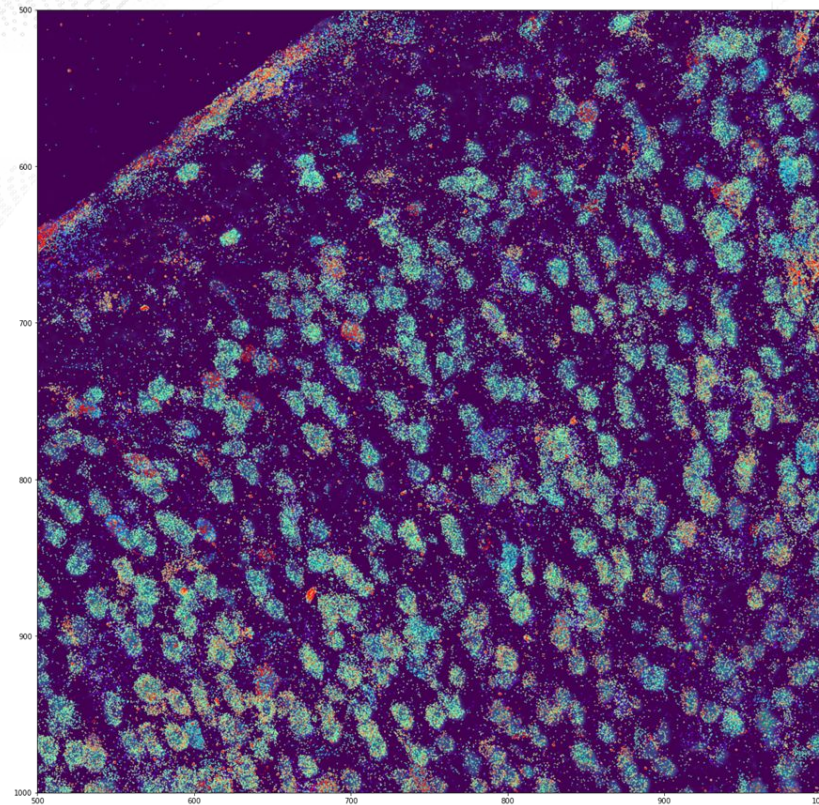
- Imaging-based methods



Poly-A image (all mRNAs)

How does the data look like?

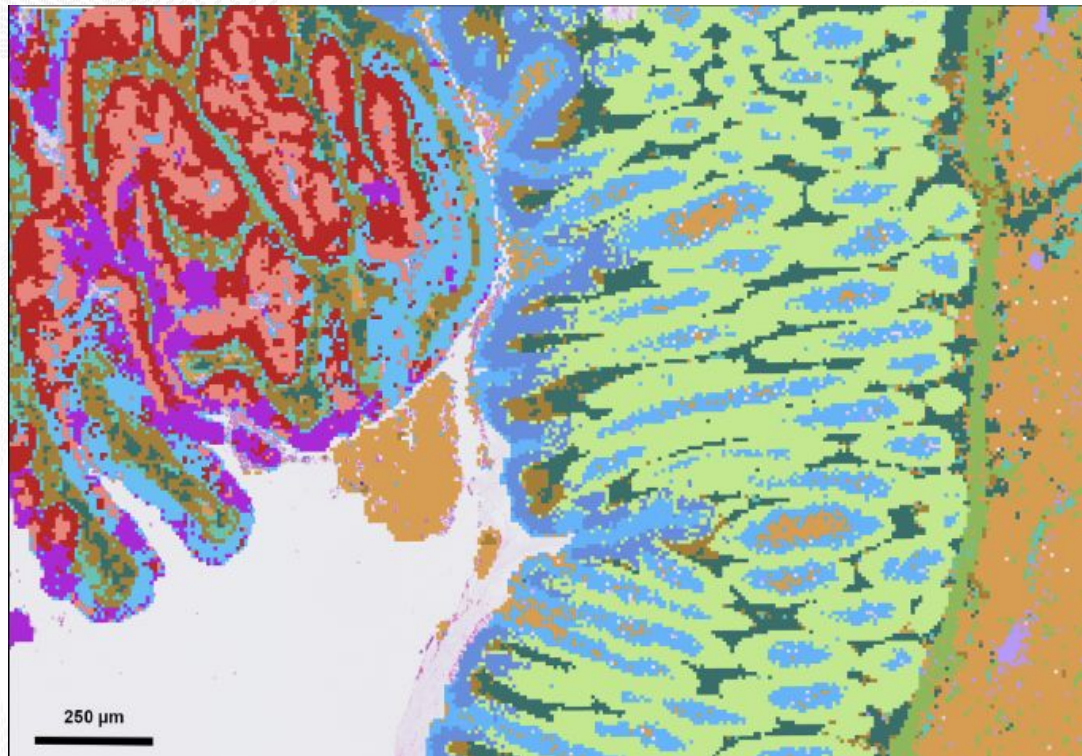
- Imaging-based methods



Scatter plot of osmFISH mRNAs
(33 genes, colored by genes)

How does the data look like?

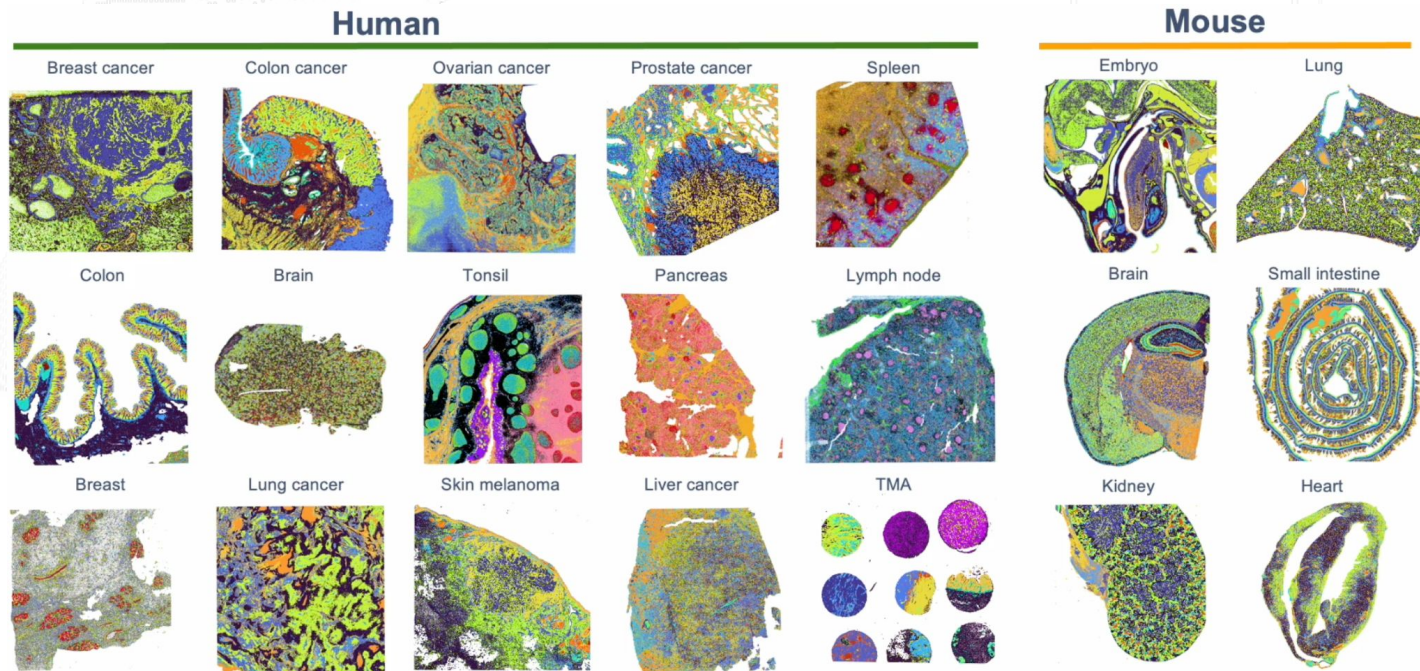
- NGS-based methods



Visium HD, from 10X website

How does the data look like?

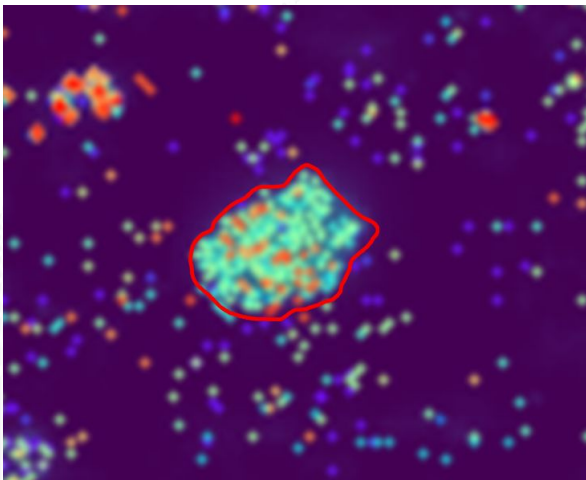
- NGS-based methods



Visium HD, from 10X website

How does the data look like?

- In the case of imaging ST, mRNA quantification is needed



1. Cell segmentation
2. Quantification within the segment

How does the data look like?

- If you need to do DEG analysis:
 - The counts are UMIs, thus no need to correct length bias, as you have done with the TPM/FPKM normalization - just use raw counts!
 - Consider pseudobulking them!

How does the data look like?

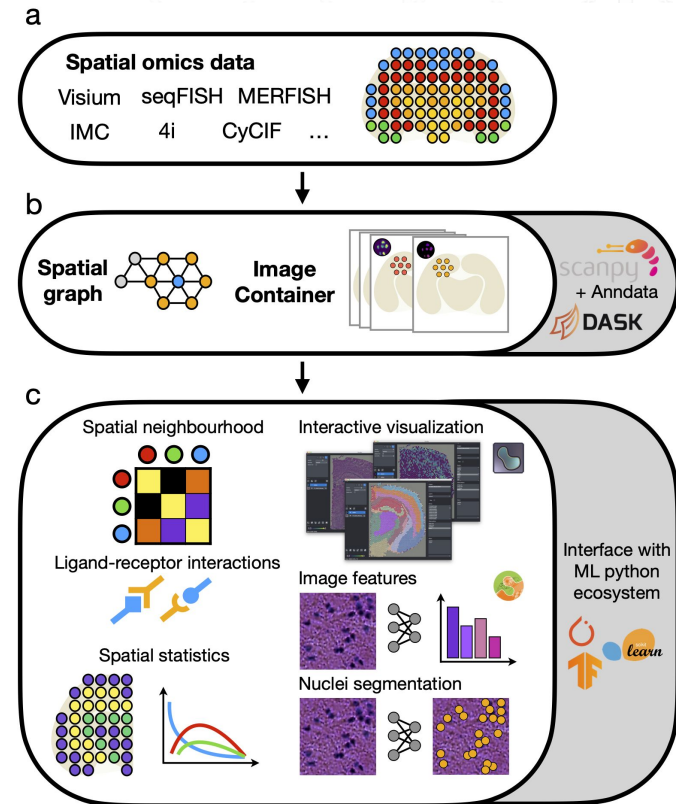
- If you want to find cell-types by clustering,
 - Try “library size normalization”!
 - Why should we do this?
 - Because gene/protein expression depends on cell size
 - <https://www.sciencedirect.com/science/article/pii/S016895251200100X>

How does the data look like?

- If you want to find cell-types by clustering,
 - Then log-transform!
 - Why should we do this?
 - As it resembles scRNA-seq data, the distribution is expected to follow negative binomial

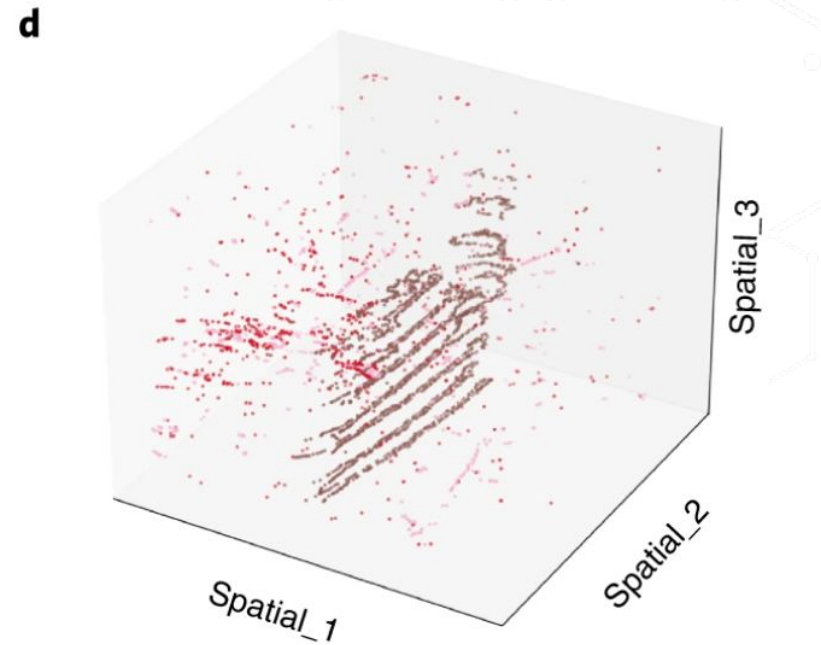
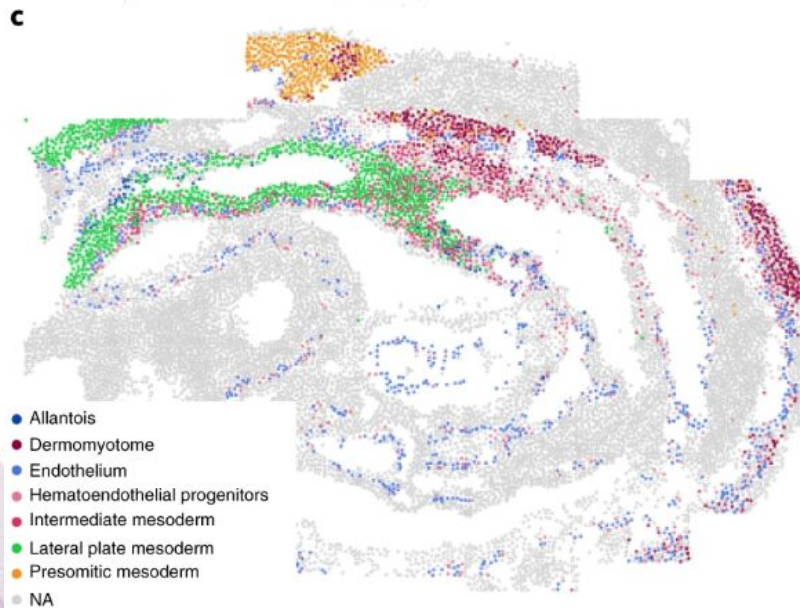
Which algorithm should I use?

- SquidPy
 - You can use SquidPy for general analysis of your spatial data
- <https://squidpy.readthedocs.io/en/stable/index.html>



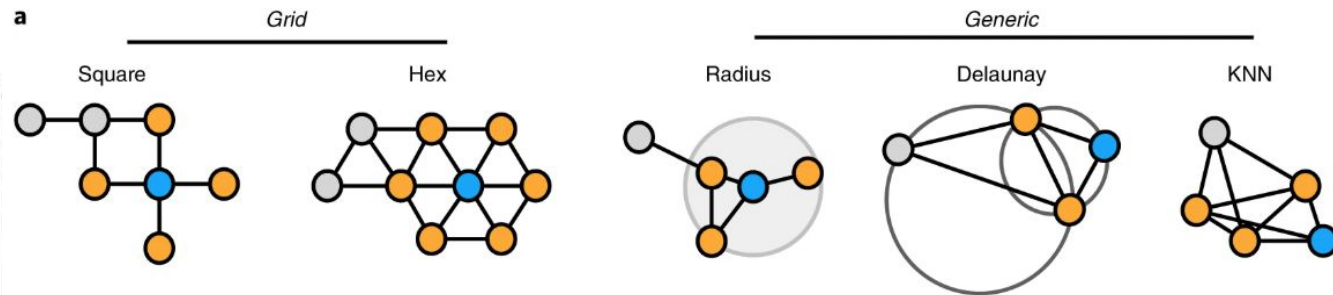
Which algorithm should I use?

- SquidPy
 - Visualization of clusters (analyzed using Scanpy)



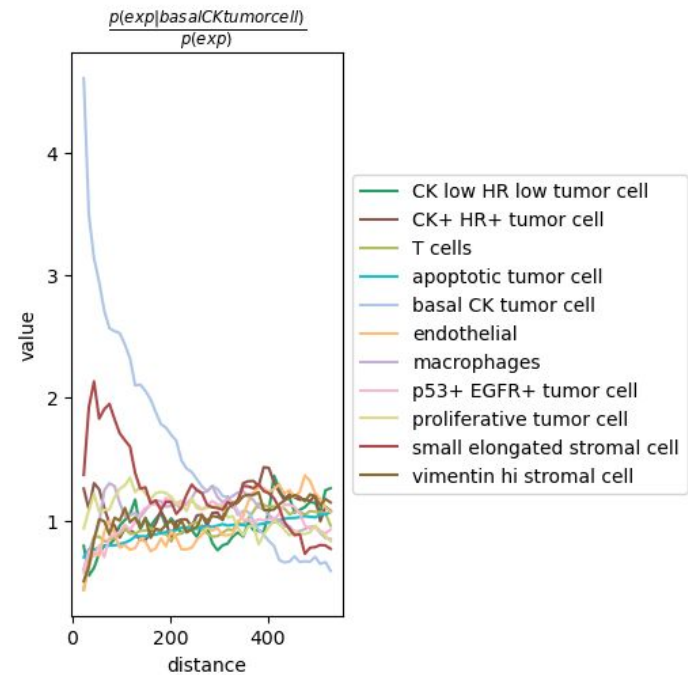
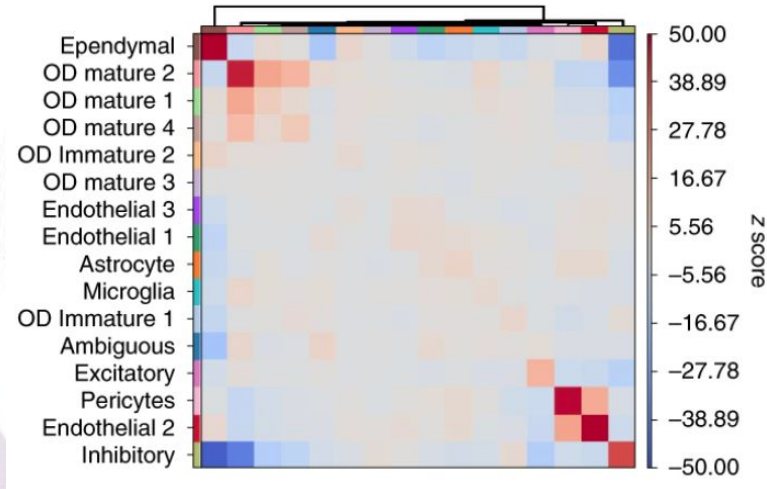
Which algorithm should I use?

- SquidPy
 - Graph-based measures to find spatial patterns



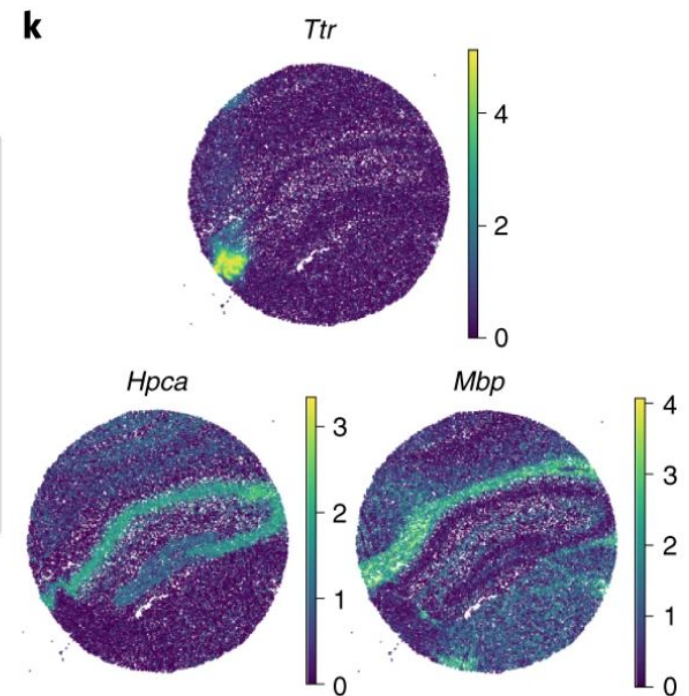
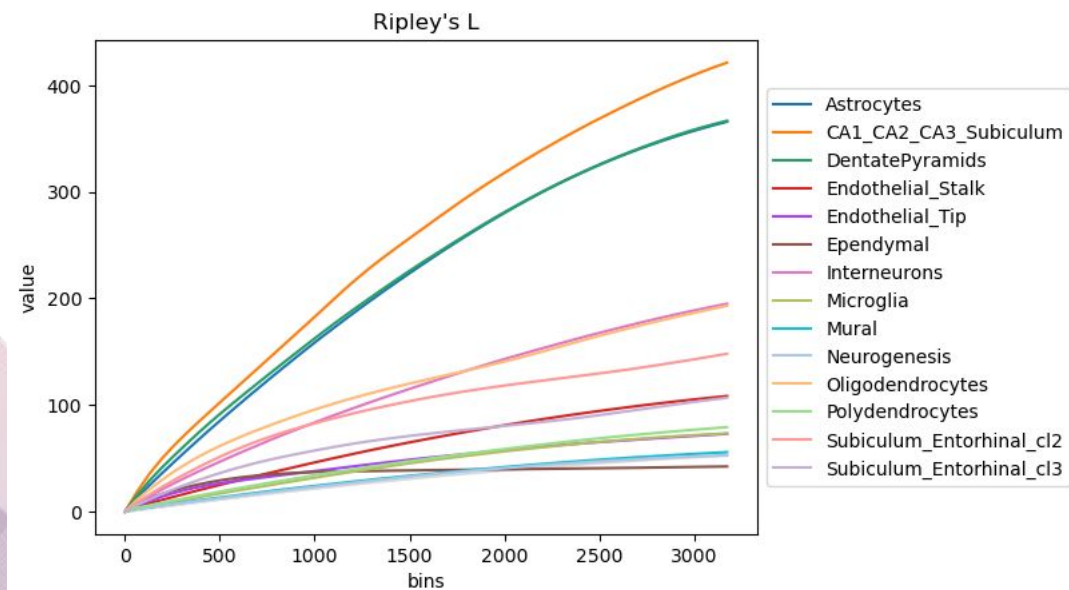
Which algorithm should I use?

- SquidPy
 - Spatial enrichment / co-occurrence analysis



Which algorithm should I use?

- SquidPy
 - Spatial pattern finding (Moran's I, Ripley's L)

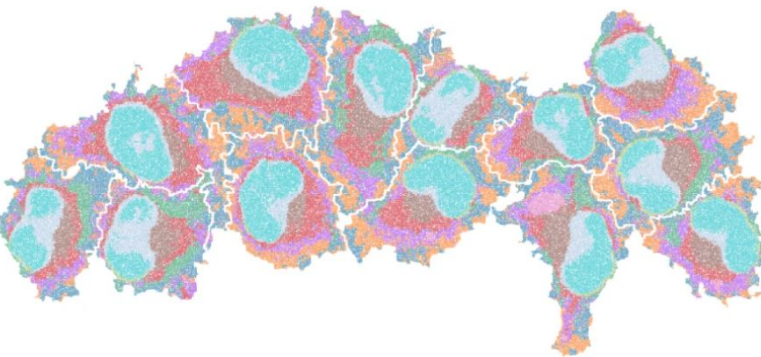


Which algorithm should I use?

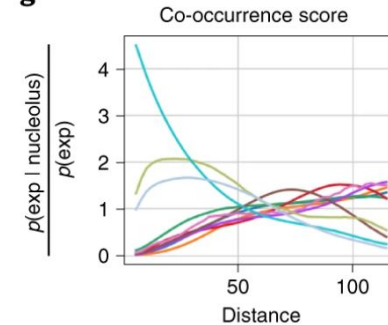
- SquidPy
 - Subcellular spatial analysis

f

• Cell_periphery_1
• Cell_periphery_2
• Cell_periphery_3
• ER_mitochondria_1
• ER_mitochondria_2
• Endosomes_Golgi_1
• Endosomes_Golgi_2
• Nuclear_envelope
• Nucleolus
• Nucleus

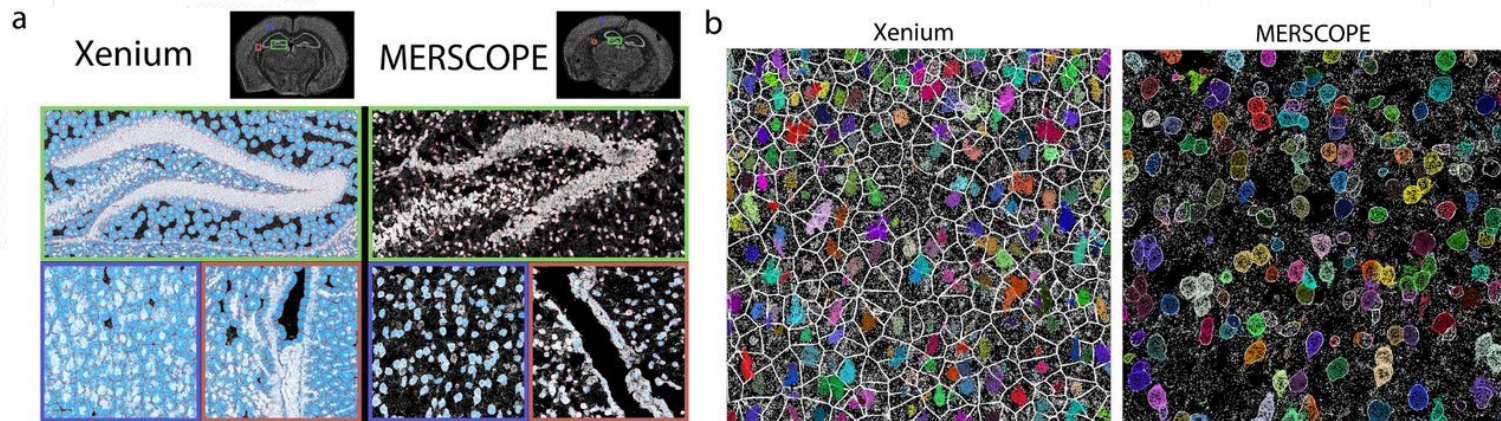


g



Which algorithm should I use?

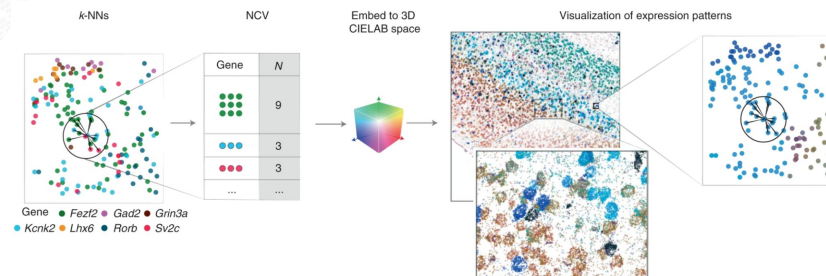
- Cell segmentation
 - For FISH/ISS-based (MERSCOPE, Xenium) or high resolution ST data (STOmics, Visium HD), you need to segment cells



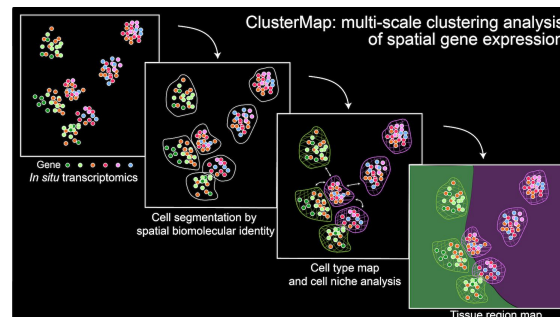
- However, the quality of reference segmentation is often limited

Which algorithm should I use?

- Cell segmentation (for imaging-based methods)
 - You can try below methods:
 - Baysor (<https://github.com/kharchenkolab/Baysor>)



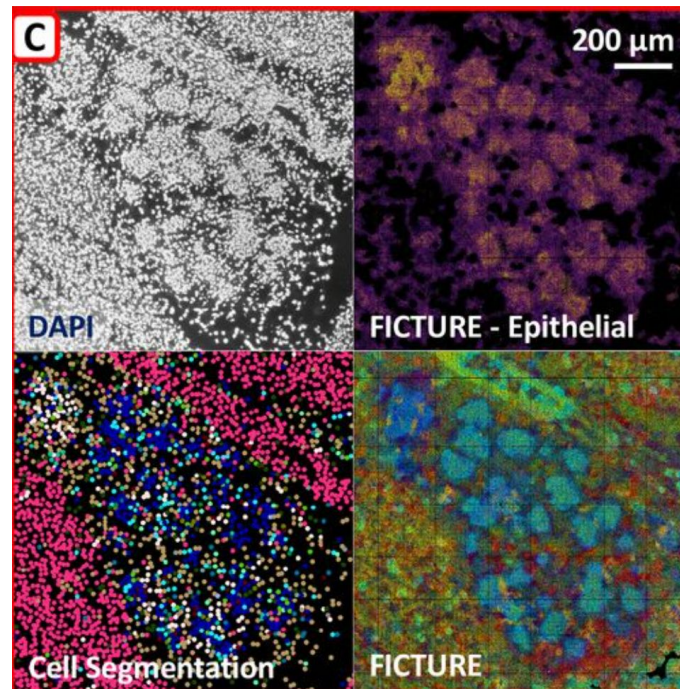
- Clustermmap (<https://github.com/wanglab-broad/ClusterMap>)



Which algorithm should I use?

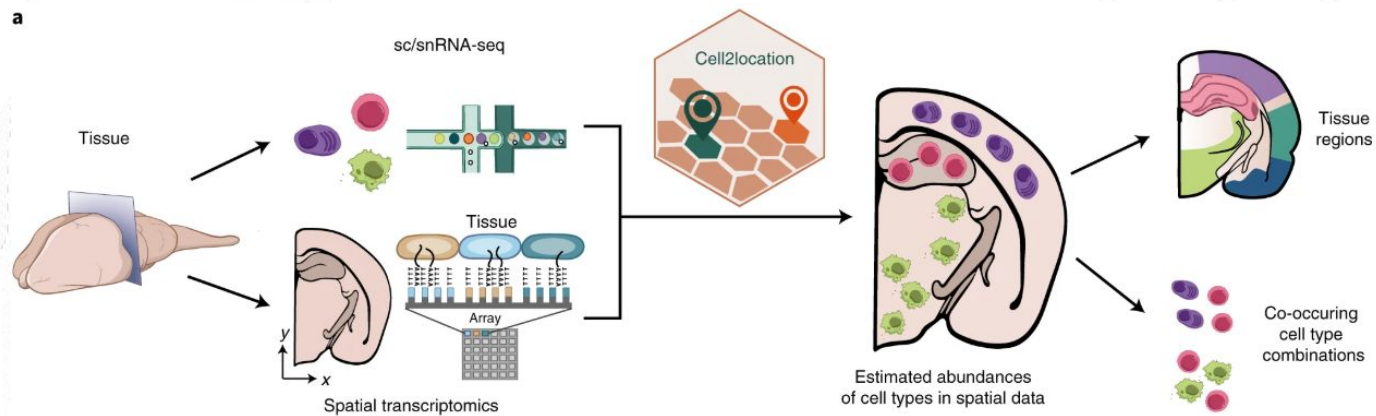
- Cell segmentation (for NGS-based methods)
 - You can try FICTURE, but it does not give you boundaries

(<https://www.nature.com/articles/s41592-024-02415-2>)



Which algorithm should I use?

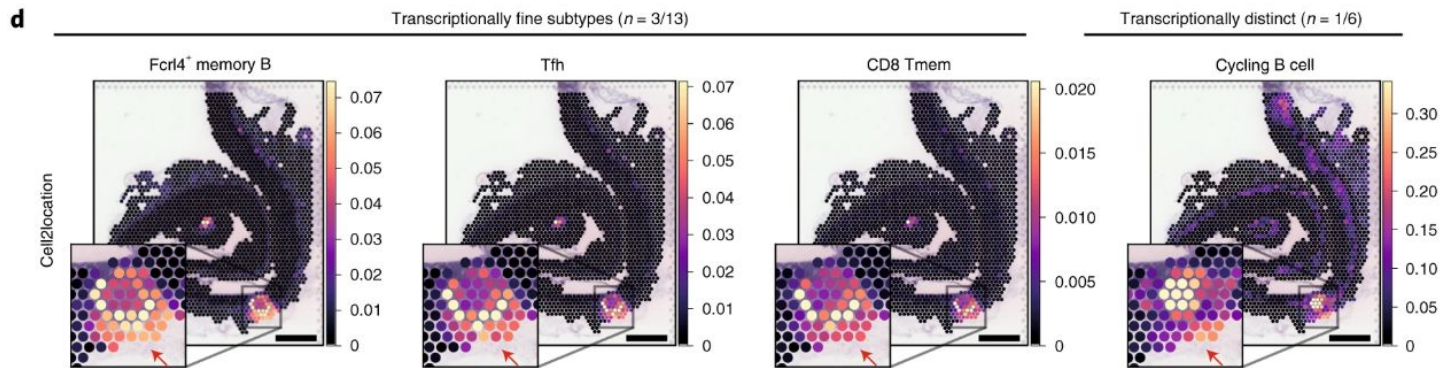
- Visium
 - Deconvolution using “Cell2location”



- It requires an additional sc/snRNA-seq data!

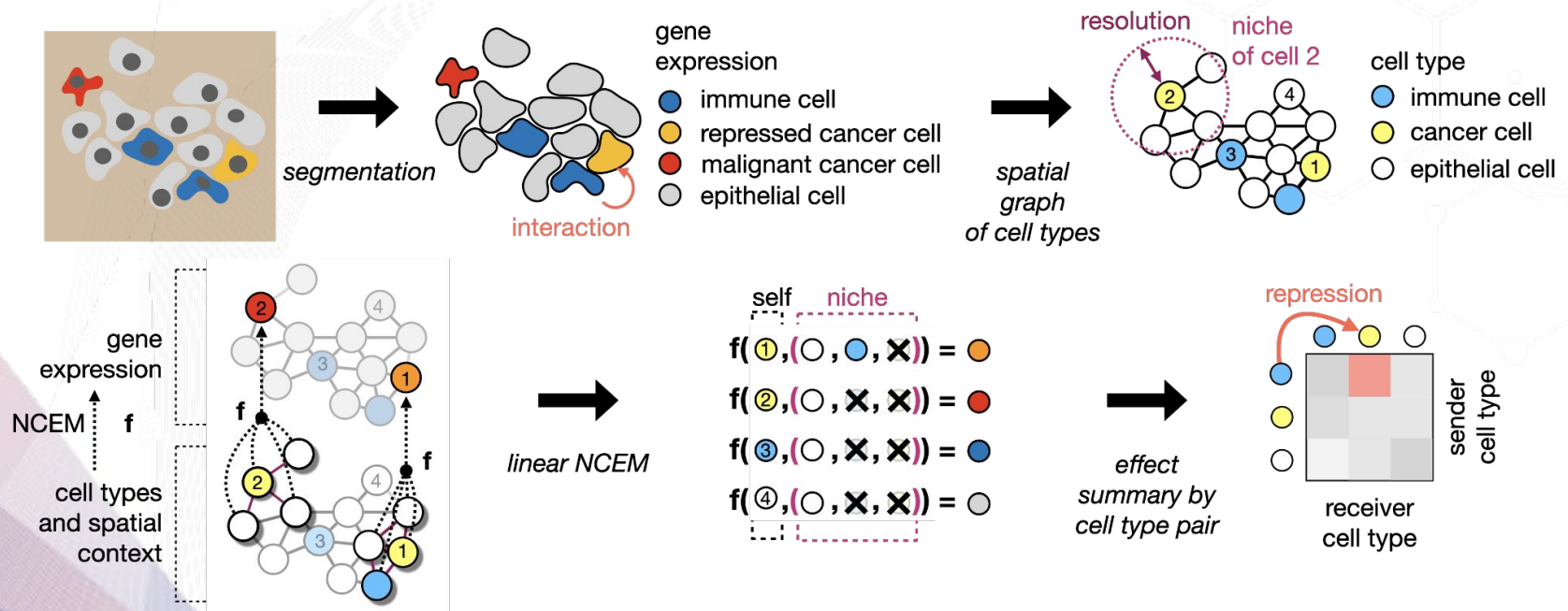
Which algorithm should I use?

- Visium
 - Cell2location enables accurate identification of spatial cell type organizations



Which algorithm should I use?

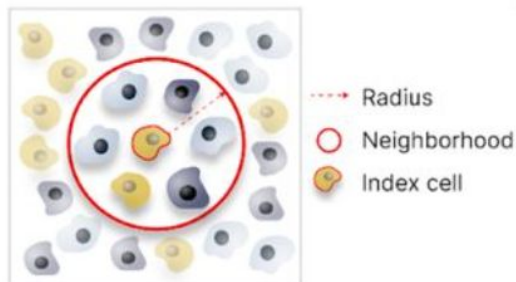
- DL/ML Tools
 - NCEM



Which algorithm should I use?

- DL/ML Tools
 - Nicheformer

Spatial cell type label transfer

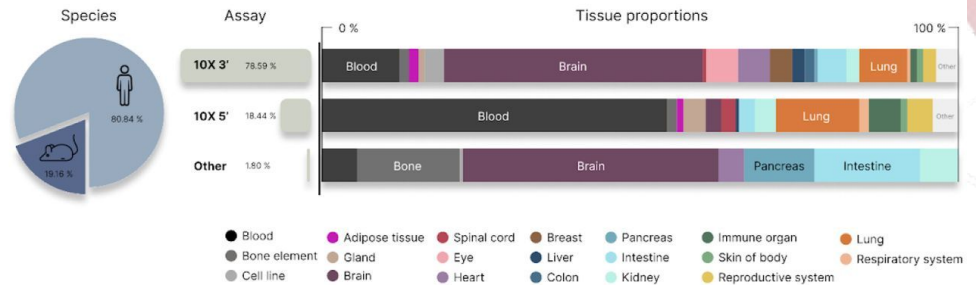


Fine-tuning task

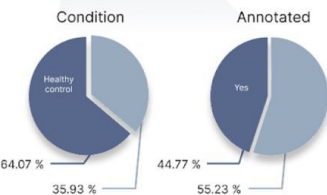
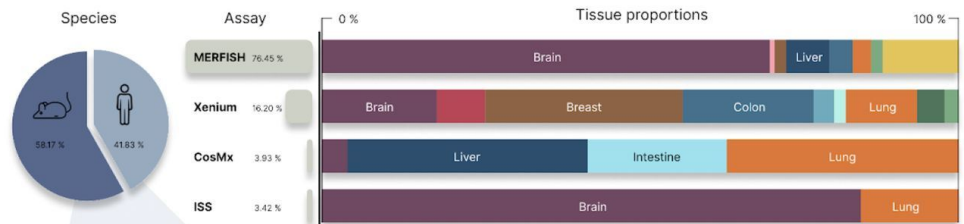
Neighborhood density: 6

Neighborhood composition: .33 .5 .17

A Dissociated collection - 57.06 million cells



B Spatial collection - 53.83 million cells



C

Nicheformer data collection schema

- Ensembl gene id
- Gene name
- Orthologous gene id



- Suspension type
- Donor id
- Condition id

Ontology term ids:

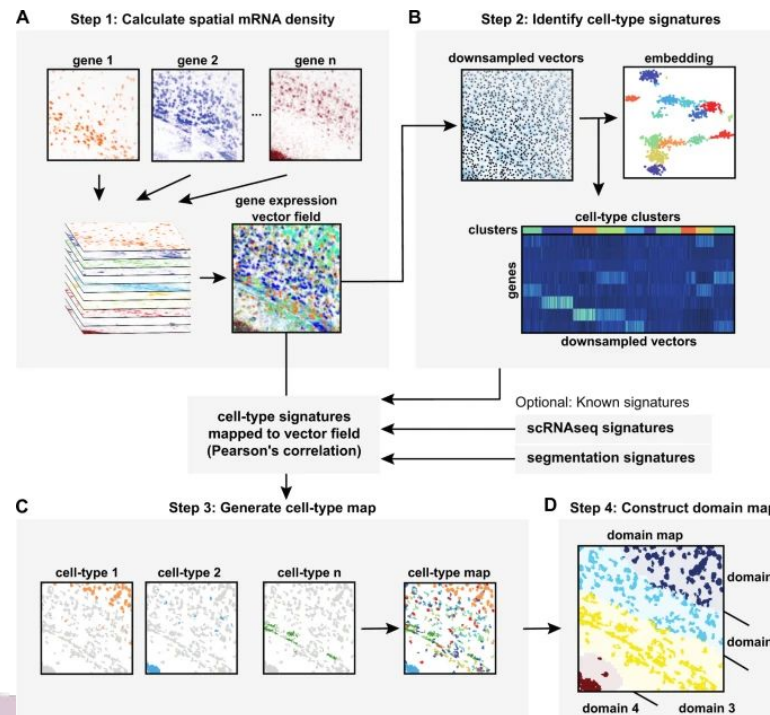
- Assay
 - Sex
 - Organism
 - Tissue
- Spatial data only:**
- Spatial coordinate x
 - Spatial coordinate y
 - Library key
 - Author cell type (optional)
 - Niche label (optional)
 - Region label (optional)

Which algorithm should I use?

- Summary
 - No perfect tool for cell segmentation yet!
 - New methods are already available!
(they would be difficult to run, though)
 - People are working with standard file formats!
 - SquidPy would be so far the best tool for beginners!

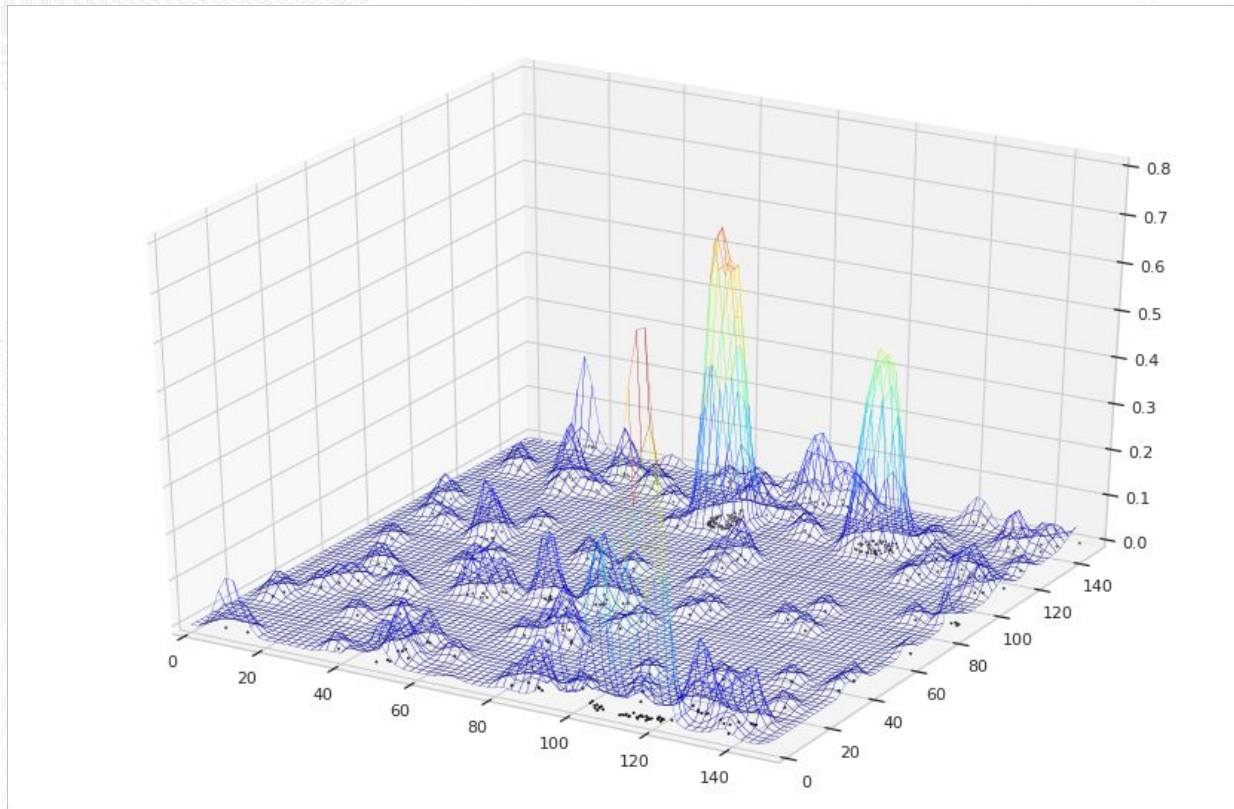
Which algorithm should I use?

- SSAM
 - As there is no perfect tool for cell segmentation, you can try “segmentation-free” cell-type analysis of ST data



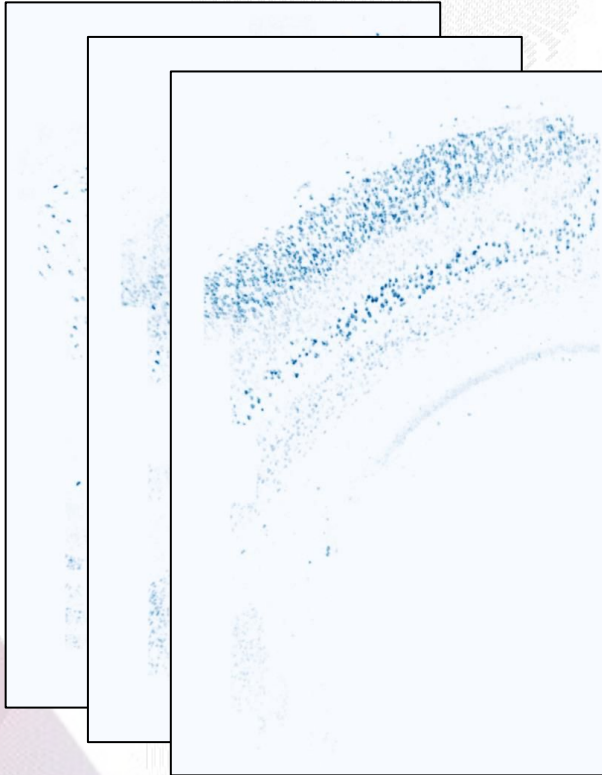
Which algorithm should I use?

- SSAM



Which algorithm should I use?

- SSAM



- Stack 33 genes
→ Vector field with 33 dimensions
- i -th gene expression (E_i):

$$\hat{E}_i(\vec{x}) = \hat{\sigma}_i(\vec{x}) N_i$$

where

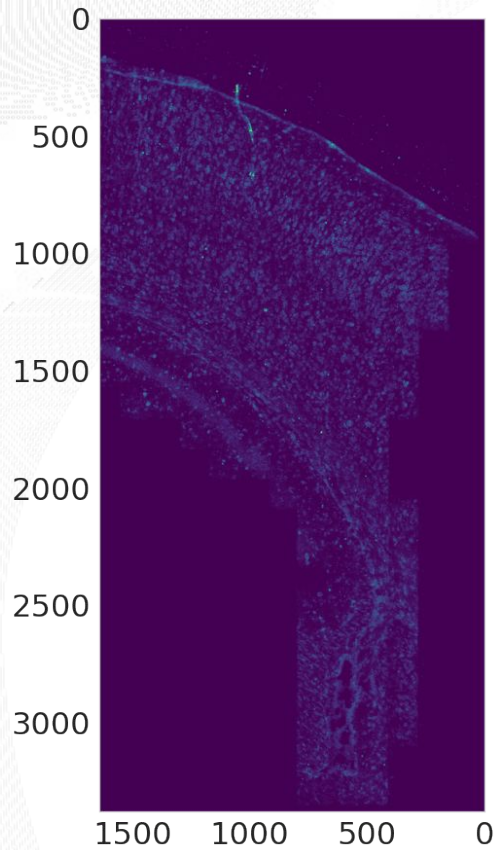
E_i : expression of i -th gene

σ_i : estimated density of i -th gene

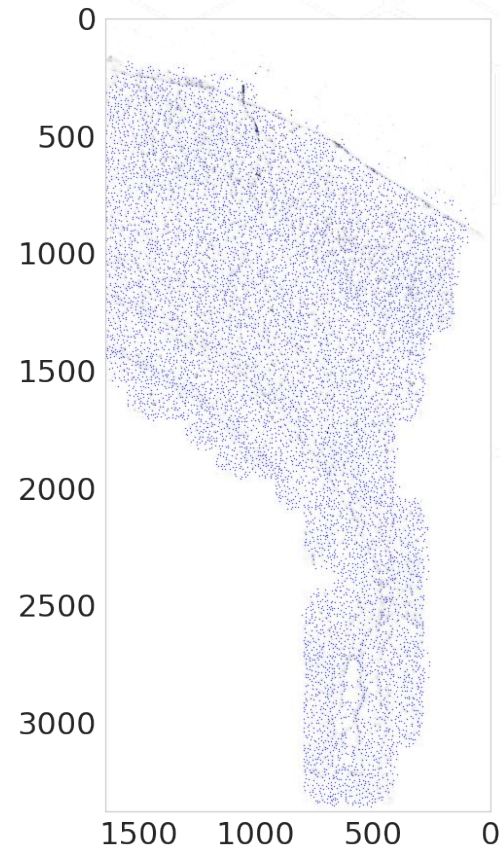
N_i : number of mRNAs of i -th gene

Which algorithm should I use?

- SSAM



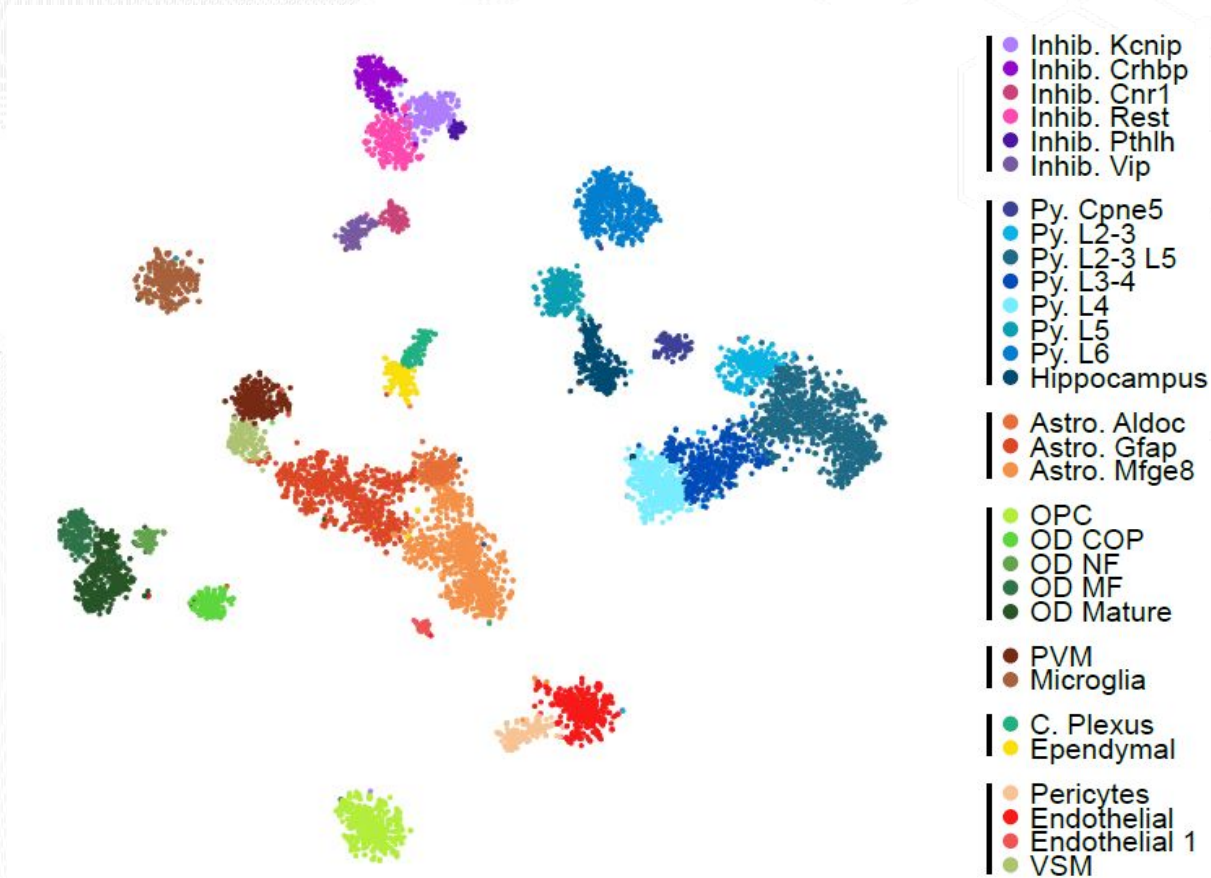
3,380 x 2,080 = 7,030,400 vectors



L1 maxima

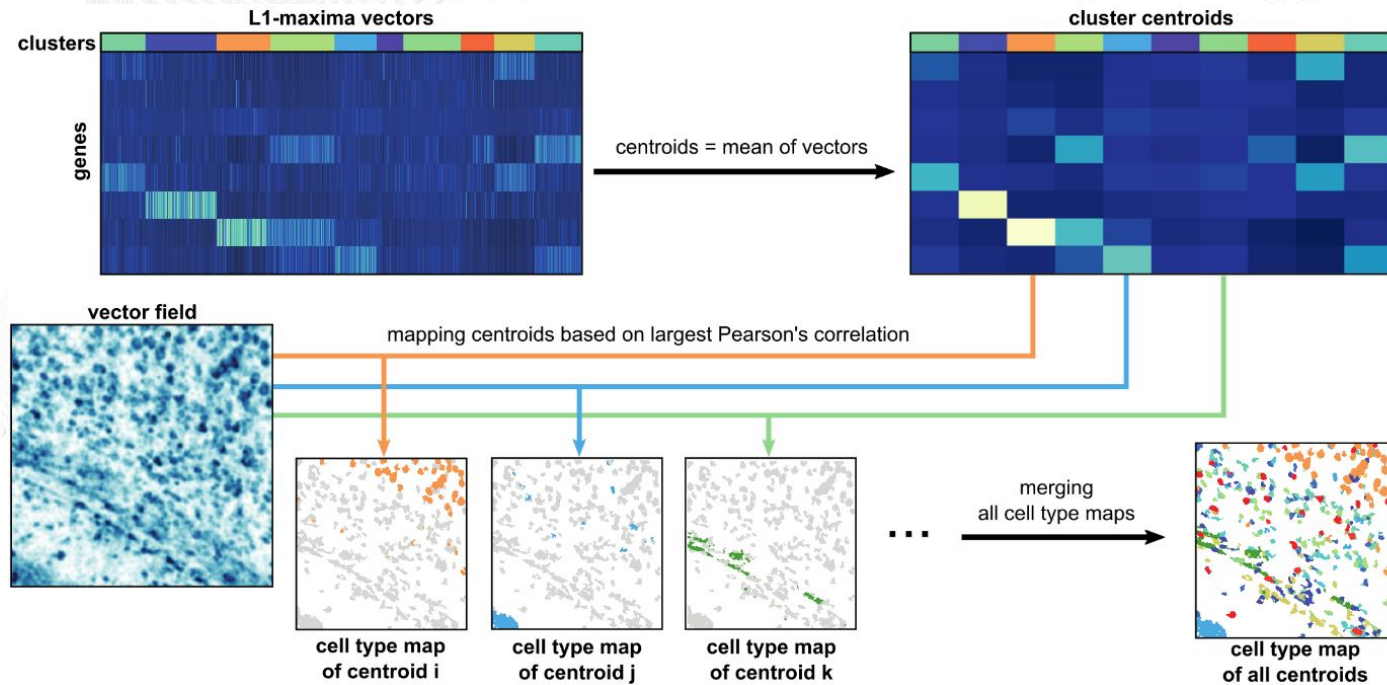
Which algorithm should I use?

- SSAM



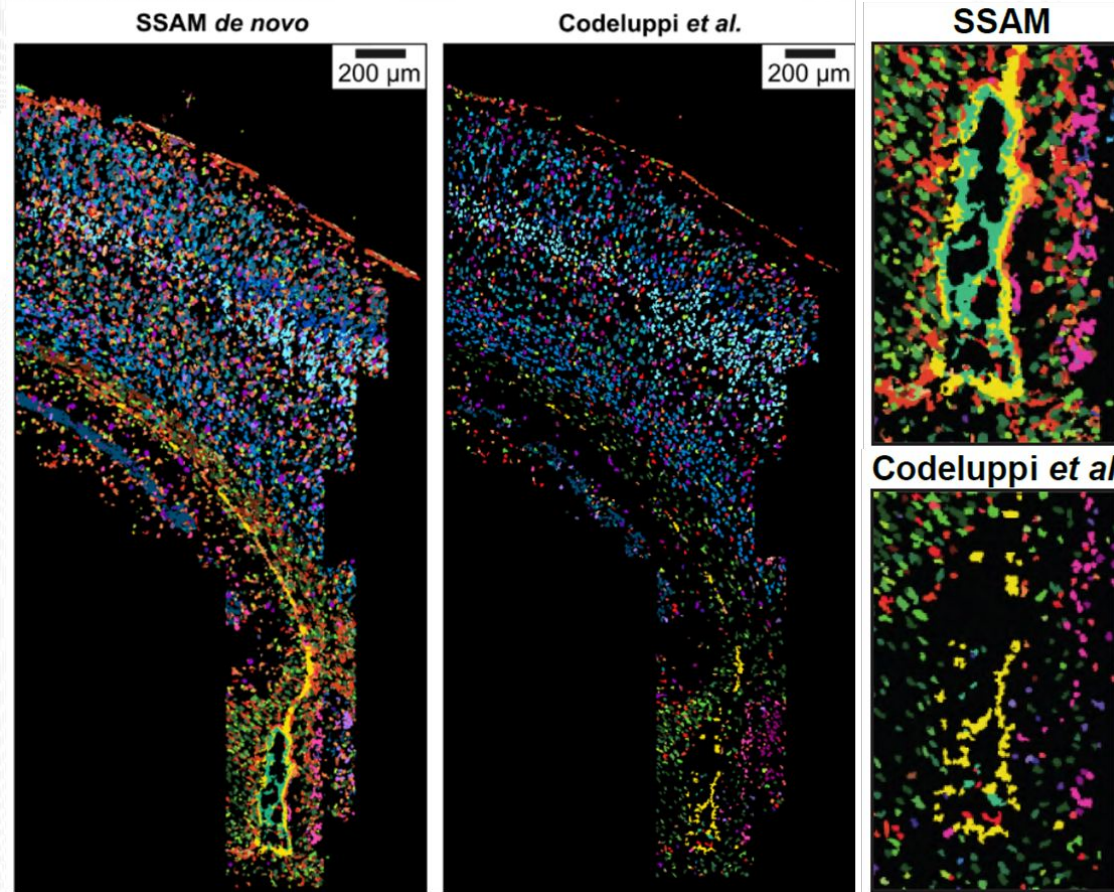
Which algorithm should I use?

- SSAM



Which algorithm should I use?

- SSAM



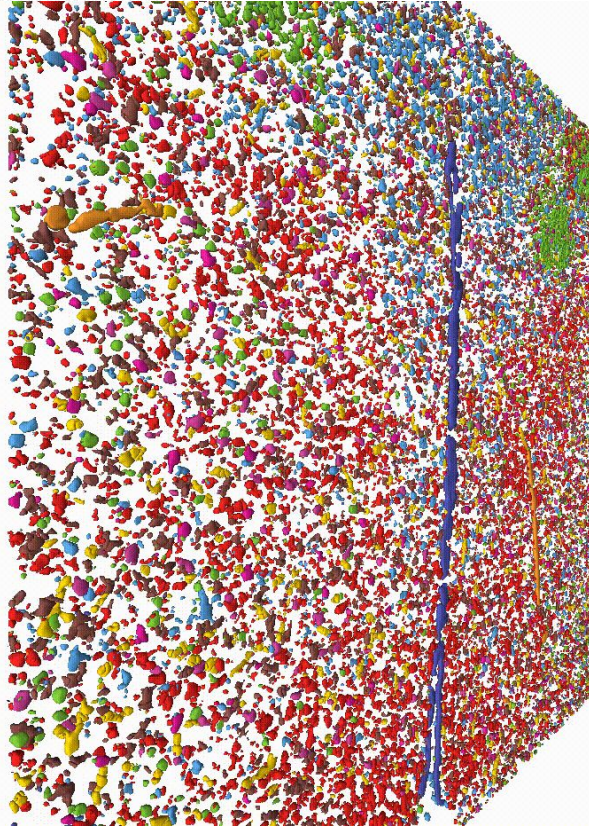
Which algorithm should I use?

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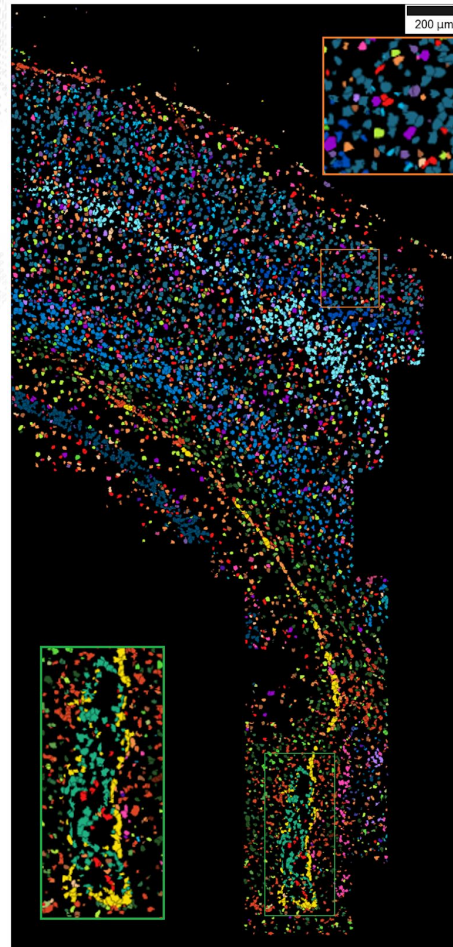
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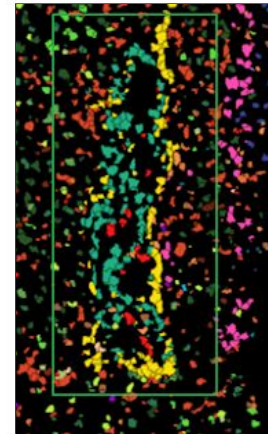


Which algorithm should I use?

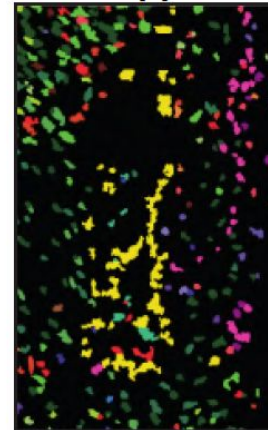
- SSAM



SSAM

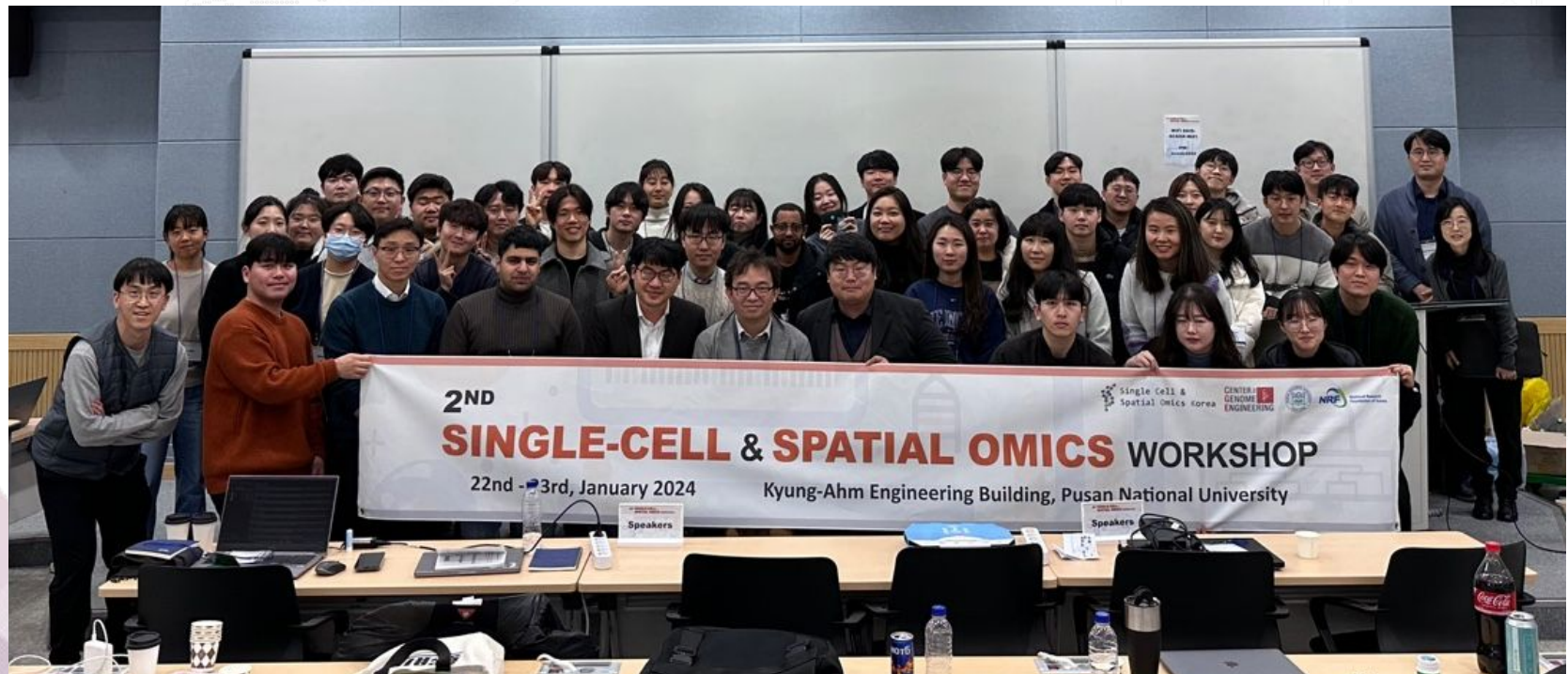


Codeluppi *et al*



Single Cell and Spatial Omics Korea (SCSOK)

- 2nd SCSOK Workshop in 2024 (Busan)



Single Cell and Spatial Omics Korea (SCSOK)

- The 3rd SCSOK Workshop in 2025
 - Dates: June 23 - 28, 2025
 - Organized by: Jeongbin Park, Junil Kim, Heetak Lee
 - Venue: Soongsil University
 - Confirmed speakers
 - Keynotes: **Nir Yosef, Jingyi Jessica Li**
 - International speakers: **Guangdun Peng, Shyam PRABHAKAR, Tim Stuart, Yasuhiro Murakawa**
 - Domestic speakers: To be announced!

Single Cell and Spatial Omics Korea (SCSOK)

- SCSOK Hackathon 2025 - 2nd edition
 - “Bring your own data; We will analyze them in a week”
 - Soon be announced at: <https://scsok.io/events>



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